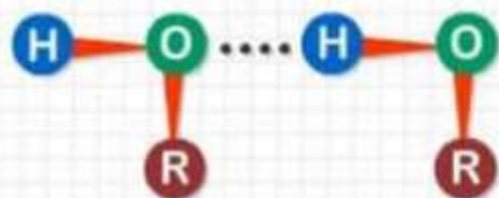


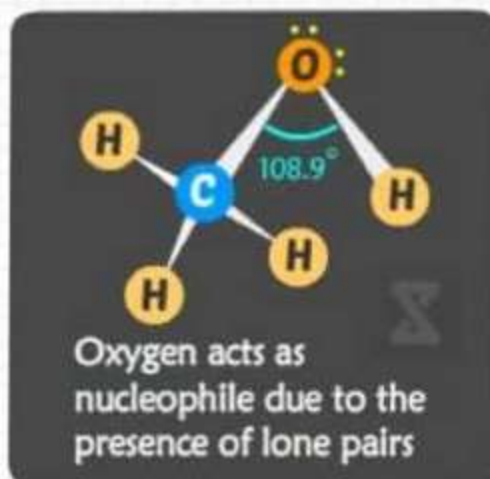


ALCOHOLS

BOILING POINT



Existence of Intermolecular Hydrogen bonding results in higher boiling point than hydrocarbons.



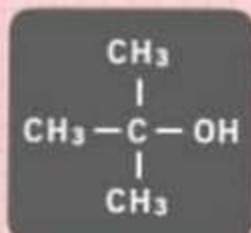
SOLUBILITY



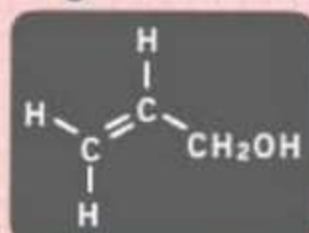
Alcohols are soluble in water, due to the presence of Hydrogen Bonds.

NOMENCLATURE

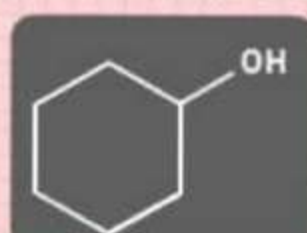
- Name the longest carbon chain that contains the carbon atom bearing the -OH group. Drop the final -e from the alkane name, and add the suffix -ol .
- Number the longest carbon chain starting at the end nearest the -OH group, and use the appropriate number, if necessary, to indicate the position of the -OH group.
- Name the substituents, and give their numbers as for an alkane or alkene.



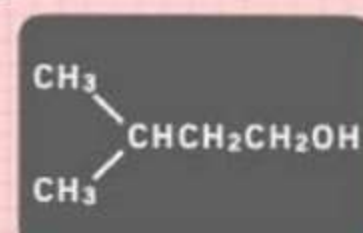
IUPAC Name
2-methyl-2-propanol
Common Name
 t -butyl alcohol



IUPAC Name
2-propen-1-ol
Common Name:
Allyl alcohol



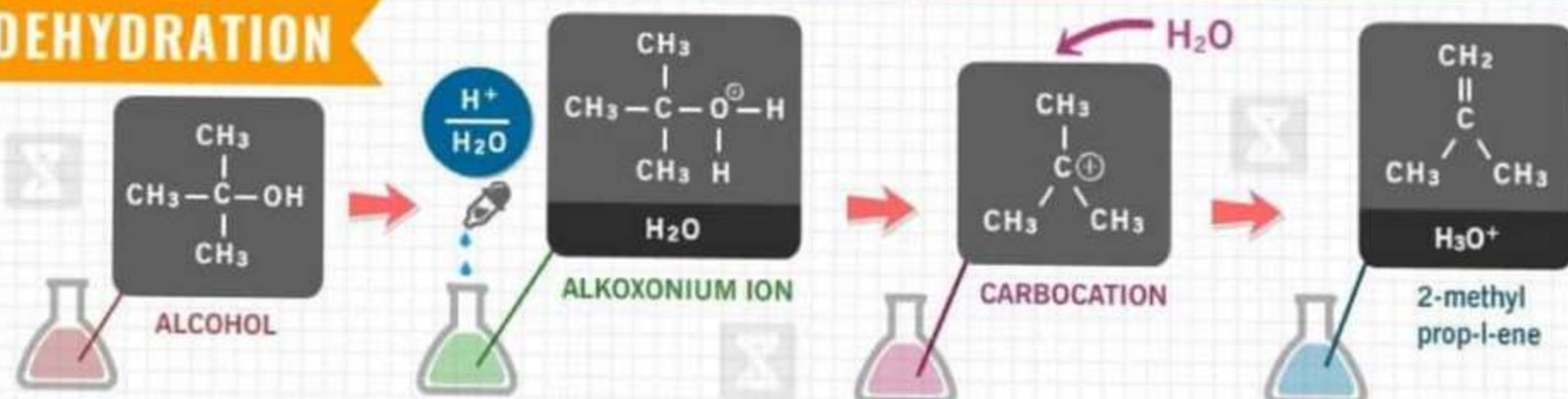
IUPAC Name
Cyclohexanol
Common Name
Cyclohexanol alcohol



IUPAC Name
3-methyl-1-butanol
Common Name
Isoamyl alcohol

CHEMICAL REACTIVITY

DEHYDRATION



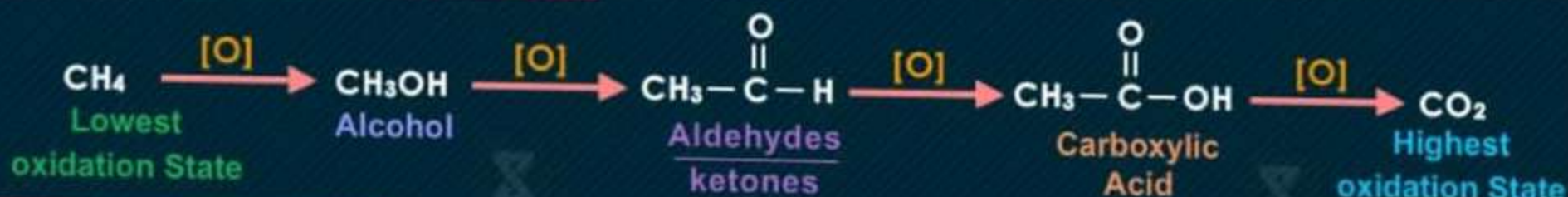
Formation of carbocation is RDS here. So always try to make a stable carbocation by rearrangement.

HOW TO PREPARE ALCOHOL?

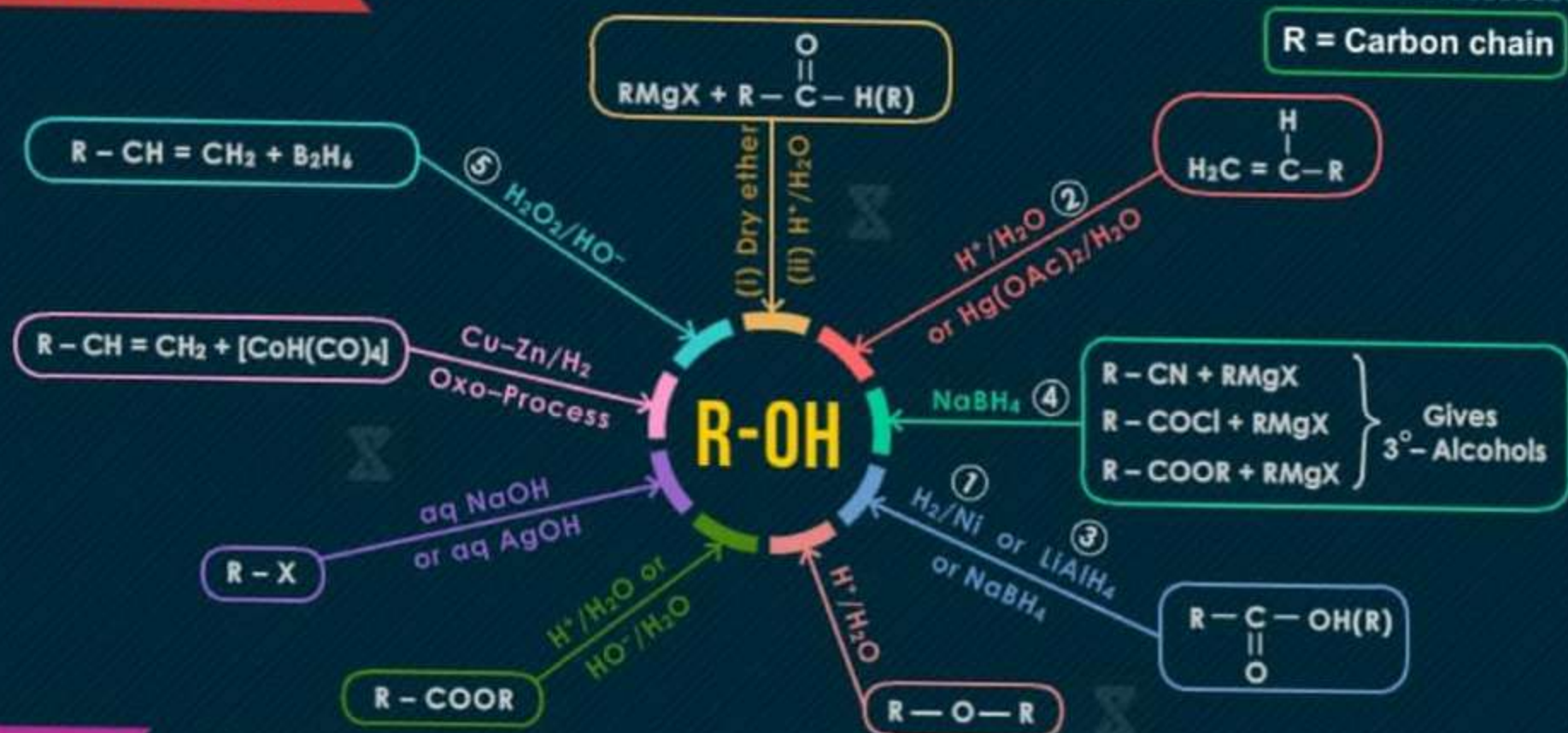


YOU NEED TO HAVE SOME **BASIC INFORMATION**

HIERARCHY OF OXIDATION



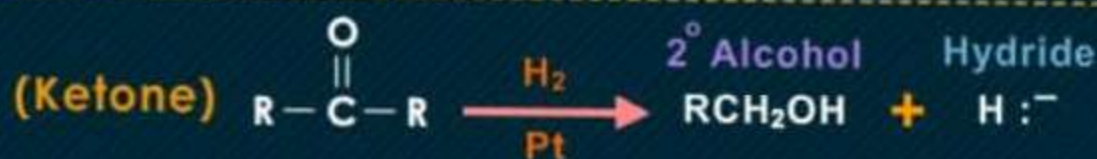
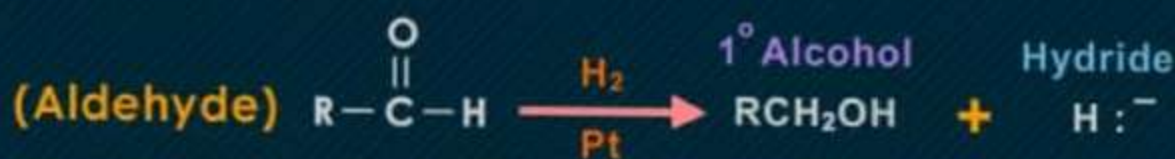
WAYS TO DO IT



Notes

- H_2/Ni reduces all reducible groups including olefinic bonds.
- Reaction proceeds via carbocation, rearrangement may accompany.
- LiAlH_4 does not reduce olefinic bonds.
- Avoid rearrangement.
- Anti-Markownikoff's hydration takes place.

CATALYTIC HYDROGENATION





ETHERS

Ethers are a class of organic compounds that contain an oxygen between two alkyl or aryl groups. They have the formula $R-O-R'$, where R and R' are alkyl groups. These compounds are used in **dye, perfumes, oils, waxes** and **industrial use**. Ethers are named as **alkoxyalkanes**.



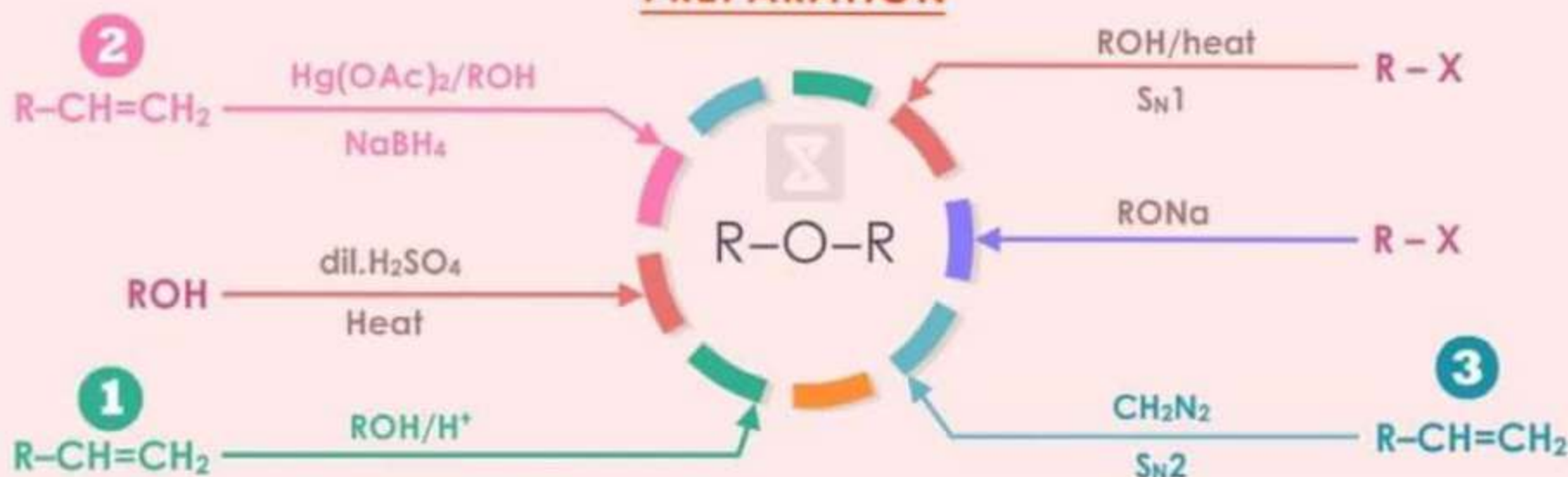
NOMENCLATURE OF ETHERS

Ethers are compounds having two alkyl or aryl groups bonded to an oxygen atom, as in the formula R_1-O-R_2 . The ether functional group does not have a characteristic **IUPAC** nomenclature suffix, so it is necessary to designate it as a substituent. To do so the common alkoxy substituents are given names derived from their alkyl component (**below**):

ALKYL GROUP	NAME	ALKOXY GROUP	NAME
CH_3-	Methyl	CH_3O-	Methoxy
CH_3CH_2-	Ethyl	CH_3CH_2O-	Ethoxy
$(CH_3)_2CH-$	Isopropyl	$(CH_3)_2CHO-$	Isopropoxy
$(CH_3)_3C-$	Tert-Butyl	$(CH_3)_3CO-$	Tert-Butoxy
C_6H_5-	Phenyl	C_6H_5O-	Phenoxy

Ethers can be named by naming each of the two carbon groups as a separate word followed by a space and the word ether. The **-OR** group can also be named as a substituent using the group name, alkoxy.

PREPARATION

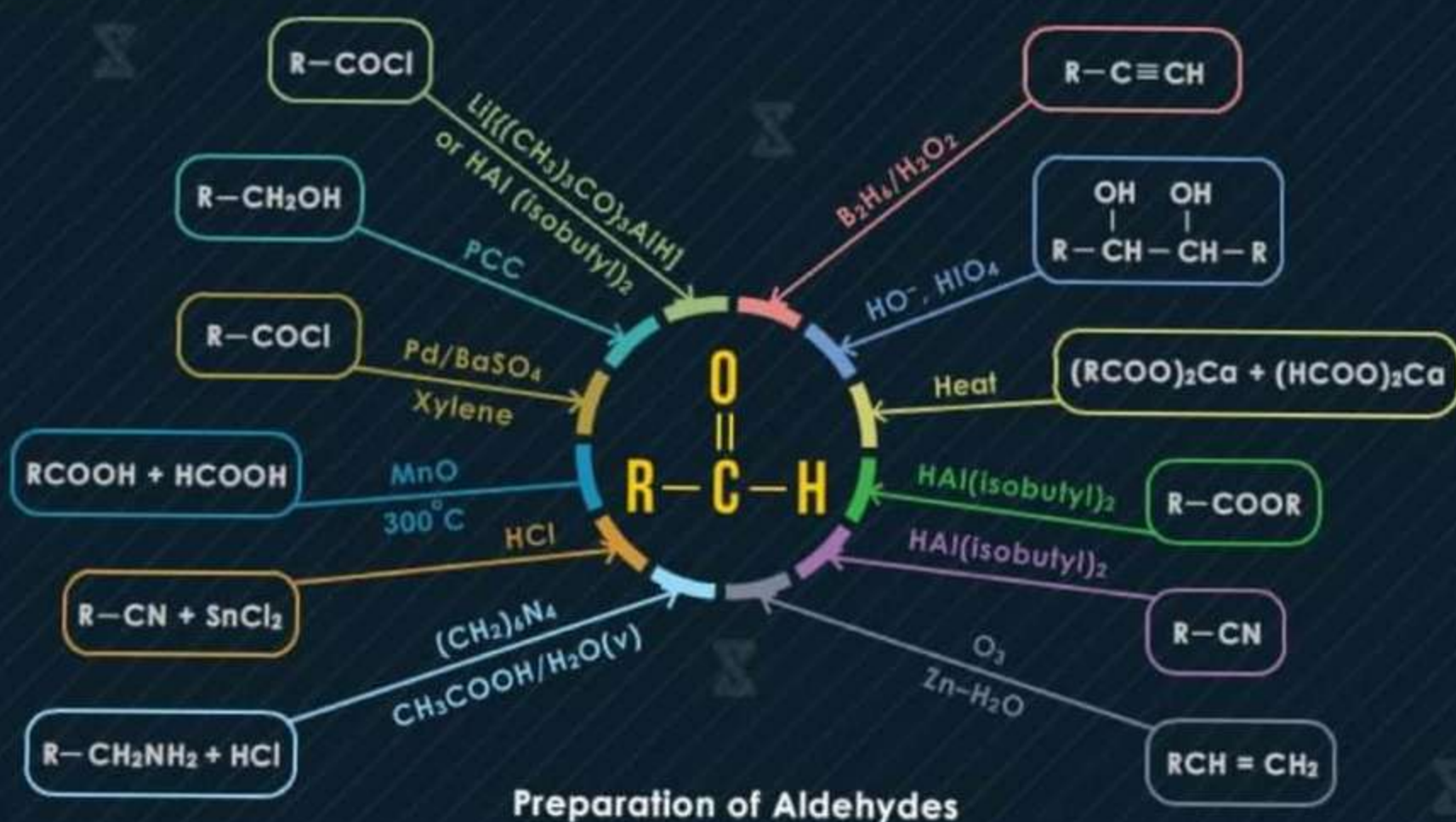


- 1 Proceeds via carbocation intermediate, rearrangement may take place.
- 2 Do not proceed via carbocation intermediate, rearrangement is avoided.
- 3 Gives methyl ether ($RCH_2CH_2OCH_3$)

ALDEHYDE & KETONES

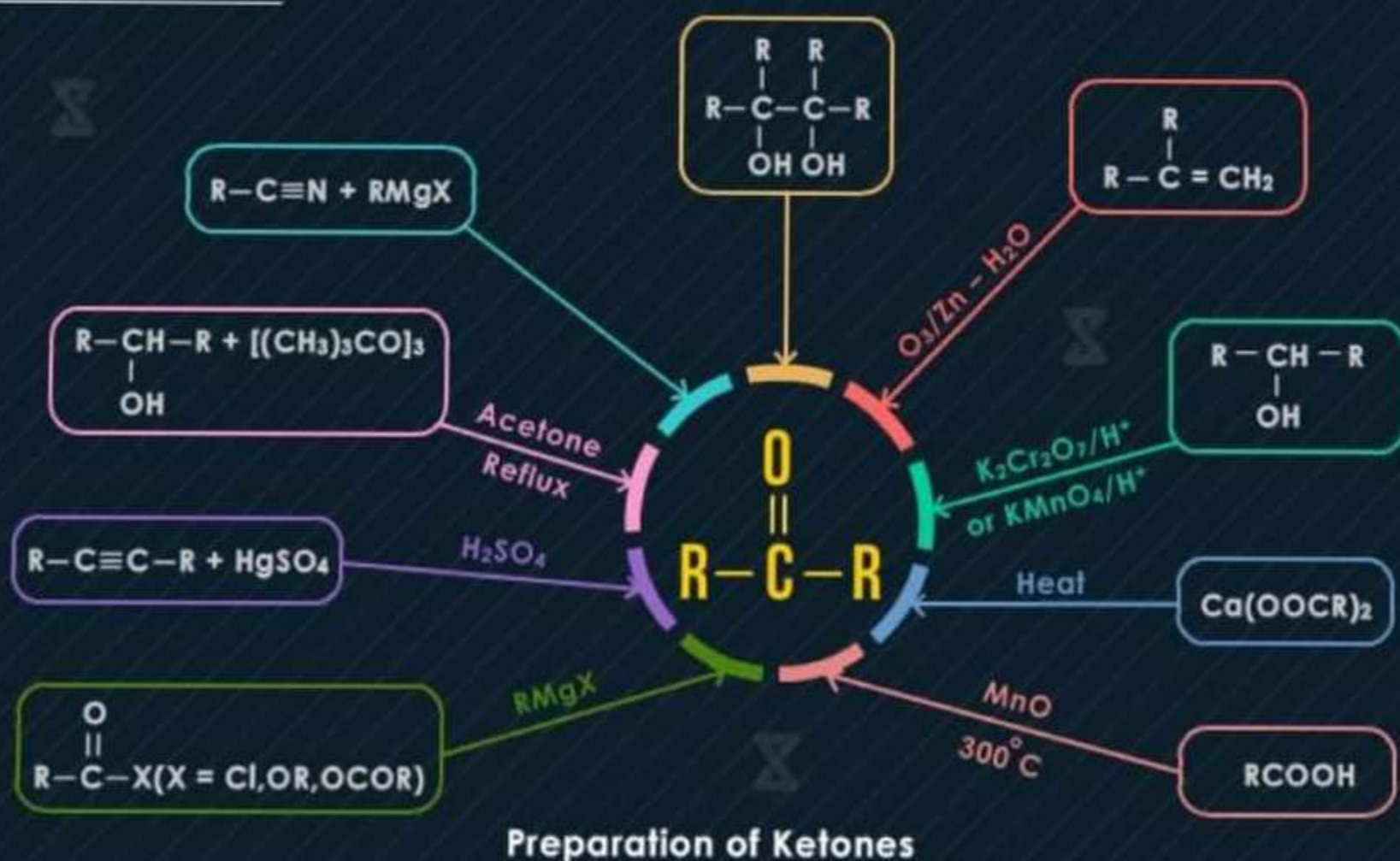
Aldehydes

PREPARATION



Ketones

PREPARATION

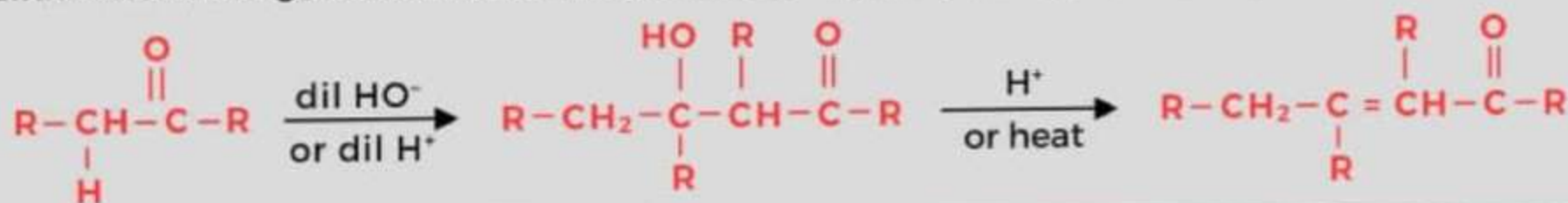


IMPORTANT REACTIONS ALDYHYDE AND KETONE

ALDOL CONDENSATION



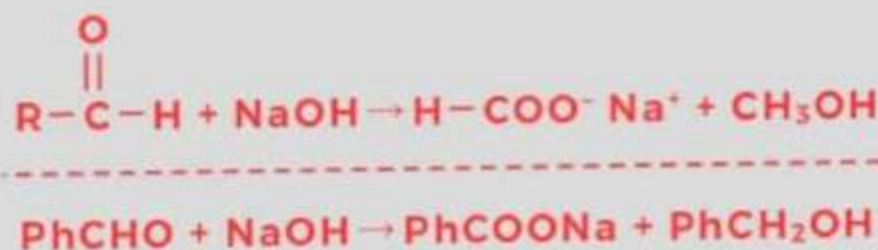
Aldehydes or ketones containing at least one α -hydrogen on treatment with dilute alkali or dilute acid undergo condensation to produce β -hydroxy aldehyde or β -hydroxy ketone.



CANNIZARO REACTION

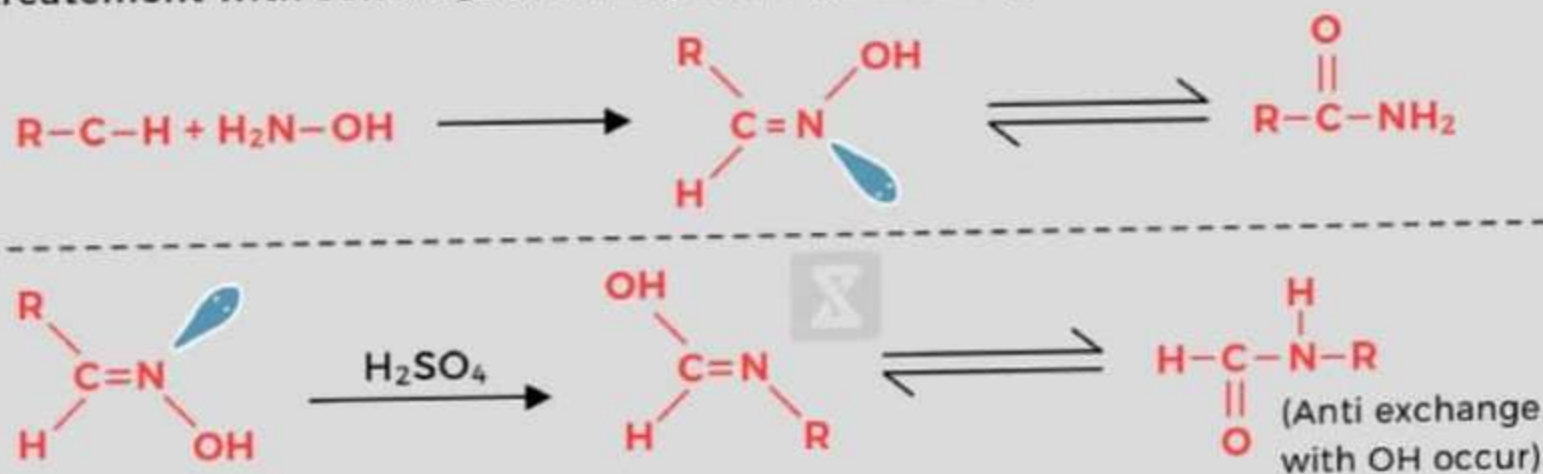


Aldehydes lacking α -hydrogen, when treated with concentrated solution of strong base, undergo mutual oxidation reaction.



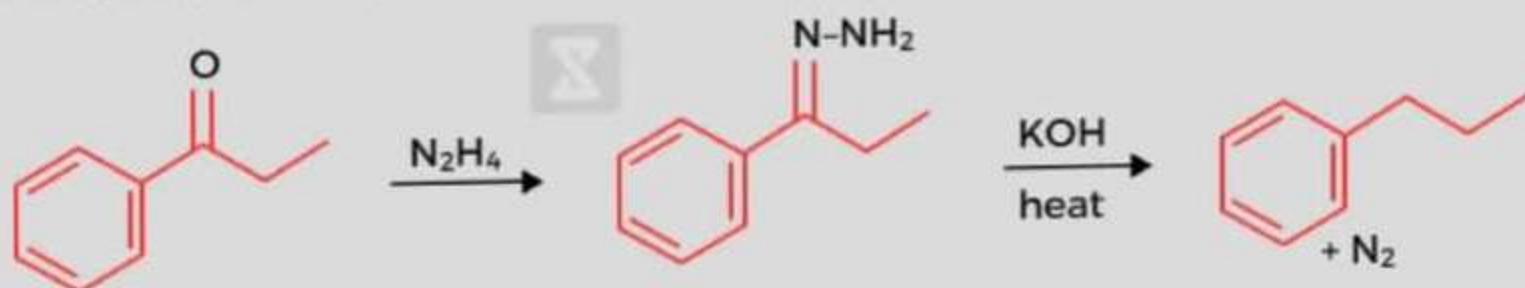
BECKMANN'S REARRANGEMENT

Aldehydes or ketones on treatment with hydroxyl amine gives oximes. Oximes on further treatment with conc. H_2SO_4 or PCl_5 undergo rearrangement as:



WOLFF – KISHNER

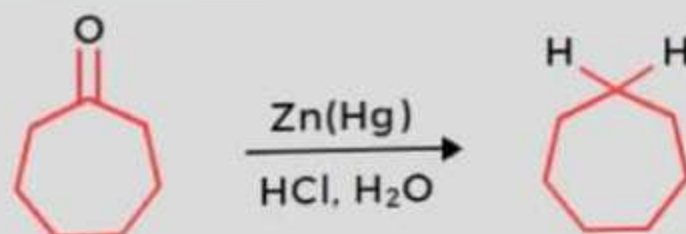
Ketone or aldehyde is converted to its hydrazone (by reaction with hydrazine) and is then treated with a strong base, which generates the reduced product.



The mechanism of hydrazone formation is analogous to imine formation.

CLEMMENSEN REDUCTION

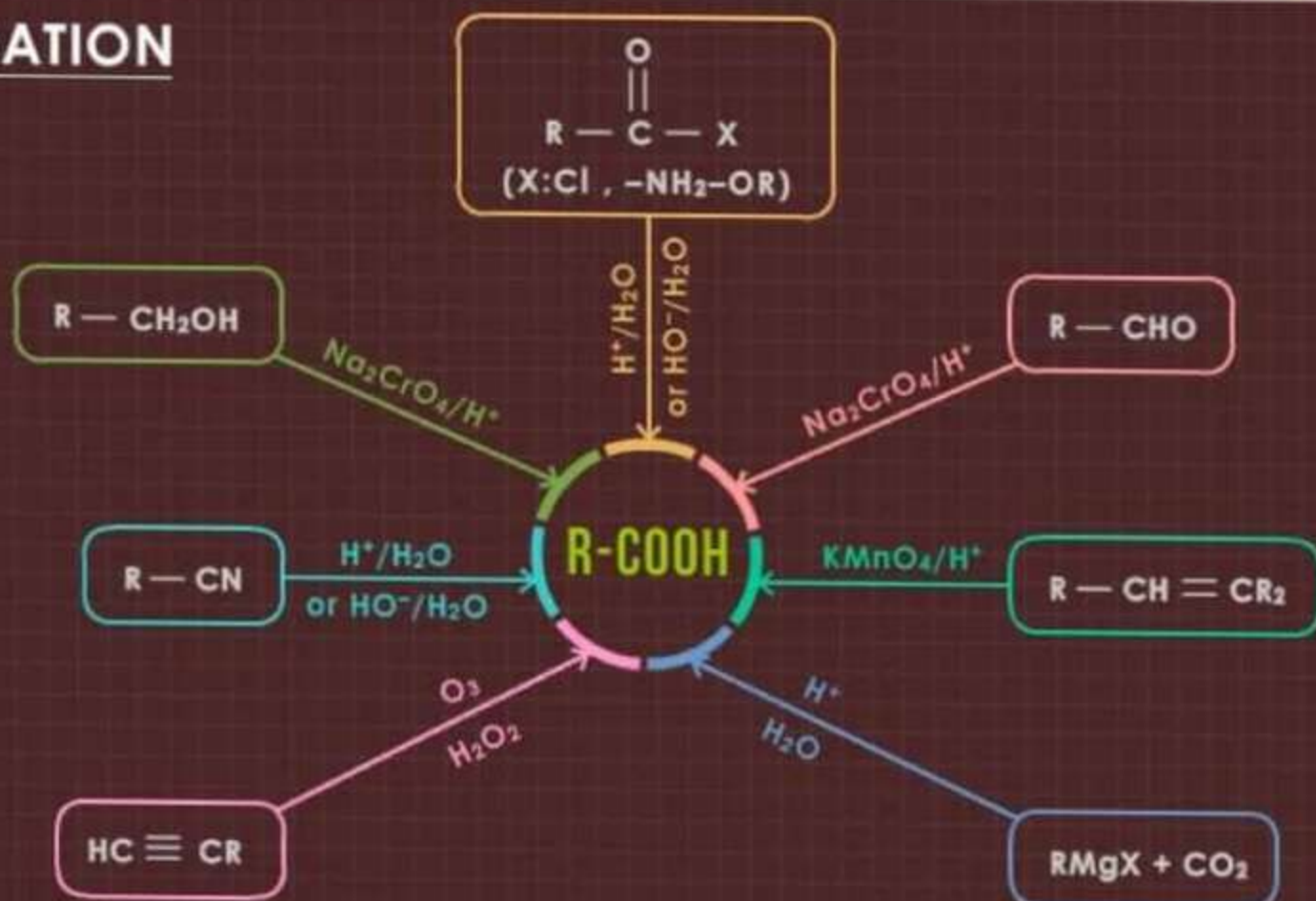
This was used in the reduction of acyl benzenes into alkyl benzenes, but it also works for other aldehydes and ketones.



CARBOXYLIC ACIDS



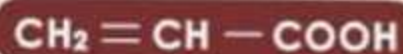
PREPARATION



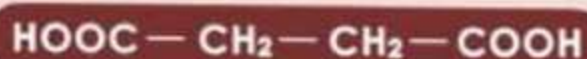
NOMENCLATURE

The **Systematic IUPAC name** consists of the corresponding hydrocarbon, in which the final -e is replaced with the suffix -oic (-dioic for dicarboxylic) and the word acid. When the carboxyl group is attached to a ring, the ending -carboxylic acid is added to the name of the parent structure. Many carboxylic acids have common names.

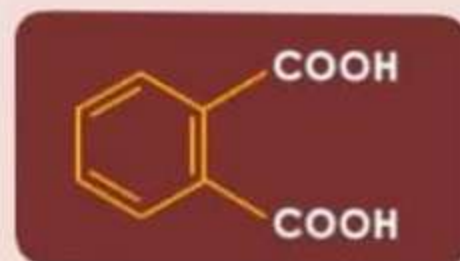
EXAMPLE :



Propen-oic acid
(acrylic acid)



Butan-di-oic acid
(succinic acid)



Benzene-1,2-dicarboxylic acid
(succinic acid)

USES

- VINEGAR CONTAINS ETHANOIC ACID**
Ethanoic acid is used in the manufacturing of rayon



- SOUR FRUITS CONTAIN CITRIC ACID**
Fruits like lemon and orange contain citric acid



- ASPIRIN IS A CARBOXYLIC ACID**
It is used for pain relief and prevention of heart attacks



- VITAMIN C CONTAINS ASCORBIC ACID**
needed to maintain health of skin, cartilage and bone



Amines

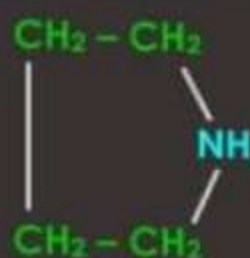
Amine are member of a family of nitrogen-containing organic compounds that is derived, either in principle or in practice, from ammonia (NH_3).

Ethylamine



An open- chain aliphatic primary amine.

Pyrolidine



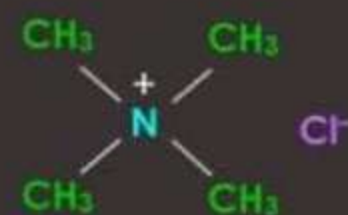
A cyclic aliphatic secondary amine.

N, N - dimethylaniline



An aromatic tertiary amine.

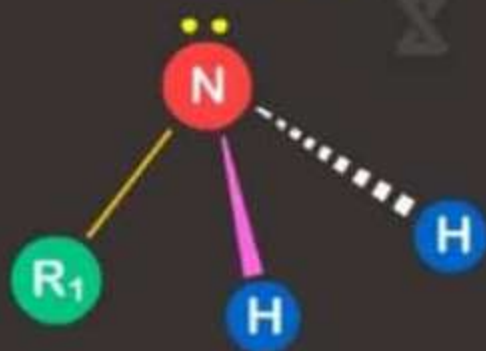
Tetramethylammonium chloride



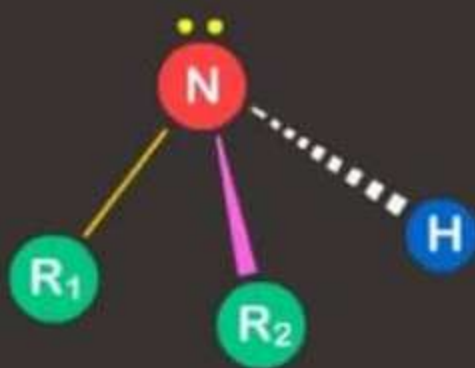
An aliphatic quaternary ammonium salt.

STRUCTURE OF AMINES

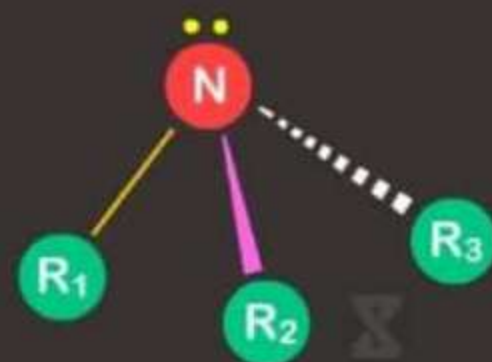
PRIMARY



SECONDARY



TERTIARY



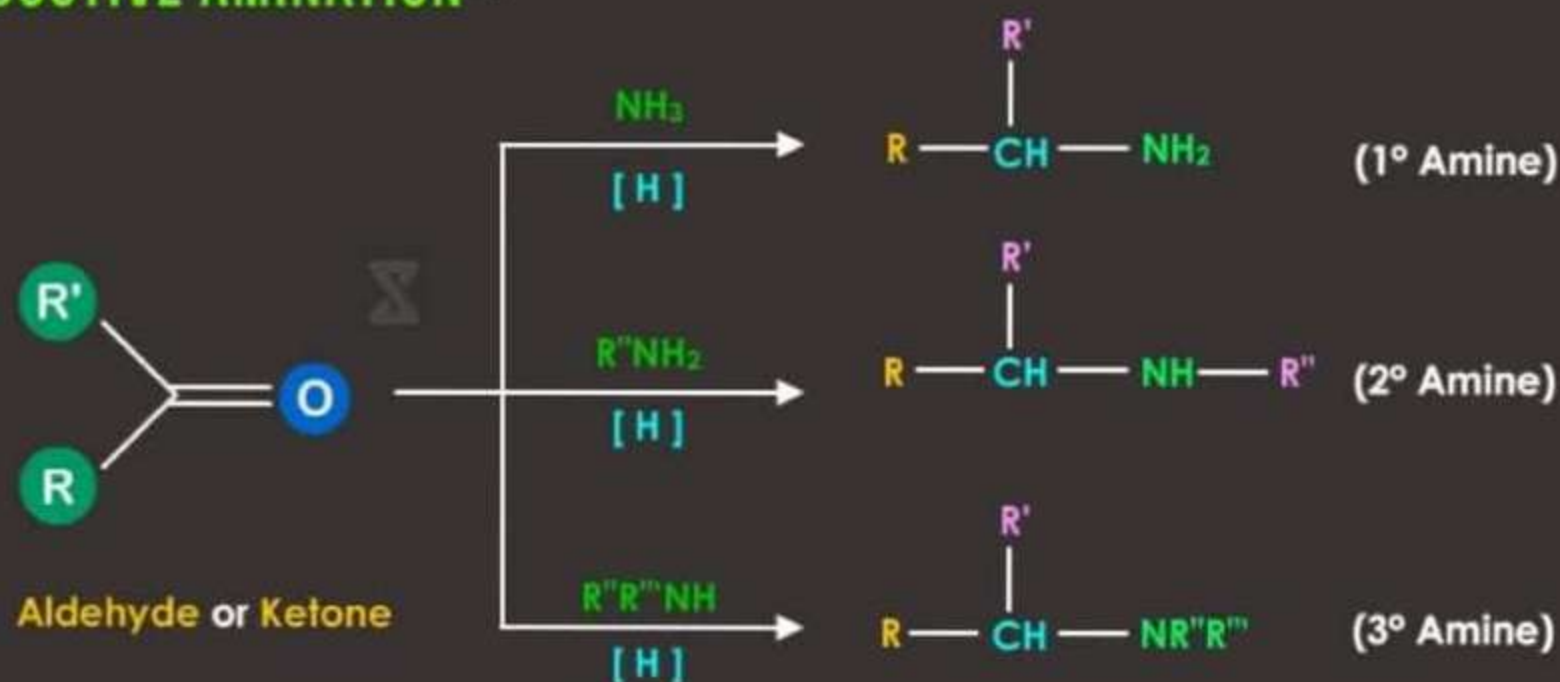
PROPERTIES

- Water Soluble
- Gaseous at room temperature
- Foul-smelling compounds
- Colourless
- High boiling point
- Burns with a yellow flame

USES OF AMINES

- Corrosion inhibitors in boilers
- Lubricating oils (morpholine)
- Stabilizers for cellulose
- Nitrate explosives (diphenylamine)
- Protectants against damage from gamma radiation (diaryl amines)

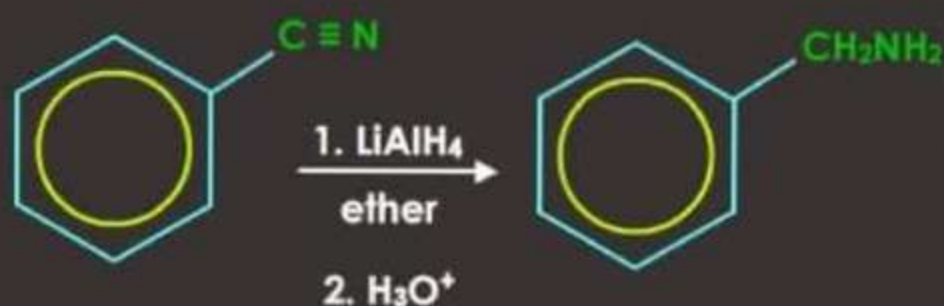
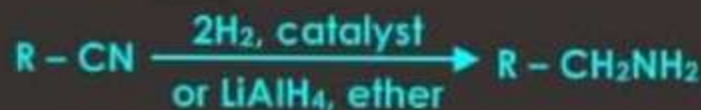
REDUCTIVE AMINATION



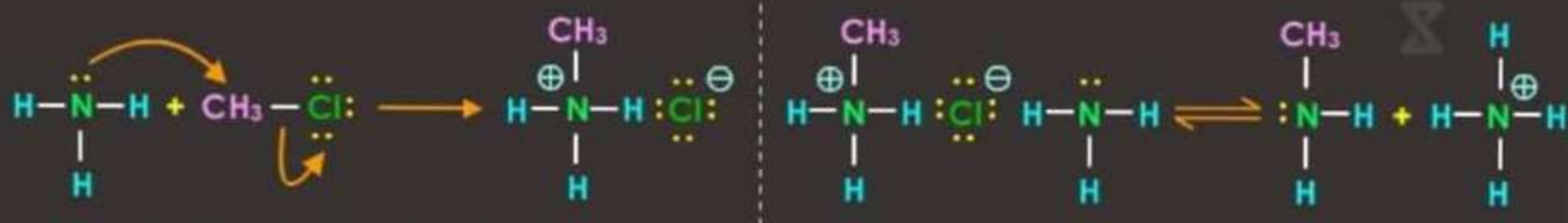
REDUCTION OF NITRILES

Nitriles can be reduced by strong reducing agent like H_2 with catalyst (example Ni or LiAlH_4) to yield primary amines via nucleophilic addition reaction.

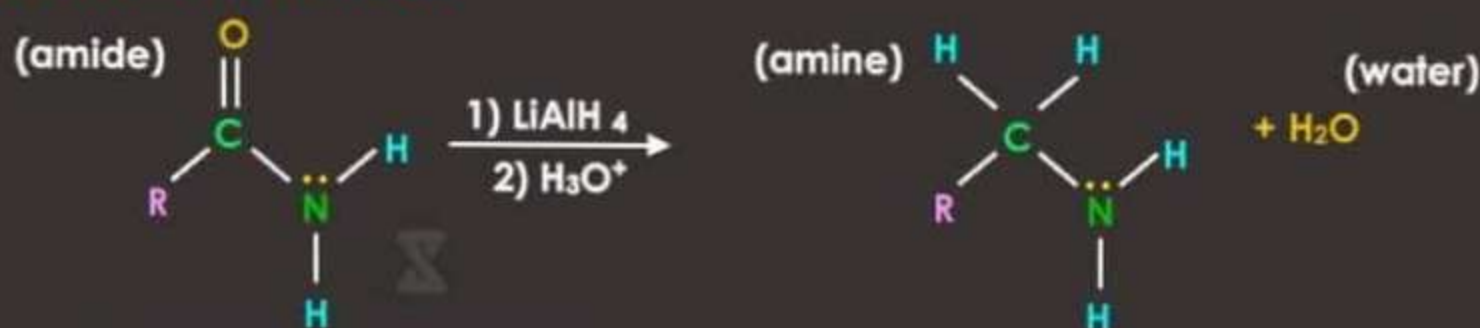
Example :



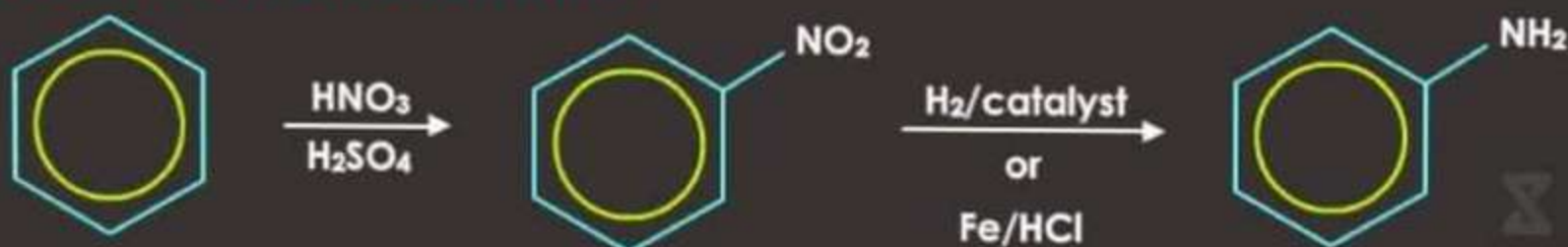
REACTION OF AMMONIA WITH ALKYL HALIDES



REDUCTION OF AMIDES



REDUCTION OF NITRO GROUPS





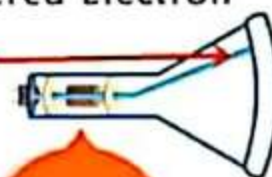
HISTORY OF ATOMIC MODEL

1885

Johann Balmer derived a formula for mathematically predicting hydrogen spectrum.

J J Thomson discovered Electron

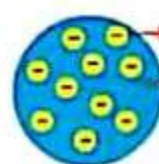
Electron beam



1897

Rutherford proposed a model where positive charge is at the center, and electron moves around in a spiral path and loses energy.

J J Thomson proposed plum pudding model



Electron
Positively charged matter

1904

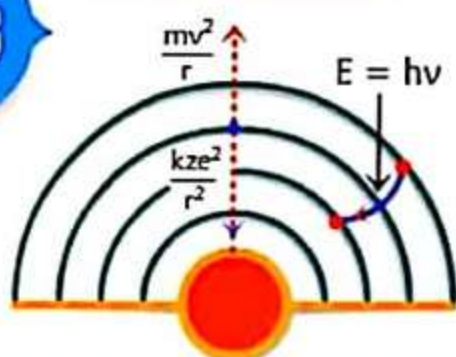
1911

Bohr's Atomic Model

1913

$$r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

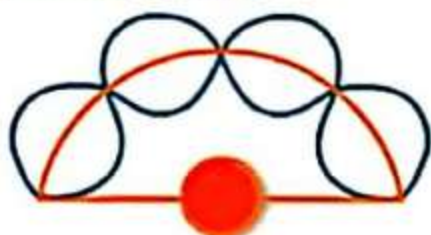
$$\frac{kze^2}{r^2} = \frac{mv^2}{r}$$



- Worked with J J Thomson and found flaws in his theory.
- He proposed electron revolves around nucleus in orbits.
- Electron is stabilized by centripetal and electrostatic forces.
- Electrons don't lose energy in an orbit.
- Electron loses or gains energy by moving across orbits.
- He proved Balmer was right by deriving his formula theoretically.
- Only applicable for one electron systems.
- Failed to predict dual nature of electron.

1923

De Broglie introduced the concept of dual nature in electrons. He used Einstein's $E = mc^2$ and proposed any moving particle or object has an associated wave.



Erwin Schrodinger developed electron cloud model using de Broglie and Bohr's atomic model. He and Heisenberg determined the regions in which electron would be likely found. He introduced one concept of orbitals.

1925



HYDROGEN SPECTRUM

Bohr's model

Niels Bohr proposed a model for the hydrogen atom that explained the spectrum of the hydrogen atom. The Bohr model was based on the following assumptions.

- The electron in a hydrogen atom travels around the nucleus in a circular orbit.
- The energy of the electron in an orbit is proportional to its distance from the nucleus. The further the electron is from the nucleus, the more energy it has.
- Only a limited number of orbits with certain energies are allowed. In other words, the orbits are quantized.
- The only orbits that are allowed are those for which the angular momentum of the electron is an integral multiple of Planck's constant divided by 2π .

$$L = \frac{nh}{2\pi} \text{ (where } h = \text{planck's constant)}$$

- Light is absorbed when an electron jumps to a higher energy orbit and emitted when an electron falls into a lower energy orbit.
- The energy of the light emitted or absorbed is exactly equal to the difference between the energies of the orbits
- When electron in an excited atom comes back from higher energy level (n_2) to lower energy level (n_1) then it emits a photon, having energy equal to difference in energy levels.

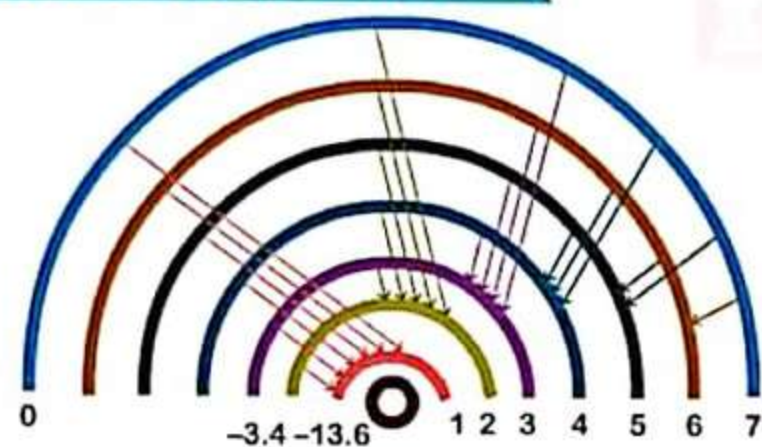
○ $h\nu = \Delta E = E_{n_2} - E_{n_1}$



Wavelength or wave no. of any line of any one electron species can be calculated as

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right], \quad \frac{hc}{\lambda} = \Delta E$$

Different series



Successes

- Combining successfully Rutherford's solar system's model, with the Planck hypothesis on the quantified energy states at atomic level and Einstein's photons
- Explaining the atomic emission and absorption spectra
- Explaining the general features of the periodic table
- First working model for the atom

LYMAN SERIES

$n_f = 1$

$n_i = 2, 3, 4, 5, \dots$

ULTRAVIOLET

BALMER SERIES

$n_f = 2$

$n_i = 3, 4, 5, 6, \dots$

VISIBLE

PASCHEN SERIES

$n_f = 3$

$n_i = 4, 5, 6, 7, \dots$

INFRARED

BRACKETT SERIES

$n_f = 4$

$n_i = 5, 6, 7, \dots$

INFRARED

PFUND SERIES

$n_f = 5$

$n_i = 6, 7, \dots$

FAR INFRARED

HUMPHREY SERIES

$n_f = 6$

$n_i = 7, 8, \dots$

FAR INFRARED

QUANTUM NUMBER

THE ELECTRON'S ADDRESS



GPS is used to track anyone at any place on the earth.

n, ℓ, m

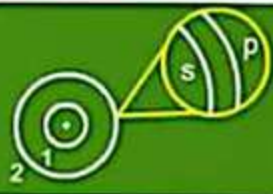
Similarly Quantum Numbers are used to identify the position of an electron in an atom.

QUANTUM NUMBER



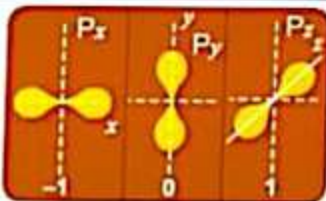
Principal Quantum Number (n)

Represents the orbit number in an atom. It is denoted by letter ' n '.



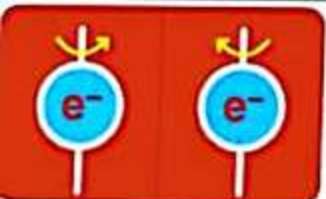
Azimuthal Quantum Number (ℓ)

Represents the shape of an orbital in atom. it is denoted by letter ' ℓ ' and its value vary from 0 to ' n '.



Magnetic Quantum Number (m_ℓ)

Represents the orientation of an orbital in the space. it is denoted by letter ' m_ℓ ' and its value vary from ' ℓ ' to ' $-\ell$ '.



Spin Quantum Number (m_s)

Represents the spin of an electron. It is denoted by m_s and each electron has an orbital either $\frac{1}{2}$ or $-\frac{1}{2}$.

DIFFERENCE BETWEEN BOTH ATOMIC MODELS

Bohr's model is a 2-Dimensional model. Therefore he used only **Principal quantum number (n)** to identify the position of an electron in an atom.



Schrodinger's model is a 3-Dimensional model. Therefore he used **n, ℓ, m** to identify the position of an electron in an atom.



Hide Something at $n = 1, \ell = 1, m_\ell = 0, m_s = \frac{1}{2}$

Quantum numbers have some restrictions. It's not possible to find an electron at every possible combination of n, ℓ, m, s . So you will never find an electron at above point.

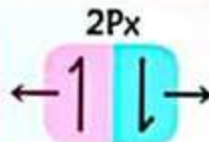
No two electrons in an atom have same Quantum Number.

$n = 2$

$\ell = 1$

$m = 0$

$s = -\frac{1}{2}$



$n = 2$

$\ell = 1$

$m = 0$

$s = \frac{1}{2}$

CARBOHYDRATES



Carbohydrates are the **sugars**, **starches** and **fibers** found in **fruits**, **grains**, **vegetables** and **milk** products. They are called carbohydrates because at the chemical level, they contain carbon, hydrogen and oxygen.

NUTRITION FACTS

Serving size 1 Large Apple

Amount Per Serving

Calories 130 Calories from Fat 0

Total Fat 0g

Saturated Fat 0g

Trans Fat 0g

Cholesterol 0g

Sodium 0g

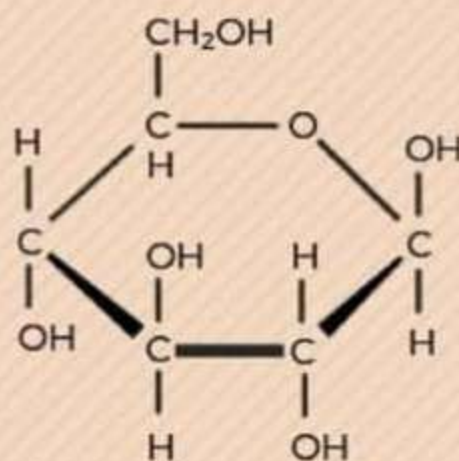
Total Carbohydrate 34g

Dietary Fiber 5g

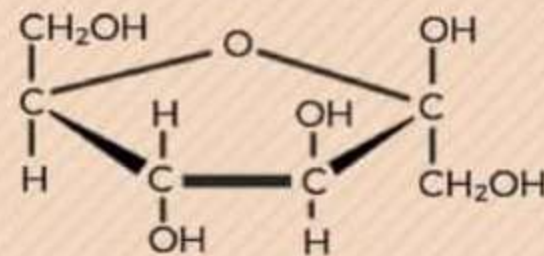
Sugars 5g

Protein 1g

GLUCOSE



FRUCTOSE



SIMPLE CARBOHYDRATES

Sources of Carbohydrates

- Fruits
- Candy
- Milk
- Sugary Beverages

CARBOHYDRATES BENEFITS

- Mental health
- Weight loss
- Good source of nutrients
- Heart health

COMPLEX CARBOHYDRATES

Grains: Grains are good source of fiber, as well as **potassium**, **magnesium** and **selenium**. Choose less processed, whole grains such as quinoa, buckwheat, and whole-wheat pasta.

Fiber-Rich Fruits: Such as **apples**, **berries**, and **bananas** (avoid canned fruit, as they usually contain added syrup).

Fiber-Rich Vegetables: Eat more of all your veggies, including **broccoli**, **leafy greens**, and **carrots**.

Beans: Aside from **fiber**, these are good sources of folate, iron, and potassium.

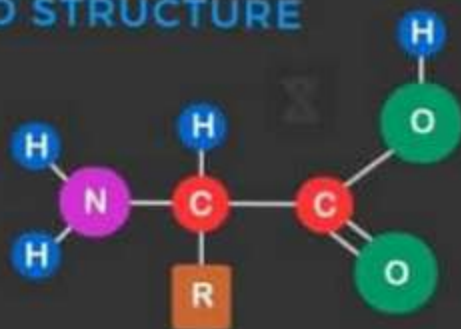
FUNCTION OF CARBOHYDRATES

Carbohydrates provide fuel for the central nervous system and energy for working muscles. They also prevent protein from being used as an energy source and enable fat metabolism.

AMINO ACIDS

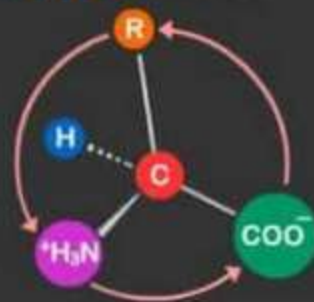
GENERAL AMINO ACID STRUCTURE

At **physiological** condition, the head group is positively charged and the tail group is **negatively charged**.



AMINO ACID STEREOCHEMISTRY

Most natural **amino acids** have the S or L stereochemistry about the α carbon. Notice the direction of the arrows from the highest atomic number atoms to the lowest. This is the same direction as when you would write the top part of the letter **S** or the direction that you would write an **L**.



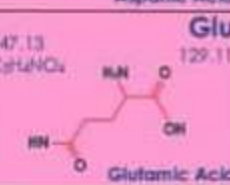
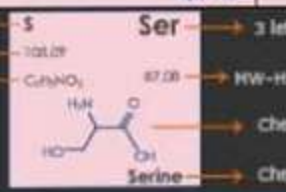
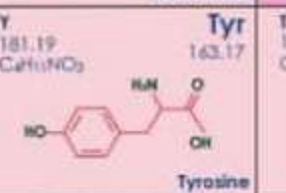
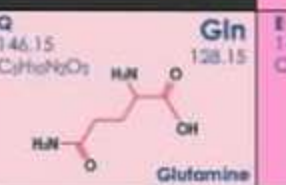
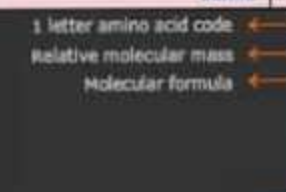
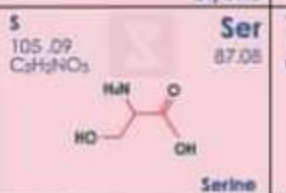
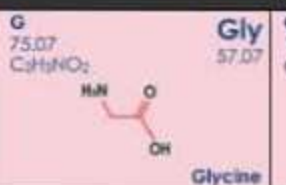
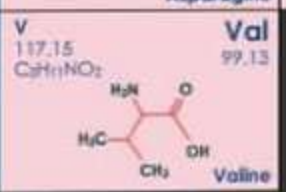
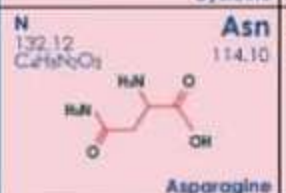
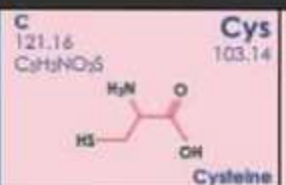
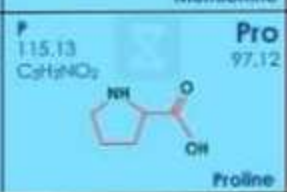
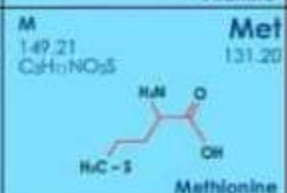
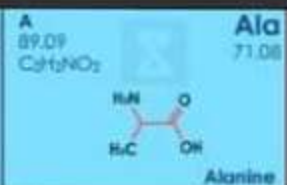
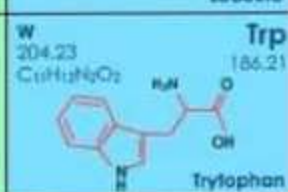
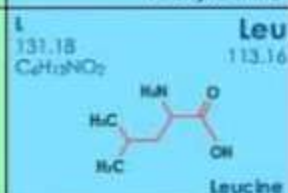
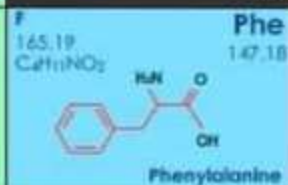
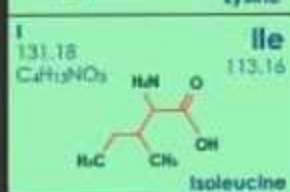
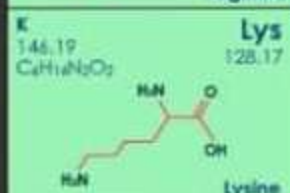
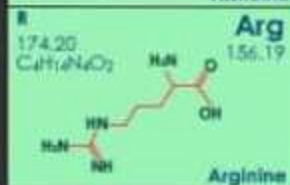
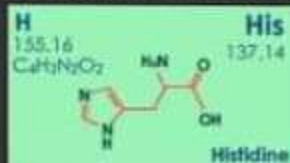
PERIODIC CHART OF AMINO ACIDS

● Basic

● Non-Polar (hydrophobic)

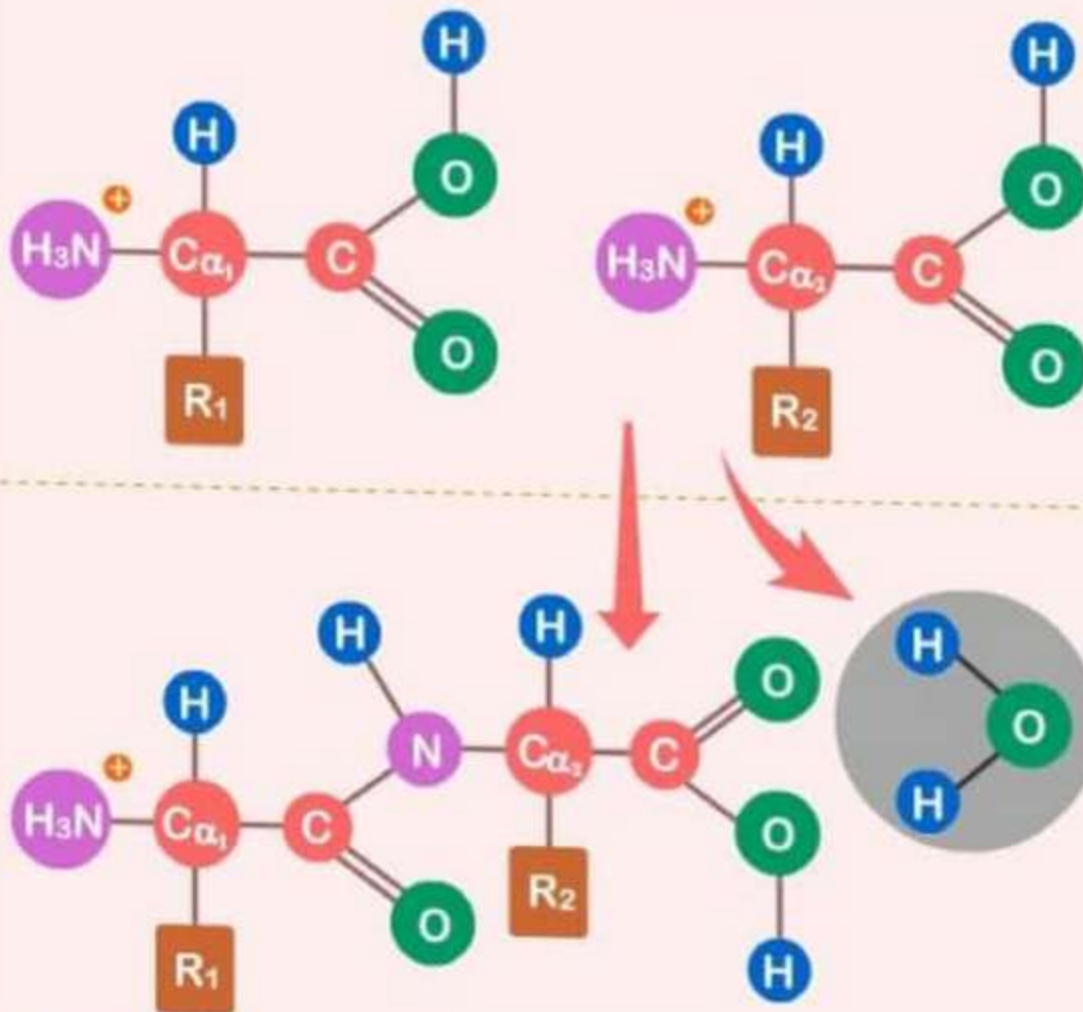
● Polar, Uncharged

● Acidic



1 letter amino acid code	Ser	3 letter Amino acid code
Relative molecular mass	105.09	
Molecular formula	$C_3H_7NO_3$	HW-H ₂ O Molecular weight in water
		Chemical structure
	Serine	Chemical name

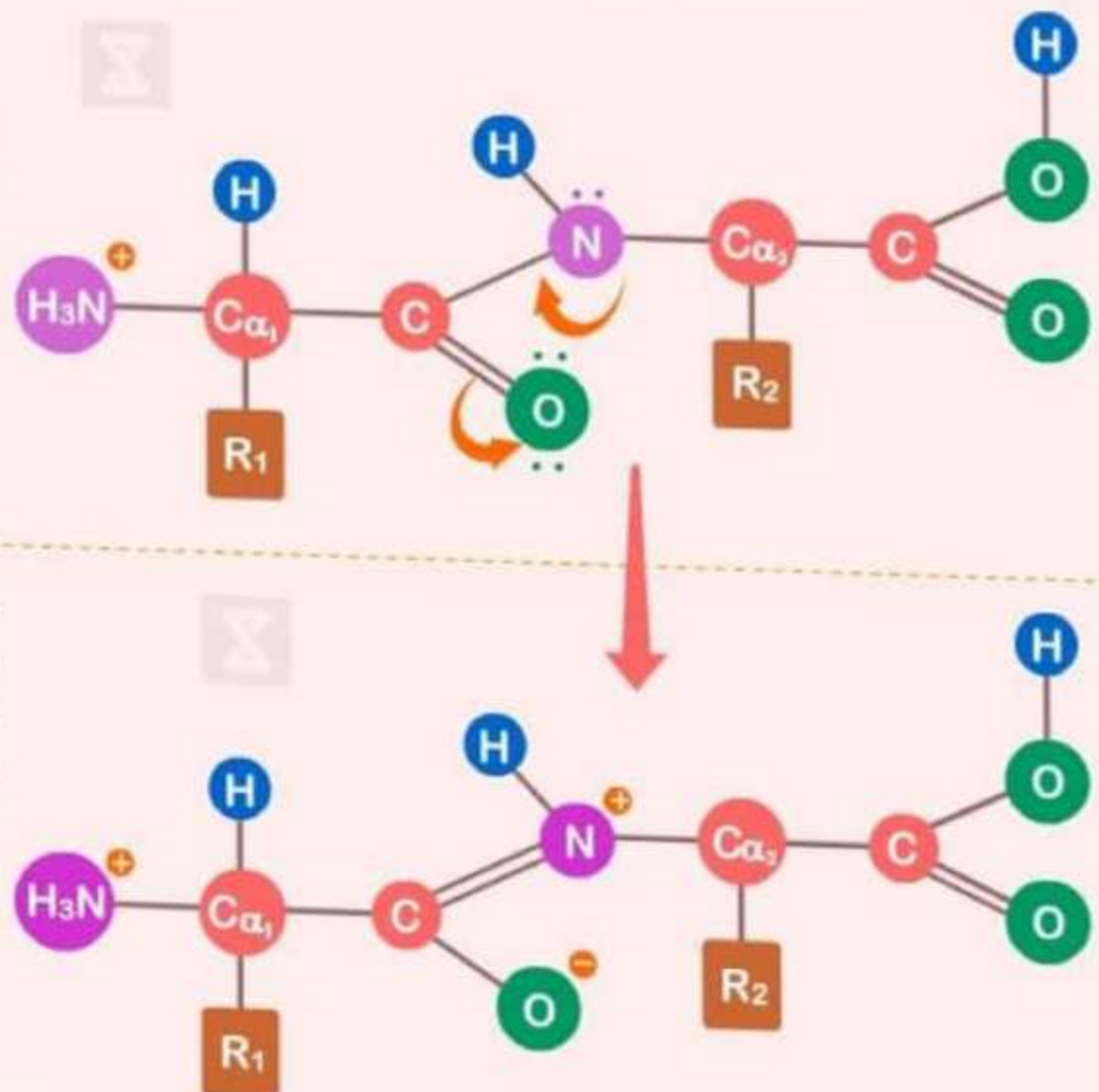
THE PEPTIDE BOND

Amino acid α_1 Amino acid α_2 

When two amino acids come together, a reaction can occur between the carboxylic tail of the leading end with the amine head of the trailing end.

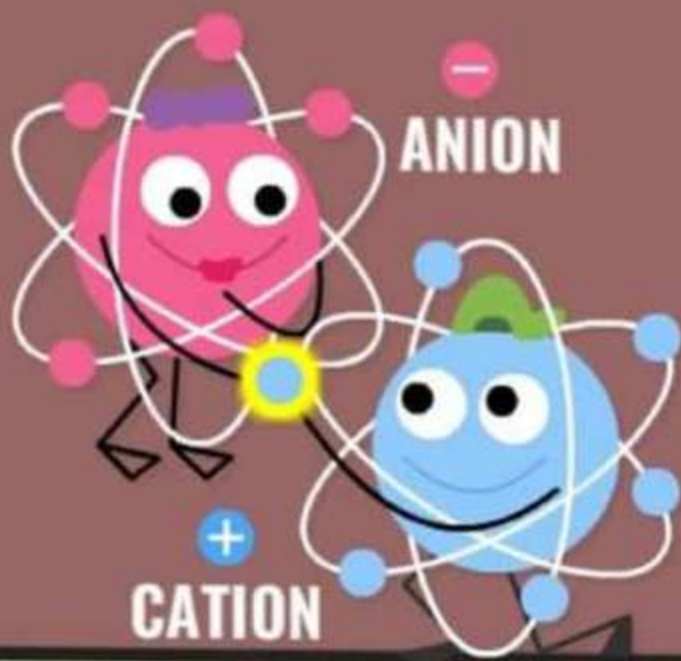
Each amino acid gives up a part of itself in a dehydration reaction and the product of their interaction is the peptide bond.

The new bond enjoys a type of resonance that strengthens it as the electrons about the nitrogen and oxygen delocalize producing a partial double bond.



The result of this resonance makes this bond difficult to rotate since the newly formed electron clouds are stabilized and do not like to bump into things.

Peptide bonds form a rigid backbone for the many proteins it is a part of.



IONIC COMPOUNDS

DEFINITION

Ionic Compound is formed due to creation of an ionic bond which is the electrostatic attraction between a **cation (+ charge)** and an **anion (- charge)** generally involving a metal and a non-metal.

NOMENCLATURE

Charges are written because ions are either positive or negative.

- Name the **cation first**, followed by the anion
- Anion must end in **ide** (drop the last few letters)
- Roman Numerals must be used for metals with more than one charge (e.g. transition metals)

Ex : NaCl = Sodium chloride

Ex : CuCl_2 = Copper (II) chloride

PROPERTIES

- High melting points
- High boiling points
- Hard and brittle
- Good insulators
- Forms crystals
- Conduct electricity when they are dissolved in water
- Ionic compounds have higher enthalpies of fusion

EXAMPLES

Some examples of Ionic compounds are **Sodium Chloride**, **Lithium Iodide**, **Potassium Iodide** and **Sodium Fluoride**.

SODIUM CHLORIDE (NaCl)

Some of sodium chloride's use includes consumption, production and is **naturally occurring**.



POTASSIUM IODIDE (KI)

Potassium iodide tablets are given to people exposed to high level of radiation.



LITHIUM IODIDE (LiI)

Lithium Iodide is commonly used in batteries, pacemakers and solar power generator.



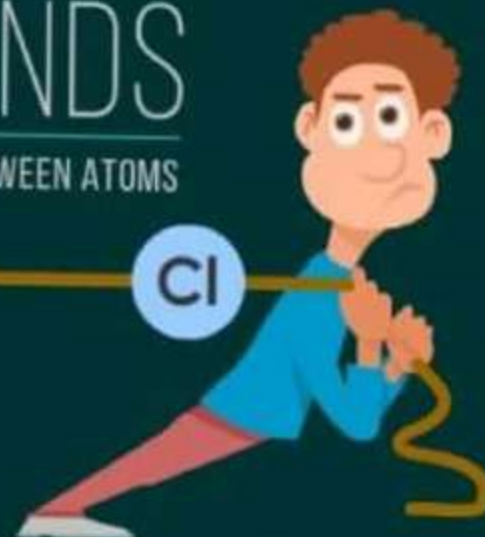
SODIUM FLUORIDE (NaF)

Sodium fluoride is used in medical treatment, water purification and cleaning solutions.



MOLECULAR COMPOUNDS

FORMED BY A COVALENT BOND, WHICH IS THE SHARING OF ELECTRONS BETWEEN ATOMS INVOLVES TWO NON-METALS



NOMENCLATURE

- Ionic charges cannot be written because it is composed of molecules, not ions.
- Prefix :- mono, di, tri, tetra, penta, hexa, hepta, octa, nona, deca.
- Name the elements in the order listed.
- Use prefixes to indicate the number of each atom of each element (mono can only be used on the second non-metal).
- The first element includes prefix + element name.
- The second element includes the prefix + the element name + ide ending (drop the last few letters).

Eg : hydrogen chloride = HCl

Eg : phosphorus pentachloride = PCl_5

PROPERTIES



LOW
MELTING
POINTS



MORE
FLAMMABLE



NOT
SOLUBLE
IN WATER



SOFTER
AND
SQUISHIER

SOME EXAMPLES OF MOLECULAR COMPOUNDS



DIHYDROGEN
MONOXIDE (H_2O)

Dihydrogen monoxide or water is vital for our survival, used in our daily needs.



SILICON
DIOXIDE (SiO_2)

Silicon dioxide is used for construction and is found naturally in sand and quartz.



CARBON
DIOXIDE (CO_2)

Some uses of carbon dioxide are carbonation of liquids and green house effect.



SULPHUR
DIOXIDE (SO_2)

Sulphur dioxide is used for food preservation and acts as a disinfectant.



HYBRIDISATION



ANALOGY FOR HYBRIDISATION



Four different sized mud balls combine to form equal shaped balls.



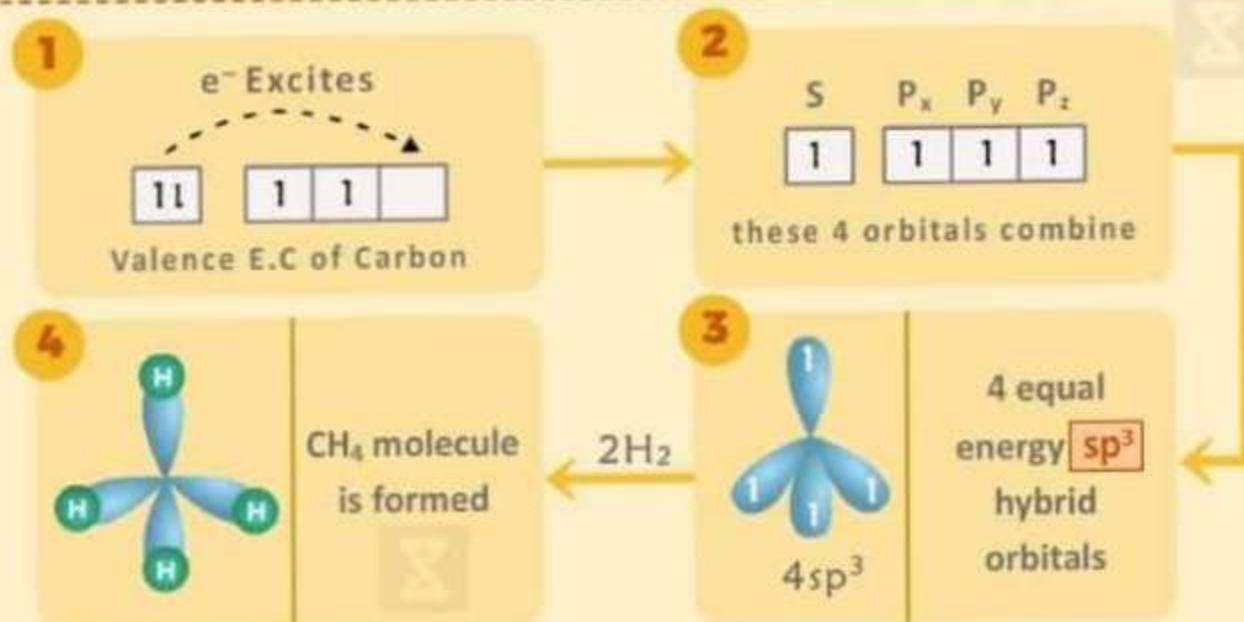
Similarly, orbitals of different energy combine to form equal energy orbitals.

HYBRID ORBITALS ARE USED IN FORMING SIGMA BONDS

Formation of CH₄



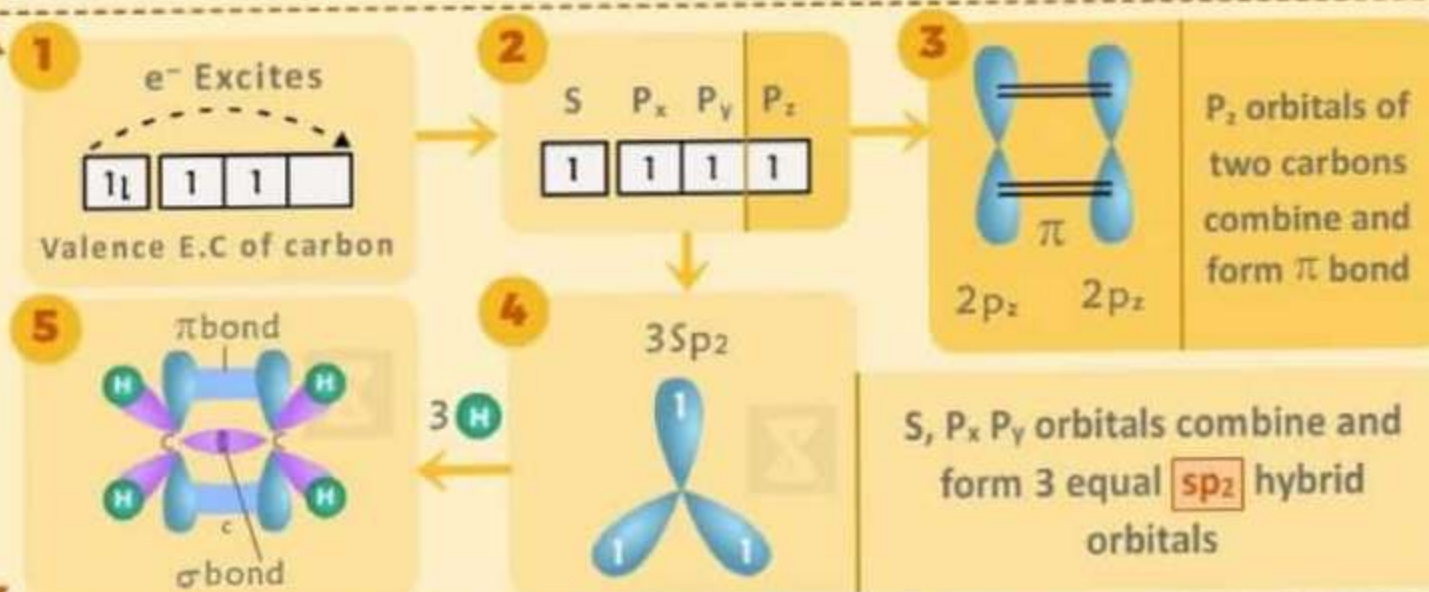
Here carbon needs to form 4 sigma Bonds.



Formation of C₂H₄



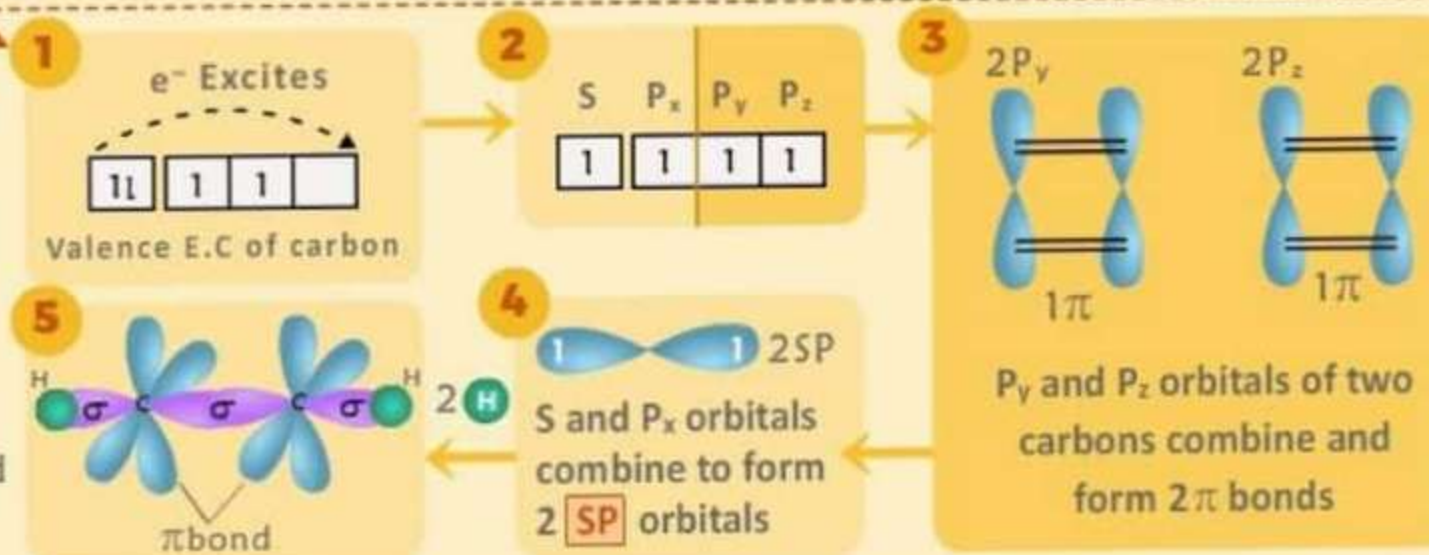
Here each carbon needs to form 3 sigma and 1 pi Bonds.



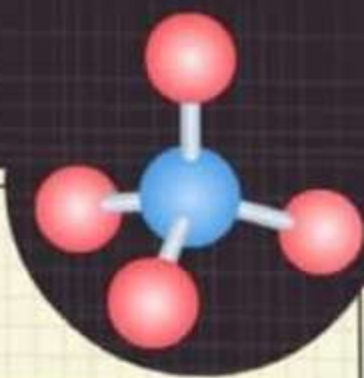
Formation of C₂H₂



Each carbon needs to form 2 sigma and 2 pi Bonds.



VSEPR & SHAPES OF MOLECULES



Bonding Pairs & Lone Pairs

Lone pairs in a molecule lie closer to the central atom, hence they repel more than a bonded pair. The order of strengths of repulsion is :

LONE PAIR/LONE PAIR > BONDED PAIR/LONE PAIR > BONDED PAIR/BONDED PAIR

Shapes with different electron Pair : (ep)

2
ep :

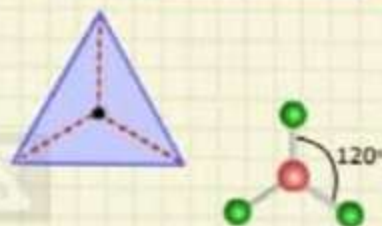
Linear



AX_2
2 bond pairs
0 lone pairs

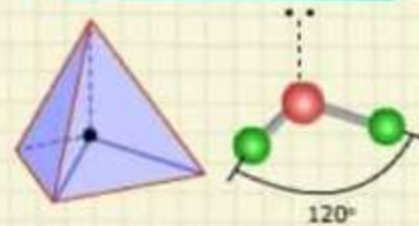
3
ep :

Trigonal-Planar



AX_3
3 bond pairs
0 lone pairs

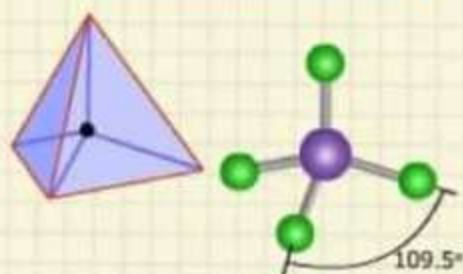
Bent



$SnCl_2$
2 bond pairs
1 lone pairs

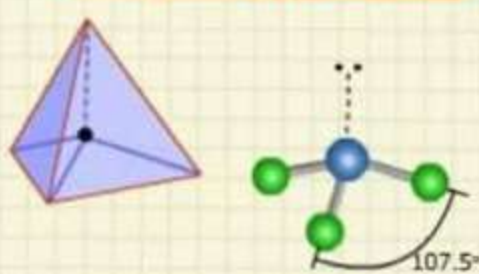
4
ep :

Tetrahedral



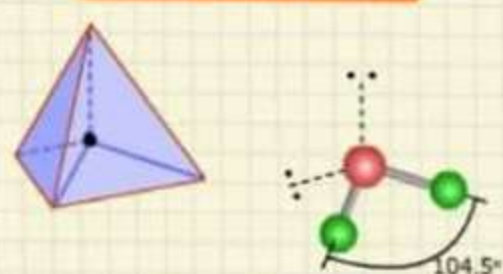
Methane (CH_4)
4 bond pairs
0 lone pairs

Trigonal pyramidal



Ammonia (NH_3)
3 bond pairs
1 lone pairs

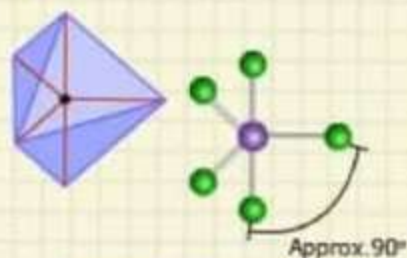
Bent



Water (H_2O)
2 bond pairs
2 lone pairs

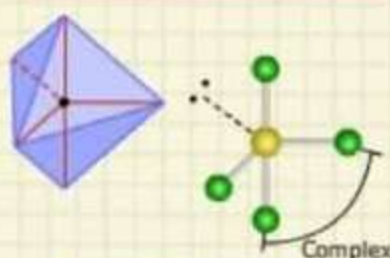
5
ep :

Trigonal Bipyramidal



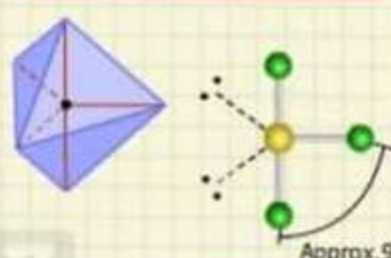
PF_5
5 bond pairs
No lone pairs

See Saw



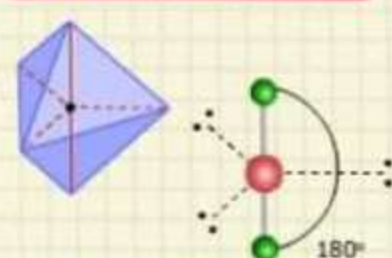
SF_4
4 bond pairs
1 lone pairs

T-Shaped



ICl_3
3 bond pairs
2 lone pairs

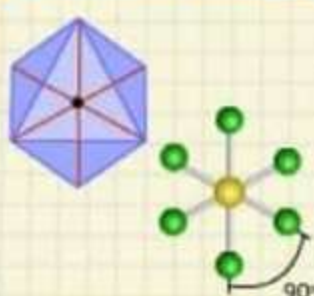
Linear



BrF_2^-
2 bond pairs
3 lone pairs

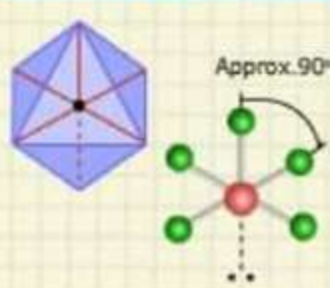
6
ep :

Octahedral



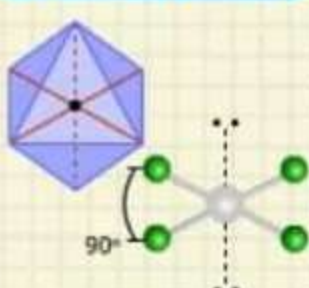
SF_6
6 bond pairs
No lone pairs

Square-pyramidal



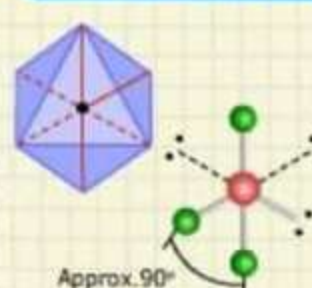
BrF_5
5 bond pairs
1 lone pairs

Square-planar



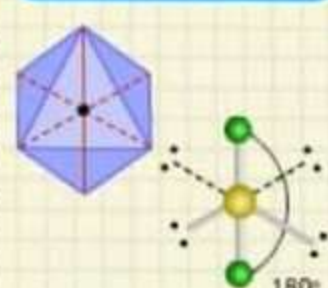
XeF_4
4 bond pairs
2 lone pairs

T-Shaped



ClF_3
3 bond pairs
3 lone pairs

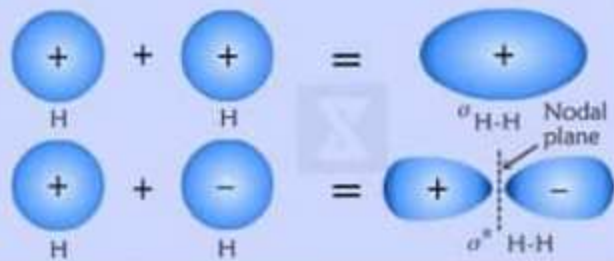
Linear



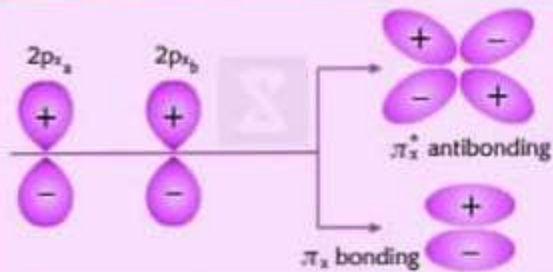
XeF_2
2 bond pairs
4 lone pairs

MOLECULAR ORBITAL THEORY

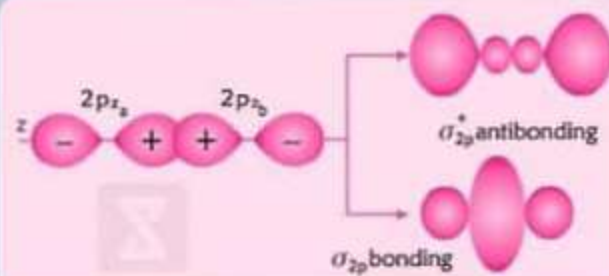
MOT explains the bonding and stability of Molecules by forming Molecular orbits



s-orbital of one atom combines with s-orbital of another atom constructively and destructively to form σ and σ^* molecular orbitals.

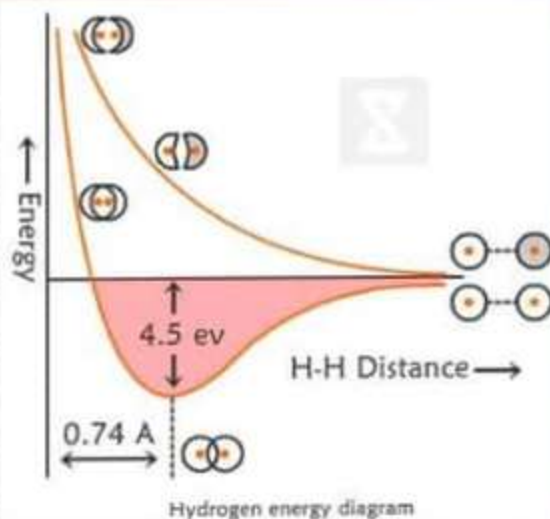
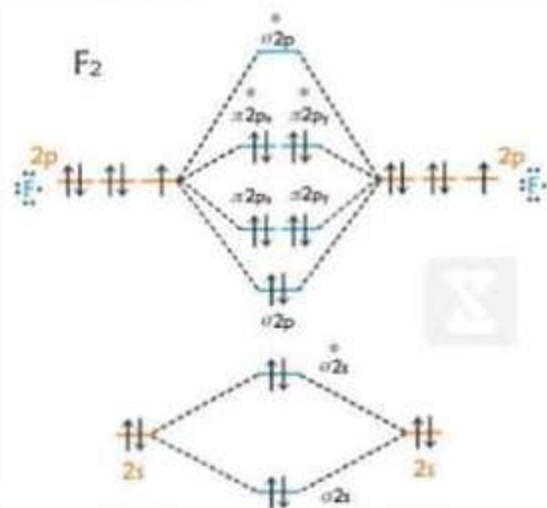


p_x orbital of one atom combines with p_x of another atom to form σ and σ^* orbitals.



p_y and p_z orbitals combine and form π and π^* orbitals.

Energy Diagram of Molecular Orbitals



Bond Order

$$\text{Bond order} = \frac{1}{2} \left[\begin{array}{l} \text{Number of} \\ \text{Bond of} \\ \text{electron in} \\ \text{bonding} \\ \text{orbitals} \end{array} - \begin{array}{l} \text{Number of} \\ \text{Bond of} \\ \text{electrons} \\ \text{in anti-bonding} \\ \text{orbitals} \end{array} \right]$$

Bond	H_2^+	H_2	He_2^+	He_2
Bond Order	$\frac{1}{2}$	1	$\frac{1}{2}$	0

The bond order must be **positive non-zero** for a bond to be stable. He_2 has a bond order of zero and that is why the He_2 molecule is not observed.

CHEMICAL EQUILIBRIUM



EQUILIBRIUM CONSTANT 'K'

For a General Reaction



The **equilibrium constant** expression is

$$K_c = \frac{[P]^p [Q]^q}{[A]^a [B]^b}$$

where K_c is the **Equilibrium Constant** (or K_p if they are all gases)

RELATION BETWEEN K_p AND K_c

For the Reaction



$$K_p = \frac{[p_C]^c \times [p_D]^d}{[p_A]^a \times [p_B]^b} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \frac{(RT)^{c+d}}{(RT)^{a+b}}$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$\text{if } \Delta n_g = 0 \Rightarrow K_p = K_c$$

Where, $\Delta n_g = (c+d) - (a+b)$

= no. of moles of gaseous products - no. of moles of gaseous Reactants

WHAT DOES THE VALUE OF 'K' MEAN ?

(a) $K \gg 1$

Reactants



Products

If $K \gg 1$, the reaction is product-favoured; product predominates at **Equilibrium**.

(a) $K \ll 1$

Reactants



Products

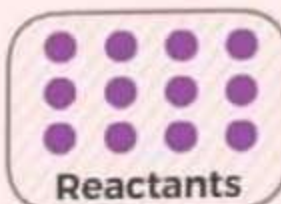
If $K \ll 1$, the reaction is reactant-favoured; reactant predominates at **Equilibrium**.

(a) $K = 1$

The **reaction lies** in the middle (mix of reactants and products)

MAGNITUDE OF 'K'

Small ($K < 10^{-3}$)



Reactants



Products

Mostly Reactants

Intermediate ($10^{-3} \leq K \leq 10^3$)



Reactants



Products

Significant amounts of reactants and products

Large ($K > 10^3$)



Reactants



Products

Mostly Products



It's Le Chatelier's Principle



If a dynamic equilibrium is disturbed by changing the conditions, the position of equilibrium moves to counteract the change.



i will
destroy your
equilibrium

i will
re-establish
it



Reactants

Change

$a A + b B$



Products

$c C + d D$

Counteract

I will increase reactant concentration



Then I will shift the reaction forward

I will steal products



I will shift the reaction forward

I will increase pressure



I will reduce number of moles

I will decrease pressure



Then I'll increase number of moles

I will heat up your exothermic reaction



I'll shift the reaction backward

I will put your endothermic reaction in ice



I'll warm it up by forward reaction

I will catalyze your reaction



Hahaha.... It won't disturb my equilibrium

I will add noble gases to your reaction



Hahaha.... It won't disturb my equilibrium

How did you bypass my tricks ?



It's Le Chatelier's principle dear !

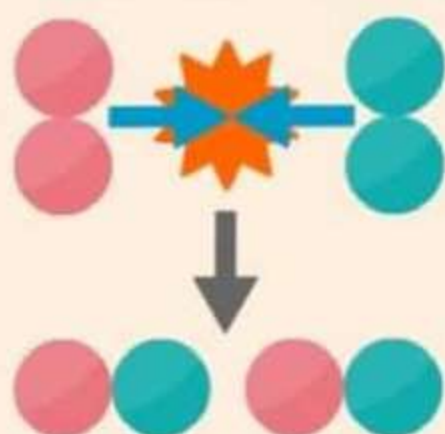
CHEMICAL KINETICS



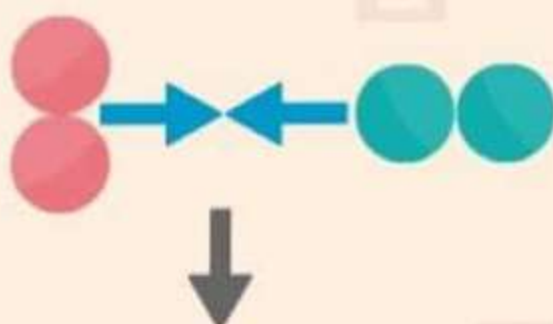
MAKING CHEMICAL REACTION HAPPEN FASTER

There are a number of different things that we can change to make a chemical reaction faster. Here, we explain the concept of collision theory, and how it can be used to explain the effects of five different factors on the rate of a chemical reaction.

COLLISION THEORY

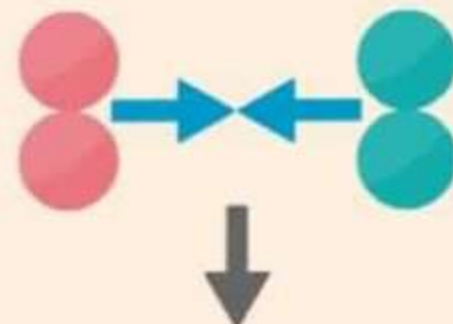


Correct Orientation
Sufficient Energy



NO RECTION

Wrong Orientation

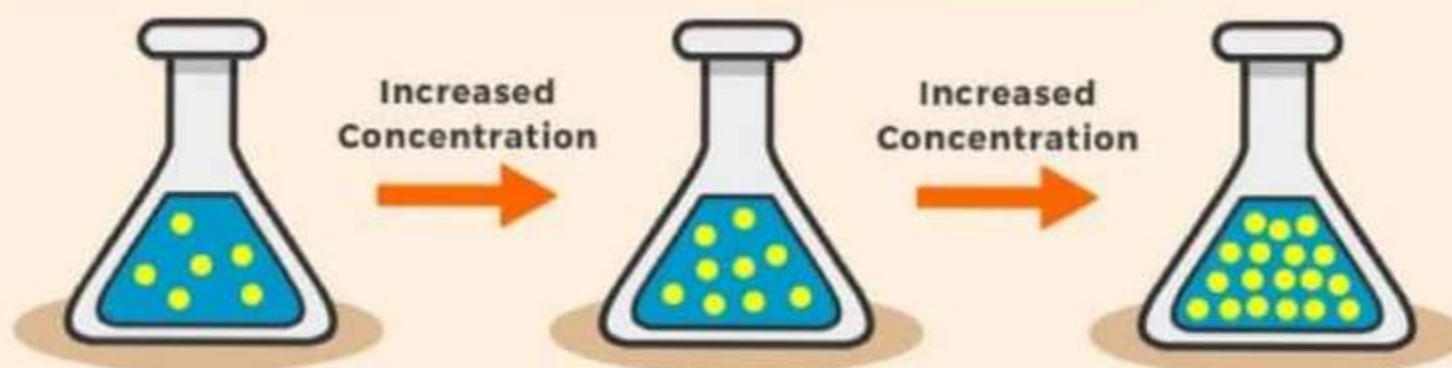


NO RECTION

Insufficient Energy

Collision theory states that for a reaction to occur, particles must collide with the **correct orientation** and with **sufficient energy** for a reaction to occur. Different factors affect the rate of the reaction by affecting the frequency of particle collisions, and the proportion of collisions that have enough energy to react.

01 INCREASE CONCENTRATION OF REACTANTS

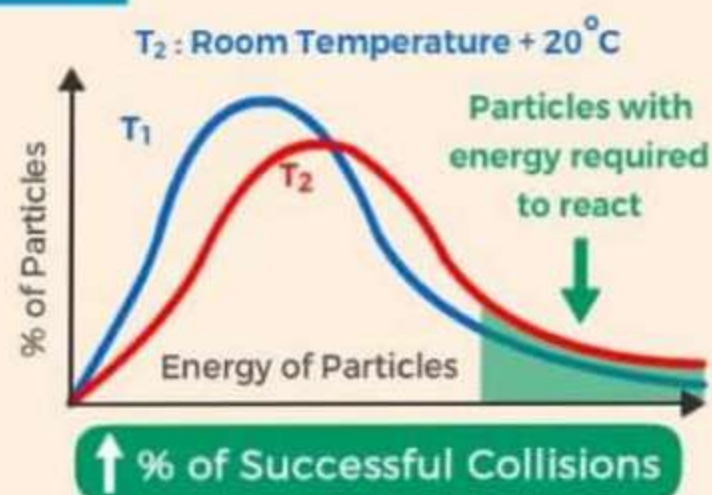
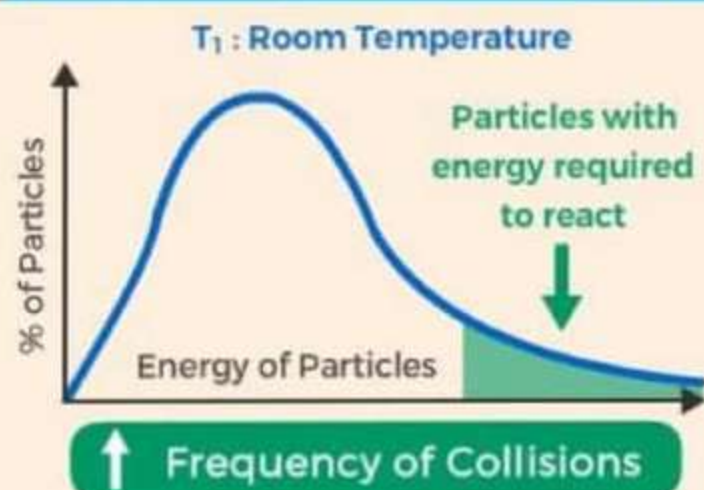


↑ Frequency of Collisions

↑ % Successful Collisions

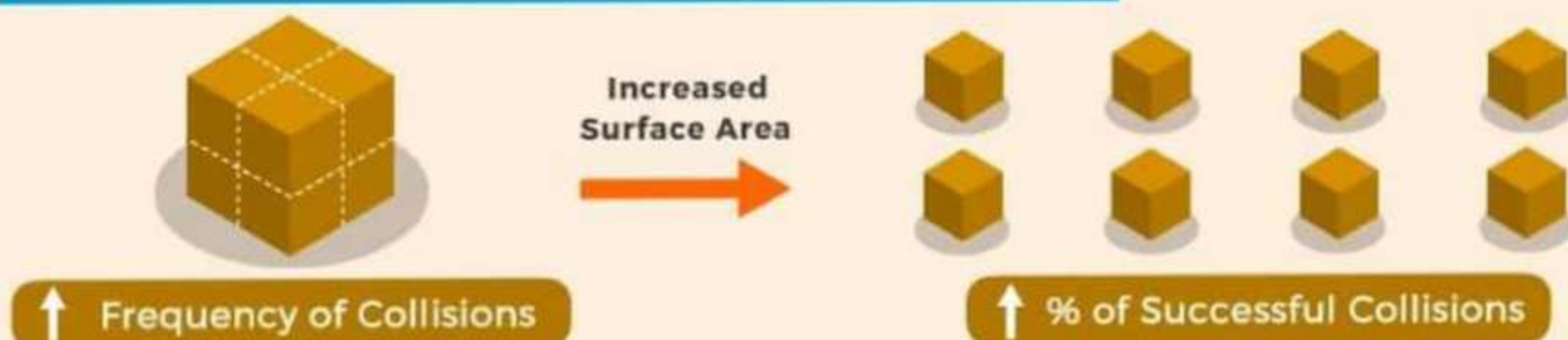
Increasing the concentration of reactants in solution increases the rate of reaction as there are a greater number of particles to react. This increases the frequency of collision between particles.

02 INCREASE TEMPERATURE OF REACTION



Increasing the temperature increases the kinetic energy of particles. This increases the frequency of particle collisions and a greater proportion of collisions will have the sufficient energy required to react.

03 INCREASE SURFACE AREA OF REACTANTS



Increasing the surface area of solid reactants increases the number of particles that are exposed and available to react and as a consequence this increases the frequency of particle collisions, thus increasing the rate.

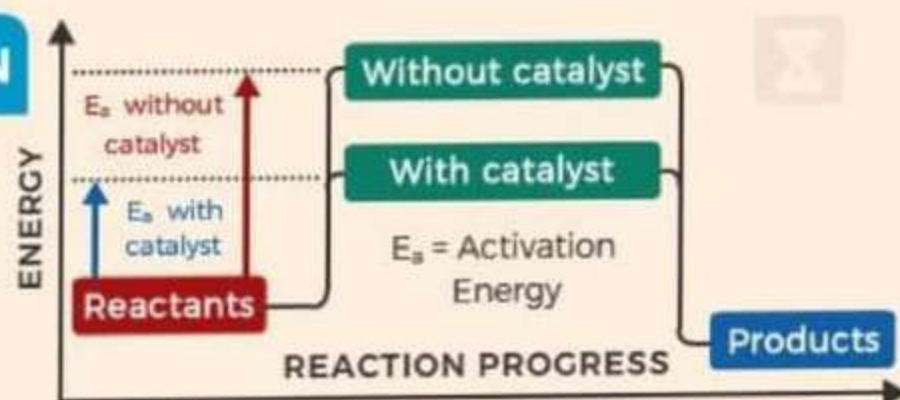
04 INCREASES PRESSURE OF REACTION



Increasing the pressure of a reaction involving gases, forces the gas particles closer together. This will increase the frequency of particle collisions and therefore increases the rate of reaction.

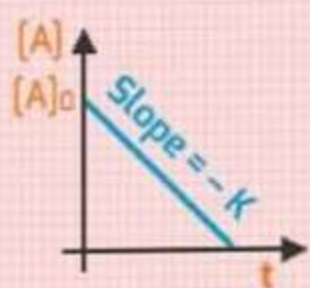
05 USE A CATALYST IN A REACTION

A catalyst provides an alternative route for the reaction with a lower activation energy. This means that particle collisions need less energy in order for a reaction to occur, thus increasing the rate of the reaction.



ORDER OF REACTION

Zero Order Reactions



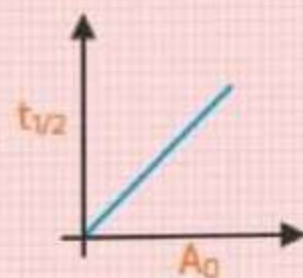
A zero order reaction has a constant rate that is independent of the concentration of the reactant(s); the rate law is simply.

$$\text{Rate} = k$$

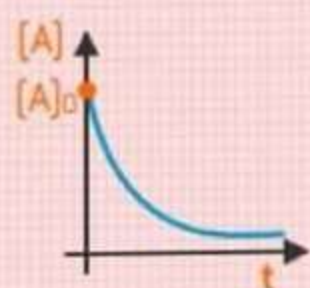
HALF LIFE ($t_{1/2}$)

Time in which half of initial amount is left.

$$t_{1/2} = \frac{[A]_0}{2k}$$



First Order Reactions



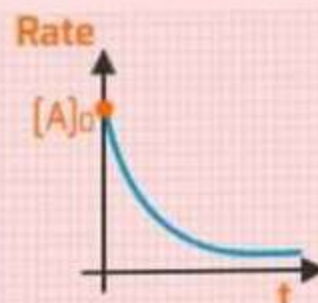
A first-order reaction is a reaction that proceeds at a rate that depends linearly on only one reactant concentration.

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

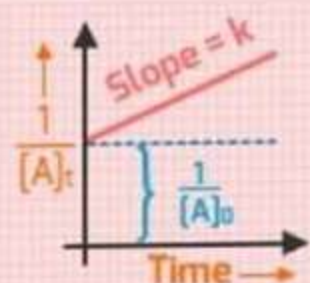
HALF LIFE ($t_{1/2}$)

$$t_{1/2} = \frac{0.693}{k}$$

$$[A] = \frac{[A]_0}{2^n}$$



Second Order Reactions

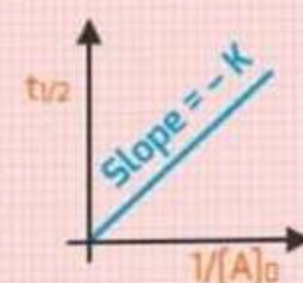


A chemical reaction in which the rate of the reaction is determined by the concentration of two chemical reactants involved or the square of the concentration of one chemical reactant.

$$k = \frac{1/[A] - 1/[A]_0}{t}$$

HALF LIFE ($t_{1/2}$)

$$t_{1/2} = \frac{1}{k[A]_0}$$

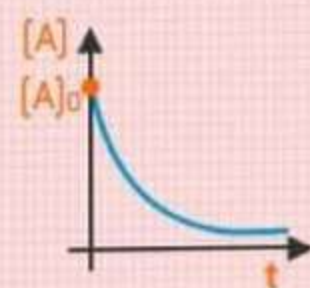


Pseudo first order reactions

If the **concentration** of a reactant remains constant (because it is a catalyst or it is in great excess with respect to the other reactants), its concentration can be included in the rate constant, obtaining a pseudo-first-order (or occasionally **pseudo-second-order**) rate equation.

For example, the hydrolysis of sucrose in acid solution rate $r = k[\text{sucrose}]$. The true rate equation is third-order, $r = k[\text{sucrose}][\text{H}^+][\text{H}_2\text{O}]$; however, the **concentrations** of both the catalyst H^+ and the solvent H_2O are normally constant, so that the reaction is pseudo-first-order.

Nth order reactions

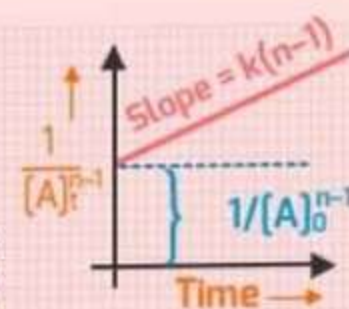


Nth order reaction is one which proceeds at the rate that depends on concentration of multiple reactants or on multiple steps.

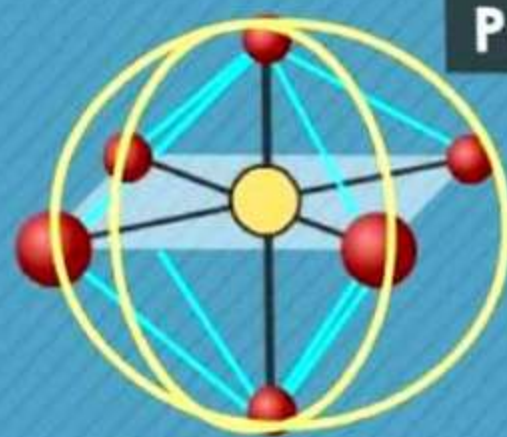
$$\Rightarrow k = \frac{1}{(n-1)t} \left[\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}} \right]$$

HALF LIFE ($t_{1/2}$)

$$t_{1/2} = \lim_{x \rightarrow n} \frac{2^{x-1}-1}{(x-1)k[A]_0^{x-1}}$$



ISOMERISM IN COORDINATION COMPOUND



STRUCTURAL ISOMERISM

1 LINKAGE ISOMERISM

This type of isomerism is shown by the coordination compounds having ambidentate ligands.

$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}$ or pentammine nitrito - N Cobalt (III) chloride and pentaammine nitrito - O' Cobalt (III) chloride.

2 COORDINATION ISOMERISM

This type of isomerism arises from the interchange of ligands between cationic and anionic complexes of different metal ions present in a complex.

$[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{CN})_6]$ and $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$

3 IONISATION ISOMERISM

This isomerism arises due to the exchange of ionisable anion with anionic ligand.

$[\text{Co}(\text{NH}_3)_5 \text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

4 SOLVATE ISOMERISM

This is also known as hydrate isomerism. In this isomerism, water is taken as solvent. It has different number of water molecules in the coordination sphere and outside it.

$[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$, $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$

5 POLYMERIZATION ISOMERISM

Polymerization isomerism, in which n varies in the complex $[\text{ML}_m]_n$, it represents an additional way in which an empirical formula may give incomplete information about the nature of complex.

$[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]_n$

$[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

$n = 1$

$[\text{Co}(\text{NH}_3)_6][\text{Co}(\text{NO}_2)_6]$

$n = 2$

$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)][\text{Co}(\text{NH}_3)_2(\text{NO}_2)_4]_2$

$n = 2$

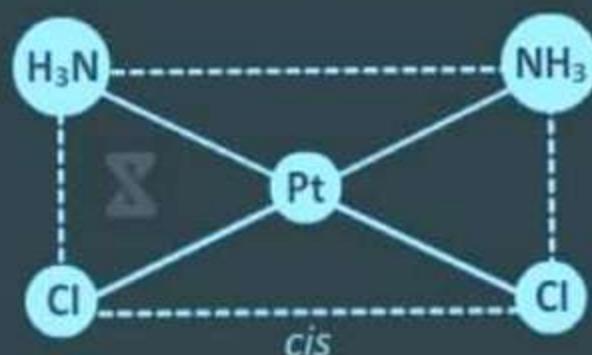
STEREISOMERISM

1 GEOMETRICAL ISOMERISM

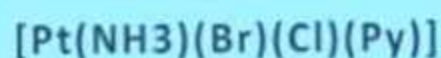
Geometrical isomers are of two types i.e., cis and trans isomers. This isomerism is common in complexes with coordination number 4 and 6.

COMPLEXES WITH COORDINATION NUMBER 4

- Tetrahedral complexes do not show geometrical isomerism.
- Square planar complexes of formula $[MX_2L_2]$ (X and L are unidentate) show geometrical isomerism. The two X ligands may be arranged adjacent to each other in a cis isomer or opposite to each other in a trans isomer.

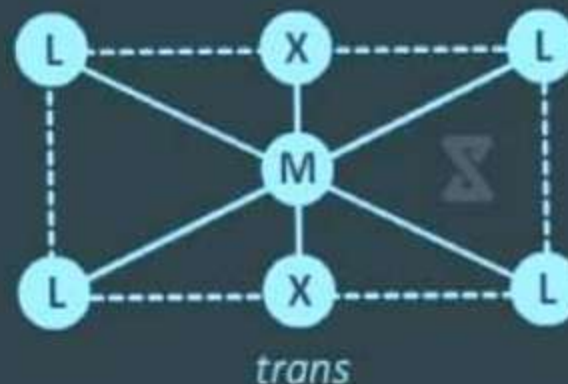
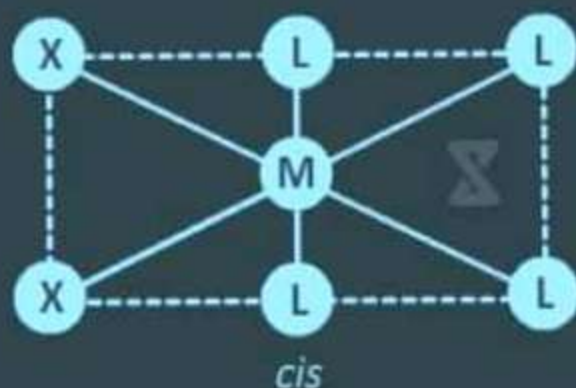
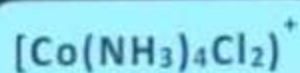


- Square planar complex of the type $[MABXL]$ (where A, B, X, L, are unidentate ligands) shows three isomers, two cis and one trans.



COMPLEXES WITH COORDINATION NUMBER 6

- Octahedral complexes of formula $[MX_2L_4]$, in which the two X ligands may be oriented cis or trans to each other



2 OPTICAL ISOMERISM

These are the complexes which have chiral structures. It arises when mirror images cannot be superimposed on one another. These mirror images are called enantiomers. The two forms are called dextro (d) and laevo (l) forms.

Tetrahedral complexes with formula $[M(AB)_2]$ show optical isomers and octahedral complexes (cis form) exhibit optical isomerism.

Colour of Co-ordination Compounds

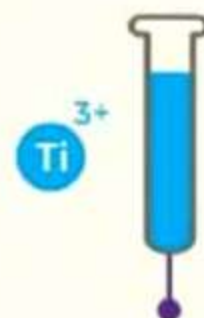
What is Colour Wheel ?

Generally white light is a mix of two opposite colors on colour wheel. These two colour are called **component colours**.

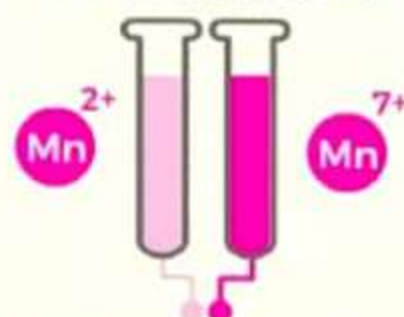


Colours shown by Metals in various oxidation states

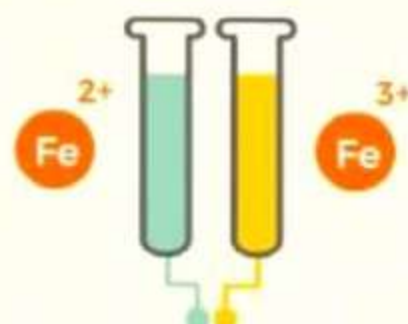
TITANIUM (Ti)



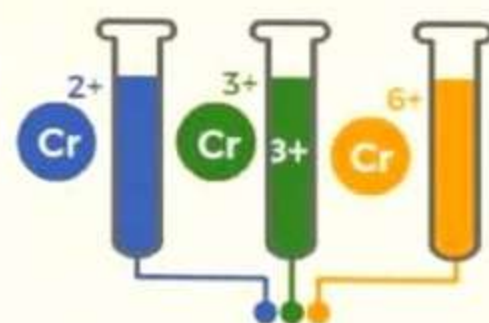
MANGANESE (Mn)



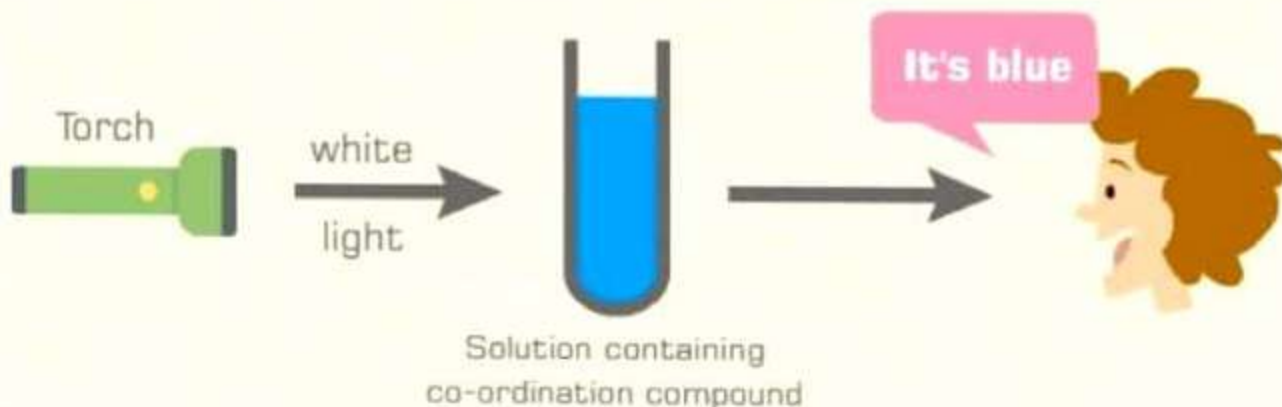
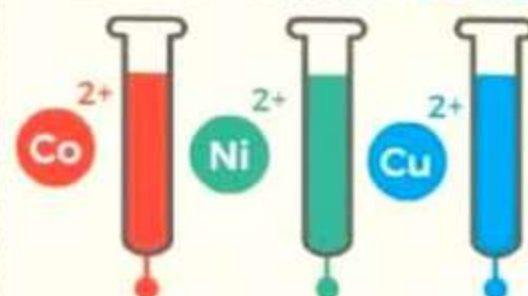
IRON (Fe)



CHROMIUM (Cr)



COBALT NICKEL COPPER



Why we are seeing Blue, Why Not Red or Green ?

Let's see what happens inside the solution containing co-ordination compound when we cast a white light on it

White light of energy E and wavelength λ is passed through the solution

In presence of ligands, d-orbitals splits



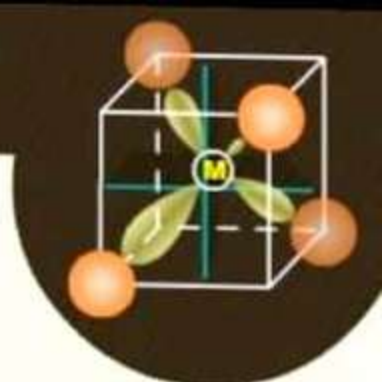
This electron absorbs the energy of wavelength λ and gets excited to d_{z^2} orbital.

If the wavelength, that got absorbed by solution represents the **orange** colour in colour wheel, then its complimentary colour, **blue** will be seen by viewer

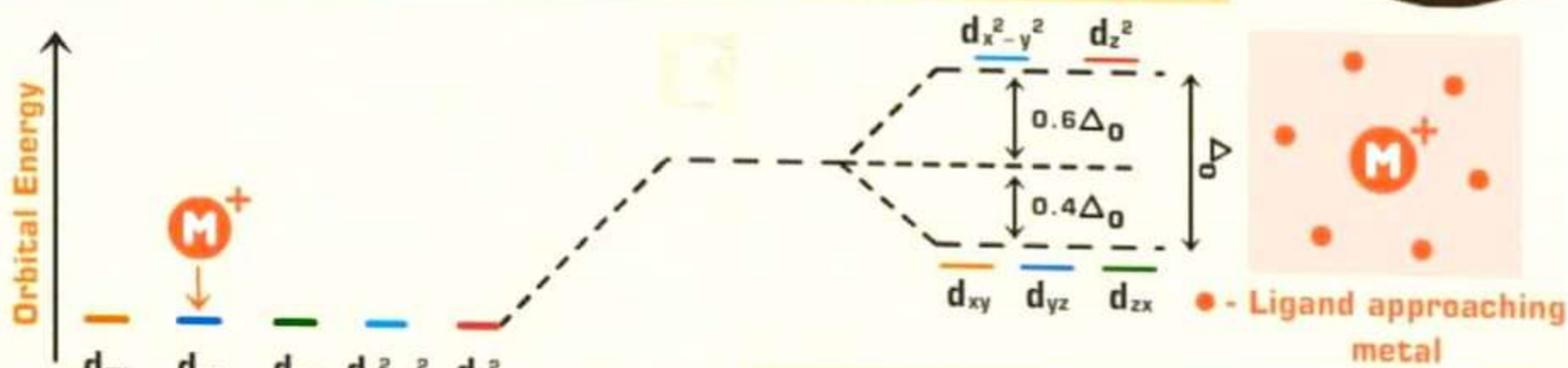


CRYSTAL FIELD THEORY

In crystal field theory, bonding between metal and ligands is purely electrostatic. Ligands are considered as negative point charges.



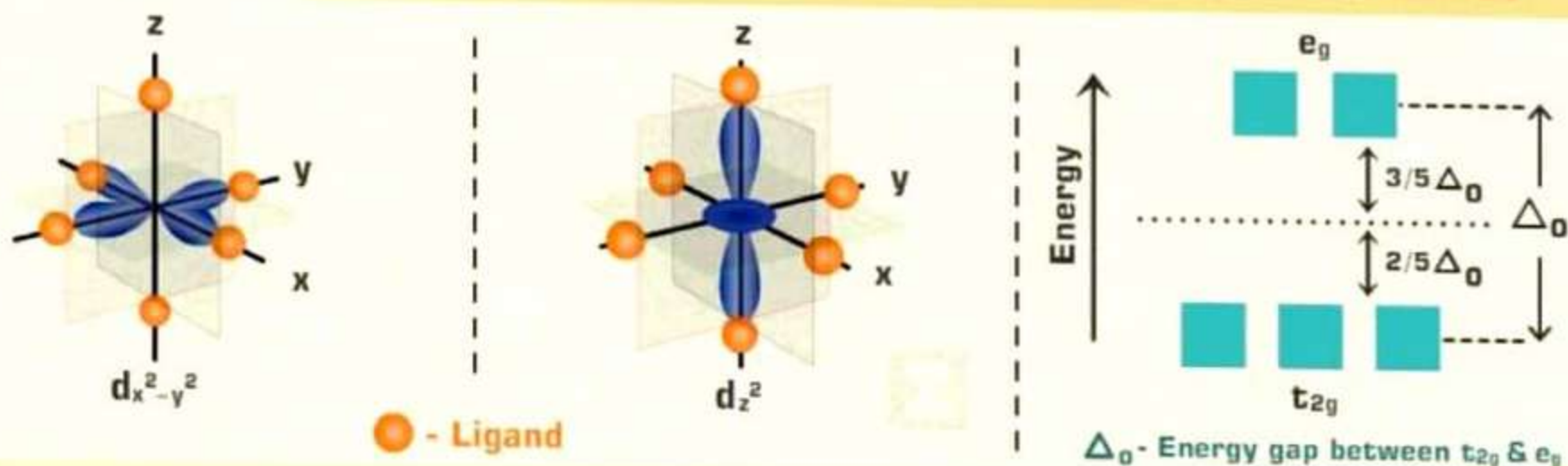
WHAT HAPPENS WHEN LIGANDS APPROACH A METAL



D-orbitals of metal ions when there are no-surrounding ligands

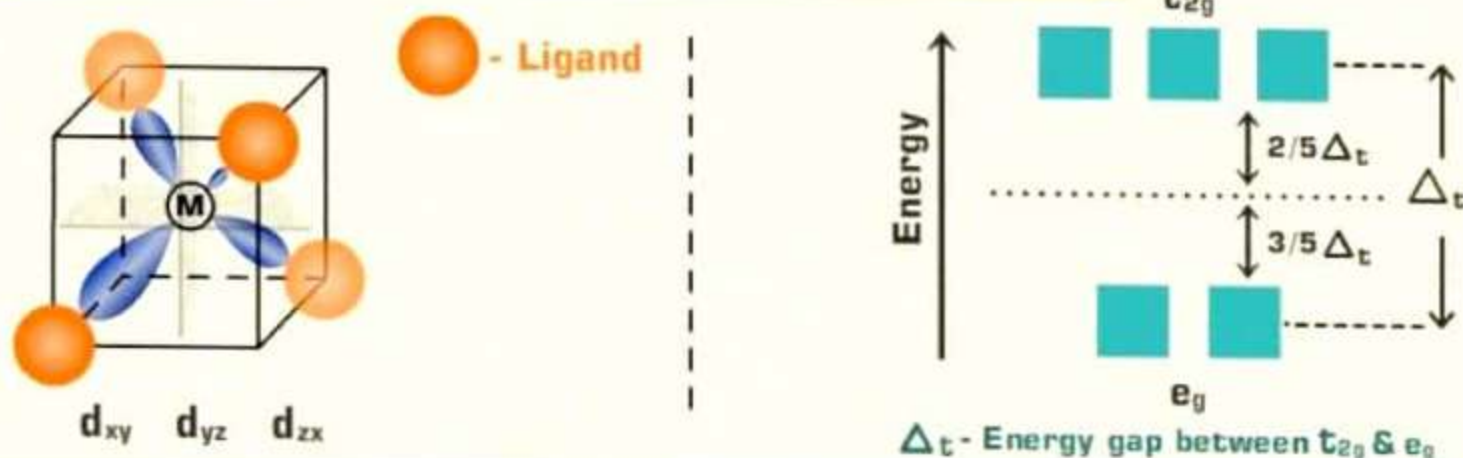
When Ligand approaches the metal ions, there will be a change in energy of electrons in d-orbitals of Metal ions.

ORBITAL SPLITTING IN OCTAHEDRAL COMPLEXES



In Octahedral complex, Ligands approach along **x, y, z** axis. **$d_{x^2-y^2}$** and **d_{z^2}** orbitals align along the axis. So the **repulsion between Orbitals and Ligands** leads to increase in energy.

ORBITAL SPLITTING IN TETRAHEDRAL COMPLEXES



In tetrahedral complexes, Ligands approach between the **x, y** and **z** axis, therefore **d_{xy}** , **d_{yz}** and **d_{zx}** has more energy than **$d_{x^2-y^2}$** and **d_{z^2}**

Strength of ligands

$\text{CO} \approx \text{CN}^- > \text{PPh}_3 > \text{NO}_2^- > \text{NH}_3 > \text{pyridine} > \text{CH}_3\text{CN} > \text{NCS}^- > \text{H}_2\text{O} \approx \text{C}_2\text{O}_4^{2-} > \text{OH}^- > \text{NCO}^- > \text{F}^- > \text{Cl}^- > \text{SCN}^- > \text{S}^{2-} > \text{Br}^- > \text{I}^- > \text{O}_2^{2-}$

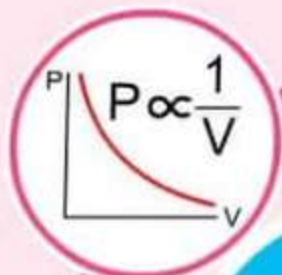
P V T GAS LAWS



Boyle's law



As water bubble rises its size increases because pressure decreases.



Charles's Law



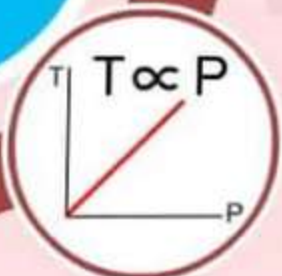
As temperature increases, volume increases



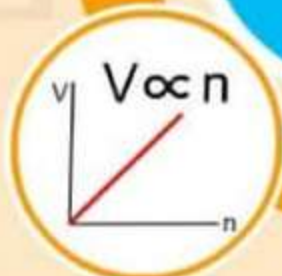
Gay Lussac's law



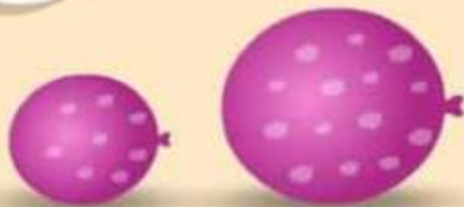
As temperature increases pressure inside the container increases and it explodes



$$PV = nRT$$



Avogadro's Law



As number of moles increase, volume increases



WHAT MAKES GASES REAL

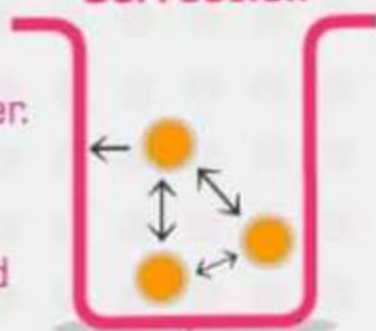


DEVIATION FROM IDEAL GAS



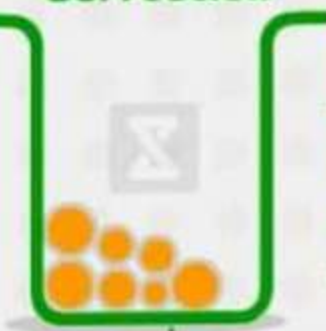
Pressure is one force applied by an ideal gas on one container. In real gas, molecules have attraction between them, thus reducing the force applied on container

Pressure Correction



$$P_{\text{real}} = P_{\text{ideal}} - \frac{an^2}{V^2}$$

Volume Correction



For ideal gas, V is equal to volume of container. However in real gases molecules occupy considerable volume

$$V_{\text{real}} = V_{\text{ideal}} + nb$$

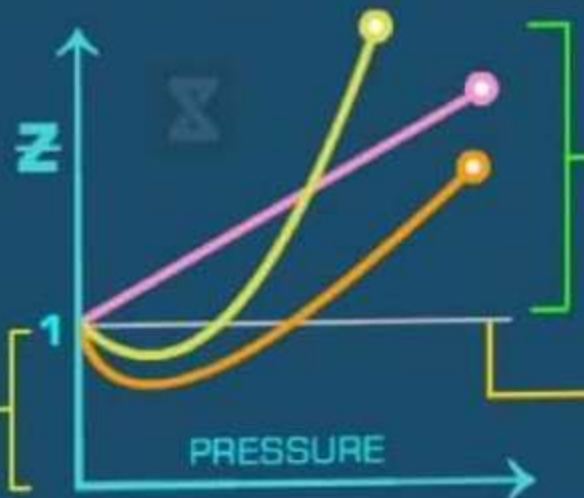
$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Vander Waal Equation

IS z A REAL DETECTOR?

$$z < 1$$

- Deviates from ideal gas behaviour
- Attractive forces dominate repulsive forces



$$z > 1$$

- Deviates from ideal gas behaviour
- Repulsive forces dominate attractive forces

$$z = 1 \Rightarrow \text{ideal gas}$$

AT LOW PRESSURE & HIGH TEMPRATURE real gas acts as ideal gas

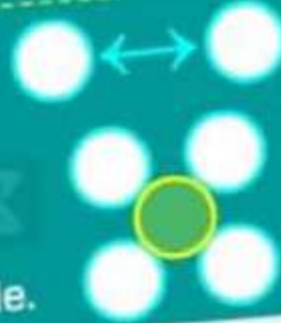
HELIUM & HYDROGEN

They are lighter gases. So force of attraction is less. therefore they have always

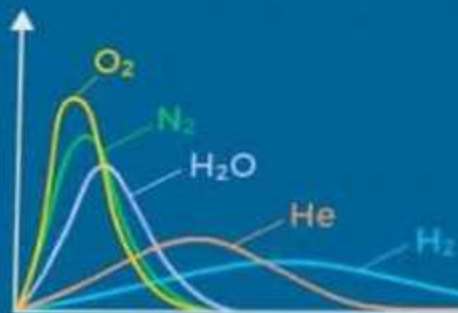
$$z > 1$$

VANDER WAAL'S CONSTANT

a - measure of average attraction of gas molecules
 b - proper volume of gas molecules. It is actually 4 times volume of single molecule.



DISTRIBUTION OF MOLECULAR VELOCITIES



Maxwell Boltzmann stated, all molecules don't have similar velocity. Each molecule moves with different velocity.

AVERAGE VELOCITY

Average of all the velocities of gas molecules

$$A.V. = \sqrt{\frac{8RT}{\pi M}}$$

MOST PROBABLE VELOCITY

Velocity of Maximum number of molecules

$$M.P.V. = \sqrt{\frac{2RT}{M}}$$

AREA

Area under these graphs gives total kinetic energy of gas molecules

$$A = \int F.v dv$$

F = fraction of molecules

Velocity

Energy

E_a

This is activation energy (E_a) barrier, Molecules having more energy than this participate in reaction.

As temperature increases, fraction of molecules having more energy than Activation energy (E_a) increases.

Fraction of molecules having more Kinetic energy than Activation energy (E_a).

ROOT MEAN SQUARE VELOCITY

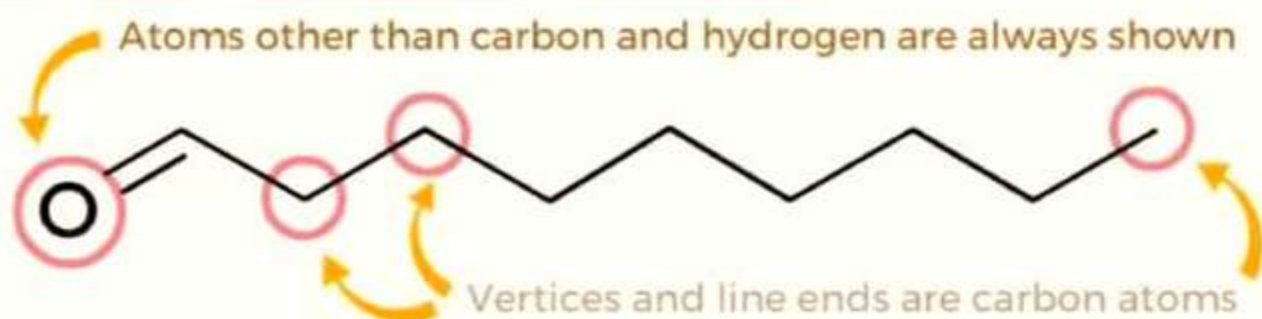
Average velocity may be zero, because velocity is vector, $RMS = \sqrt{\frac{3RT}{M}}$ so we use root mean square velocity.

NOMENCLATURE

A BASIC GUIDE TO DECODING ORGANIC COMPOUNDS NAME

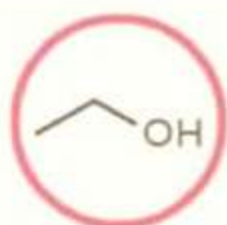
Name of an organic molecule can be long and look like a confusing mix of words and numbers. However, it follows a particular set of rules which allow it's structure to be decoded from it's name.

ORGANIC COMPOUND REPRESENTATION



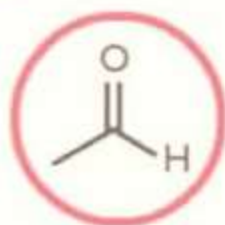
Organic molecules are usually represented using a skeletal formula. The line ends and vertices represent carbon atoms. Hydrogen atoms are implied. Atoms other than carbon and hydrogen are always shown.

FUNCTIONAL GROUPS



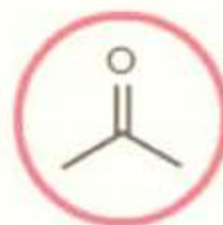
ALCOHOL

Suffix:-ol
e.g.-ethanol



ALDEHYDE

Suffix:-al
e.g.-ethanal



KETONE

Suffix:-one
e.g.-propanone

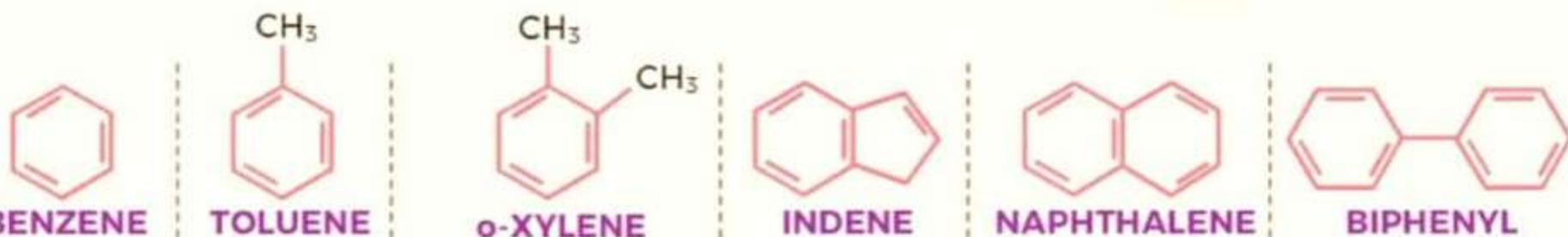


AMINE

Suffix:-amine
e.g.-ethanamine

A molecule's functional group is the group of atoms that give its chemical properties and reactivity. It's usually indicated by a suffix at the end of the name, with a number indicating its position.

AROMATIC COMPOUNDS



Aromatic compounds also known as **arenes** or **aromatics**, are chemical compounds that contain **conjugated** planar ring systems with delocalized **pi-electron**. They should satisfy **Hückel's rule**.

NOMENCLATURE

BOND TYPES

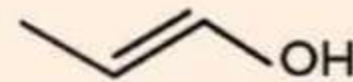
Carbon atoms can be linked by single bond, double bond or even triple bond. The name of the molecule reflects the type of bond present.



BUTANE



BUT-1-ENE



PROP-1-EN-1-OL



BUT-2-ENE



BUT-2-YNE



PROP-2-EN-1-OL

-an-present in name $\Rightarrow$ molecule contains only **single bonds**

-en-present in the name $\Rightarrow$ molecules contains **at least 1 double bond**

-yn-present in name $\Rightarrow$ molecules contains **at least 1 triple bond**

For double and triple bond, number indicates the position of bond.

PARENT CHAIN

NUMBER OF CARBON IS DENOTED BY PREFIX

1 METH-

2 ETH-

3 PROP-

4 BUT-

5 PENT-

6 HEX-

7 HEPT-

8 OCT-

9 NON-

10 DEC-

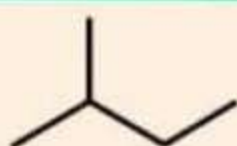


BUTANE

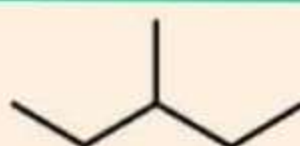
HEXANE

Prefix is part of the organic molecule's name that denotes how many carbons make up its '**parent chain**'. This is defined as the longest continuous connected chain of carbon atoms including the functional groups in the molecule.

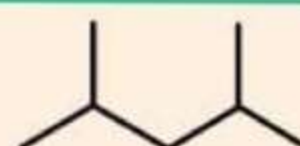
SIDE CHAINS



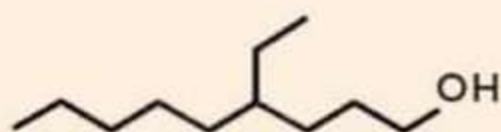
2-METHYLBUTANE



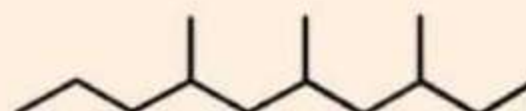
3-METHYLPENTANE



2,4-DIMETHYLPENTANE



4-ETHYLNONAN-1-OL



3,5,7-TRIMETHYLDECANE

Molecules can have one or more carbons that aren't part of the **parent chain**, referred to as '**side chains**'. A number is added to show the location of the side chain on the parent chain.

FUNCTIONAL GROUPS IN ORGANIC CHEMISTRY

ALKANE



Name:- ane
e.g.ethane

ALKENE



Name:- ene
e.g.ethene

ALKYNE



Name:- yne
e.g.ethyne

ALCOHOL



Name:- ol
e.g.ethanol

ETHER



Name:- oxy-ane
e.g.methoxyethane

HALOALKANE



Name:- halo-
e.g.chloroethane

ALDEHYDE



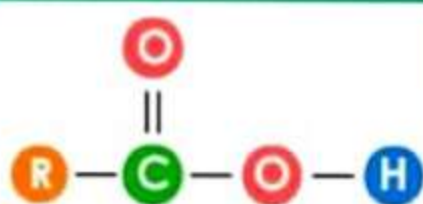
Name:- -al
e.g.ethanal

KETONE



Name:- ane
e.g.propanone

CARBOXYLIC ACID



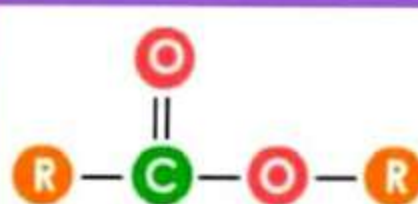
Name:- oic acid
e.g.ethanoic acid

ACID ANHYDRIDE



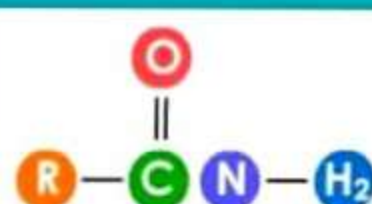
Name:- thiol
e.g.methanethiol

ESTER



Name:- -yl -oate
e.g.ethyl ethanoate

AMIDE



Name:- amide
e.g.ethanamide

ACYL HALIDE



Name:- -oyl halide
e.g.ethanoyl chloride

AMINE



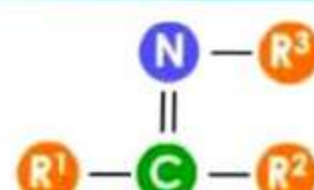
Name:- -amine
e.g.ethanamine

NITRILE



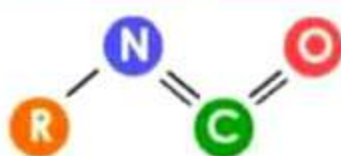
Name:- -nitrile
e.g.ethanenitrile

IMINE



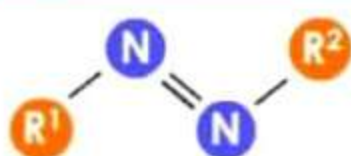
Name:- -imine
e.g.ethanimine

ISOCYANATE



Name:- -yl isocyanate
e.g.ethyl isocyanate

AZO COMPOUND



Name:- -azo-
e.g.azoethane

ARENE



Name:- -yl benzene
e.g. ethyl benzene

THIOL



Name:- thiol
e.g.methanethiol

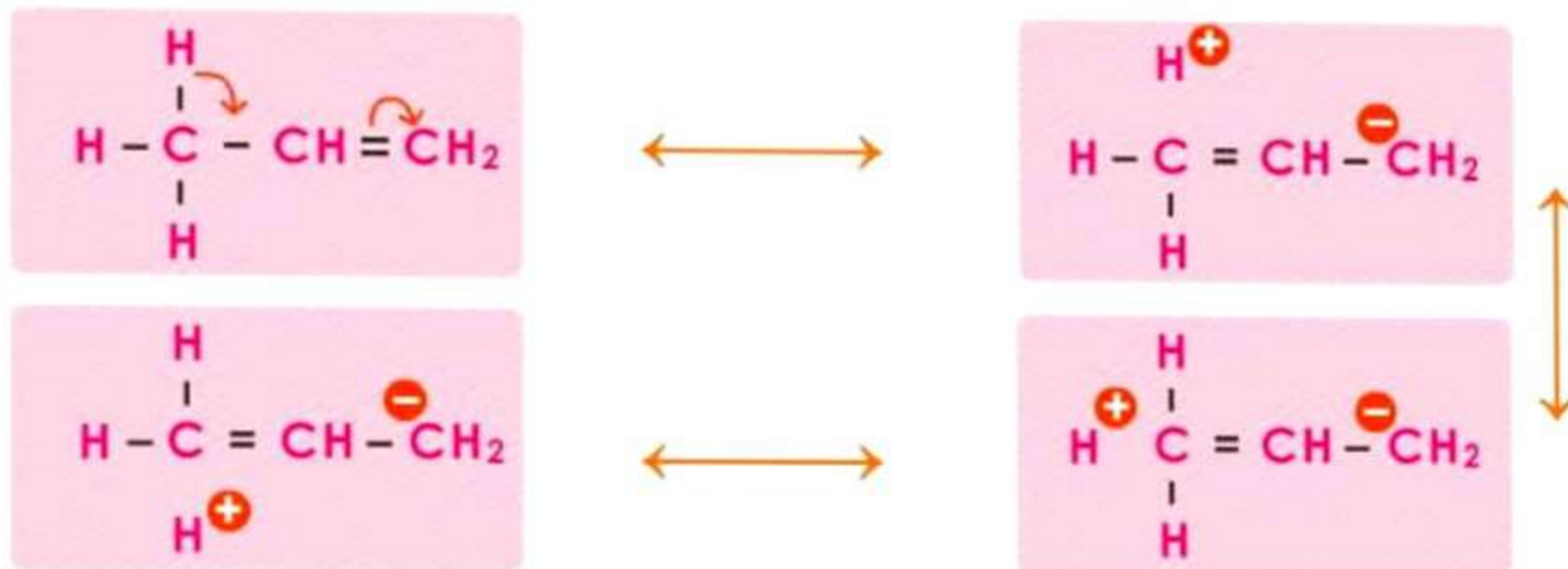
ELECTRON DISPLACEMENT

Part 1



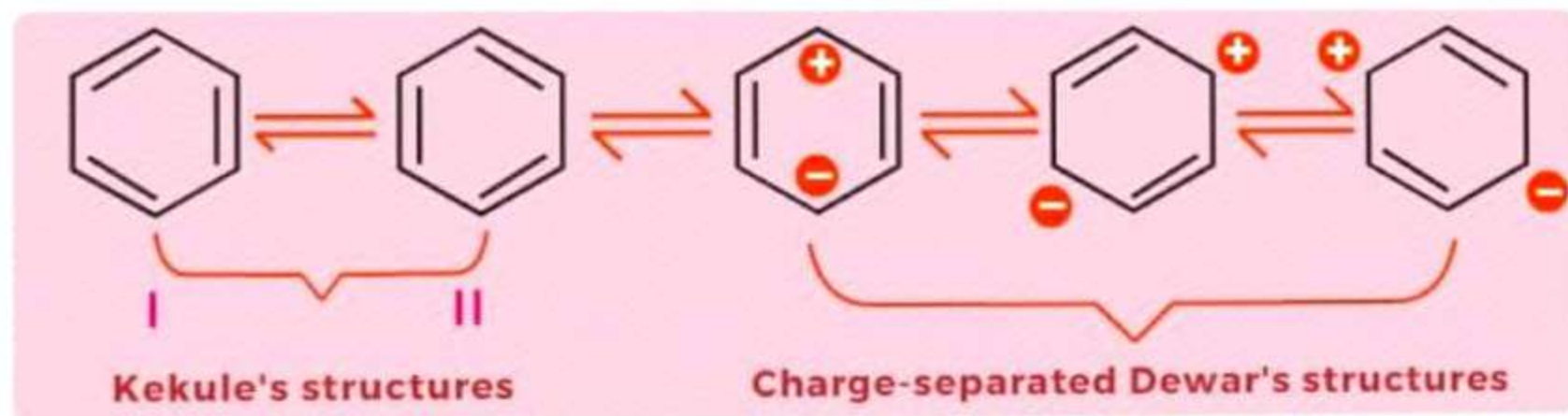
HYPER CONJUGATION

The mechanism of electron release by an alkyl group when it is attached to an unsaturated system is called Hyper conjugation. Hyper conjugation effect takes place through the interaction of σ - electrons of C-H bond with π - electrons of double bond.



MESOMERIC/ RESONANCE EFFECT

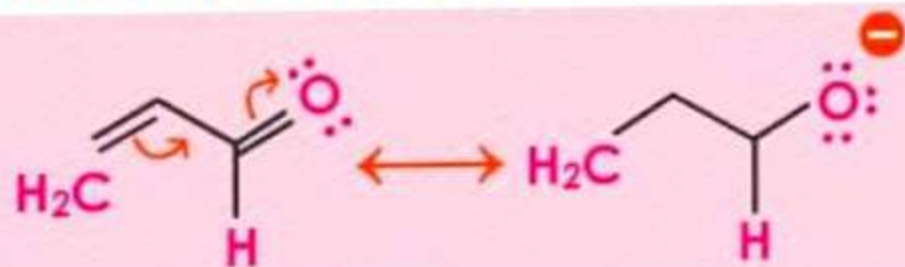
The flow of electrons from one part of a conjugated system to the other caused by phenomenon of resonance is called **resonance effect** or **mesomeric effect**.



-M or -R effect

When the electron displacement is towards the group.

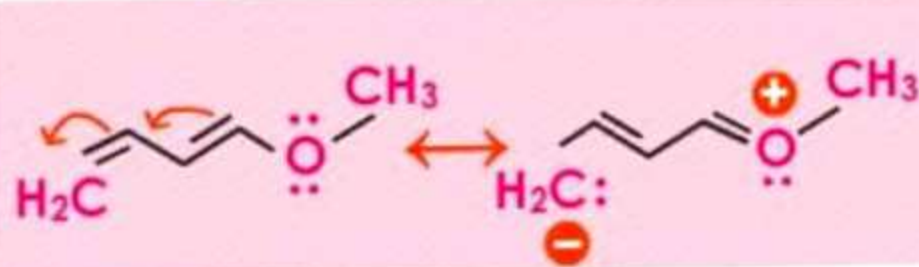
For example :- NO_2 , $-\text{CHO}$.



+M or +R effect

When the electron displacement is away from the group.

For example :- OH , $-\text{OR}$, $-\text{Cl}$



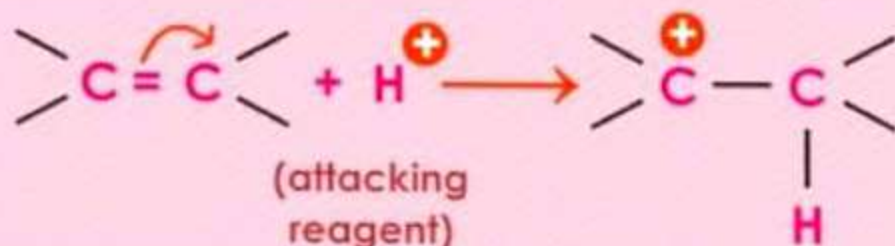
Electromeric effect is defined as the complete transfer of electrons of a multiple bond towards one of the bonded atoms at the demand of an attacking reagent.

TYPES OF ELECTROMERIC EFFECT

+E effect

When displacement of electrons is away from the atom or group.

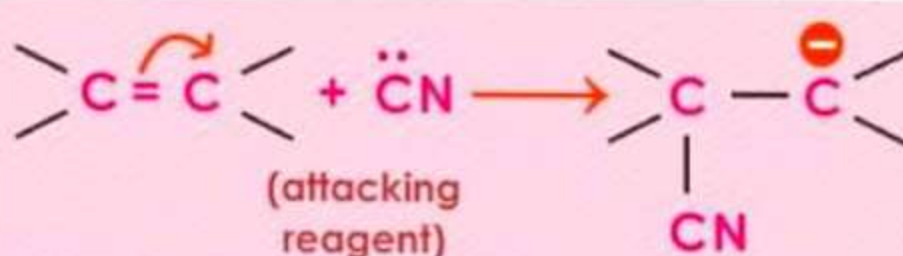
e.g : addition of H^+ to alkene.



-E effect

When displacement of electrons is towards the atom or group.

e.g : addition of cyanide ion (CN^-) to the carbonyl group.



INDUCTIVE EFFECT

Inductive effect is defined as permanent displacement of shared electron pair in a carbon chain towards more electronegative atom or group.



electron withdrawal
stabilizes
carbocation (make
less-)



electron withdrawal
destabilizes
carbanion (make
less+)



electron release
destabilizes
carbanion (make
more-)

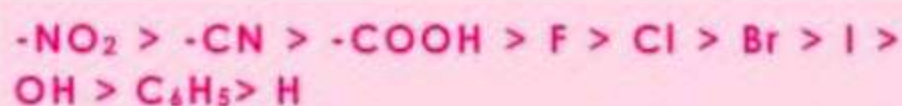


electron release
stabilizes
carbocation
(make more+)

TYPES OF INDUCTIVE EFFECT

Negative Inductive Effect (-I effect, Electron withdrawing effect)

When an electronegative atom or group (more electronegative than hydrogen) is attached to the terminal of the carbon chain in a compound, the electrons are displaced in the direction of the attached atom or group.



Positive Inductive Effect (+I effect, Electron releasing effect)

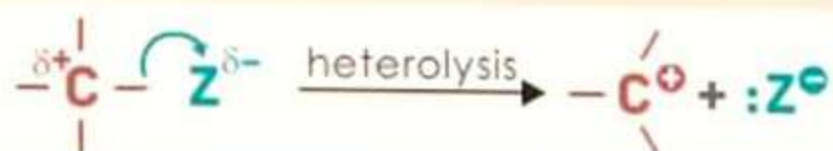
When an electropositive atom or group (more electropositive than hydrogen) is attached to the terminal of the carbon chain in a compound, the electrons are displaced away from the attached atom or group.



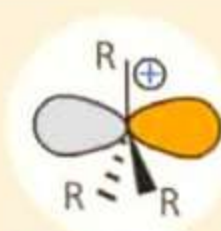
CARBOCATION & CARBANION

CARBOCATION

Ionic species with **positively** charged carbon atom.



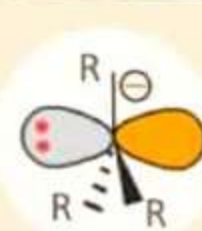
- sp^2 Hybridized
- Trigonal planar
- Incomplete octet



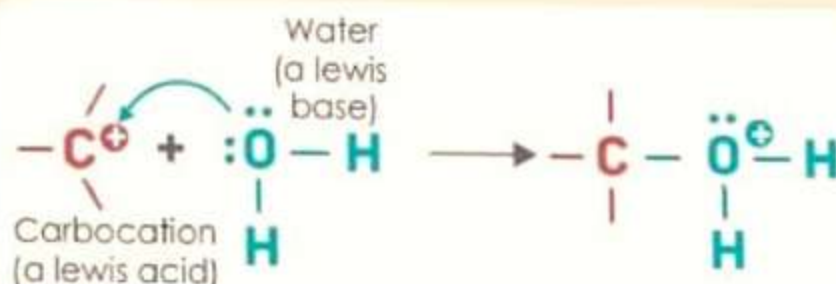
Carbocation



Carbon radical



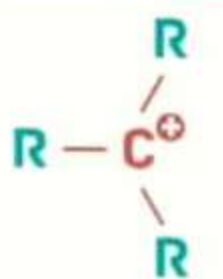
Carbanion



Carbocations are strong Lewis acids.

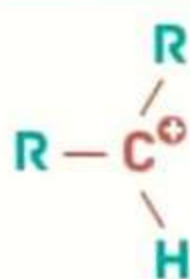
RELATIVE STABILITIES OF CARBOCATIONS

MOST STABLE



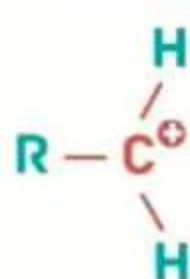
Tertiary carbocation

>



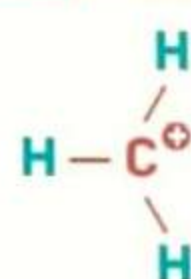
Secondary carbocation

>



Primary carbocation

>



Methyl carbocation

LEAST STABLE

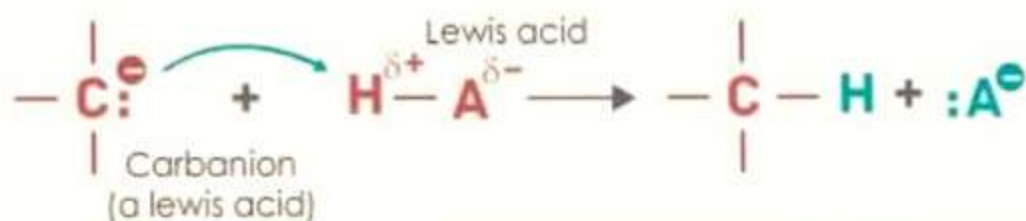
CARBANION

Ionic species with **negatively** charged carbon atom.



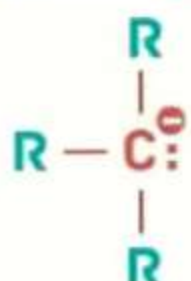
- sp^3 Hybridized
- Pyramidal
- Complete octet

Carbanions are **strong Lewis** bases and Bronsted bases.



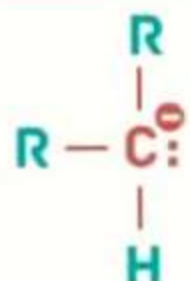
RELATIVE STABILITIES OF CARBANIONS

LEAST STABLE



Tertiary carbanion

>



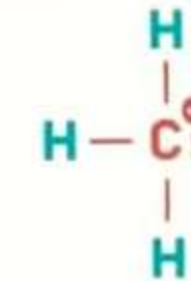
Secondary carbanion

>



Primary carbanion

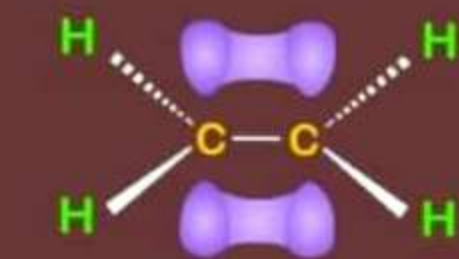
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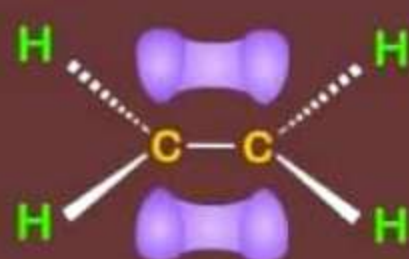
Methyl carbanion

MOST STABLE

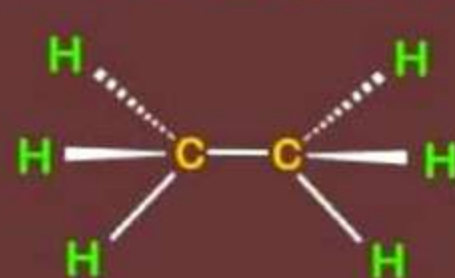
HEAT OF HYDROGENATION



Hydrogen molecules settle on the surface of the catalyst and react with metal atoms



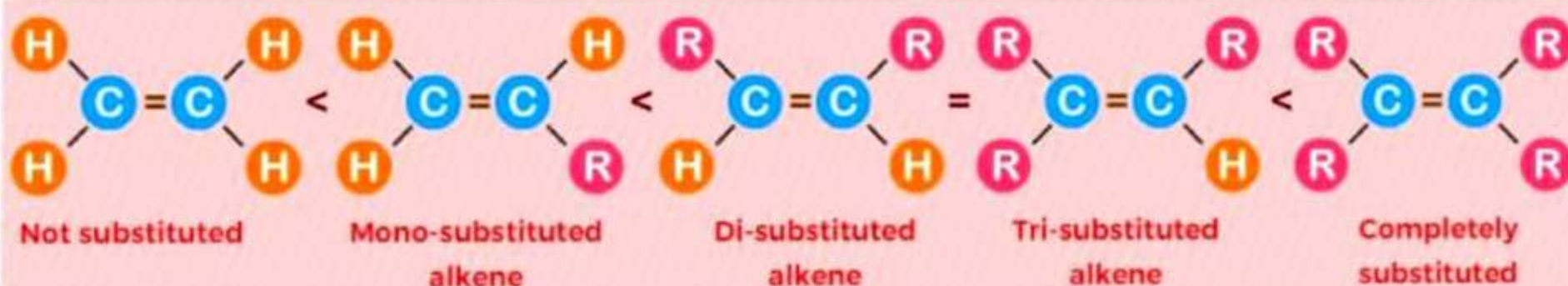
The alkene approaches the surface of the catalyst



The π bond between the two carbons is replaced by two C-H σ bonds

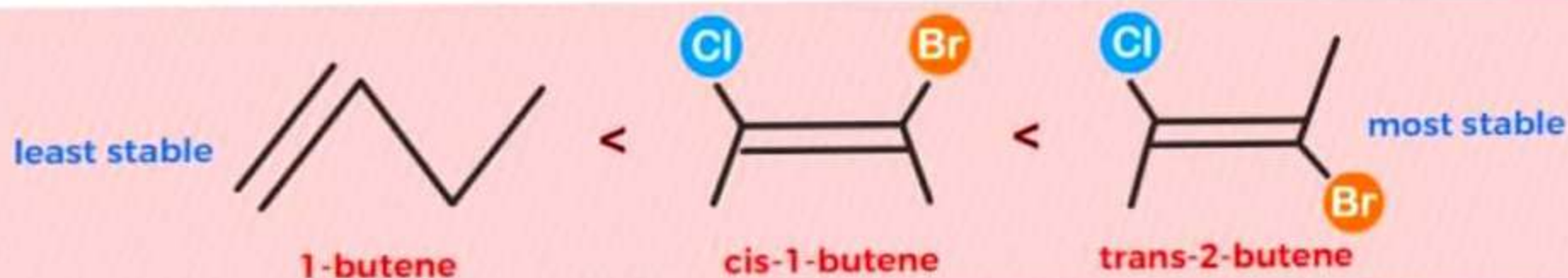
Heat of hydrogenation (symbol: ΔH_{hydro} , ΔH°) of an **alkene** is the standard enthalpy of **catalytic hydrogenation** of an alkene. Catalytic hydrogenation of an alkene is always **exothermic**. Therefore, heat of hydrogenation of alkenes is always **negative**.

least stable (higher ΔH°).....most stable (lower ΔH°)



least substituted.....most substituted

In disubstituted alkenes, **trans-isomers** are more stable than **cis-isomers** due to **steric hindrance**. Also, internal alkenes are more stable than terminal ones. See the following isomers of butene:



$\text{HOH} \propto \text{no. of } \pi \text{-bond in compound}$

If no. of π -bond is same then

$$\text{HOH} \propto \frac{1}{\text{Stability of compound}}$$

In case of alkene

$$\text{HOH} \propto \frac{1}{\text{Stability of compound}} \propto \frac{1}{\text{No. of } \alpha \text{ H}}$$



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$\text{p}K_a = -\log_{10} K_a$$

Key factors that affect the acidity



Electronegativity - The more electronegative the anionic atom in the conjugate base, the better it is at accepting the negative charge.

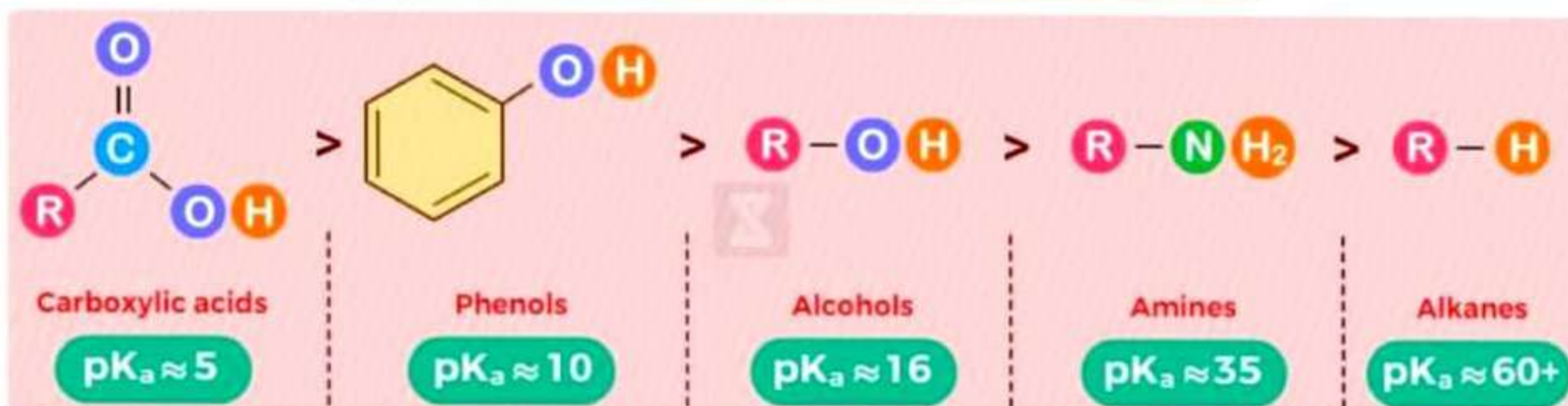


Size - The easier it is for the conjugate base to accommodate negative charge (lower charge density). The size of the group also weakens the bond H-X.



Resonance - In the carboxylate ion (RCOO^-) the negative charge is delocalised across 2 electronegative oxygen atoms which makes it more stable than being localised on a specific atom as in alkoxide (RO^-).

General acidity trend of common organic acids



Key factors that affect the basicity



Electronegativity - The more electronegative the atom donating the electrons is, the less willing it is to share those electrons with a proton, so the weaker the base.



Size - The larger the atom the weaker the H-X bond and the lower the electron density making it a weaker base.



Resonance - In the carboxylate ion (RCOO^-) the negative charge is delocalised across 2 electronegative atoms which makes it the electrons less available than when they localised on a specific atom as in the alkoxide (RO^-).

General basicity trend of some common organic bases



SUBSTITUTION REACTION (SN_1)



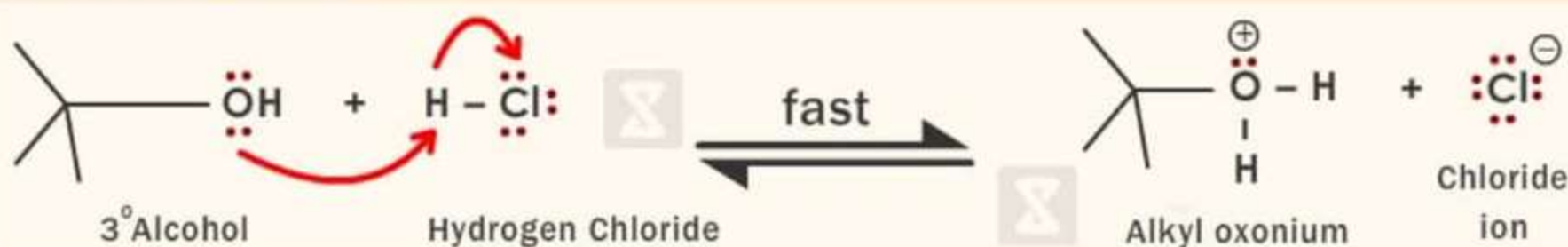
Black cat waits for the brown cat to leave and then takes the pod. Similarly in SN_1 reaction one nucleophile waits for other nucleophile to move from substrate.

WHAT IS SN_1 REACTION ?

SN_1 indicates the **unimolecular nucleophilic substitution** reaction in organic chemistry. Their **rate determining step** of the mechanism depends on the decomposition of a single molecular species. So that, the rate of a SN_1 reaction can be expressed as **rate = k [R-LG]**. Furthermore, SN_1 is a **multi-step** reaction, which forms an intermediate and several transition states during the reaction. This intermediate is a more stable carbocation and the reactivity of the molecule depends on the **R- group**.

SN_1 MECHANISM (R - OH $\rightarrow$ R - X)

PROTON TRANSFER



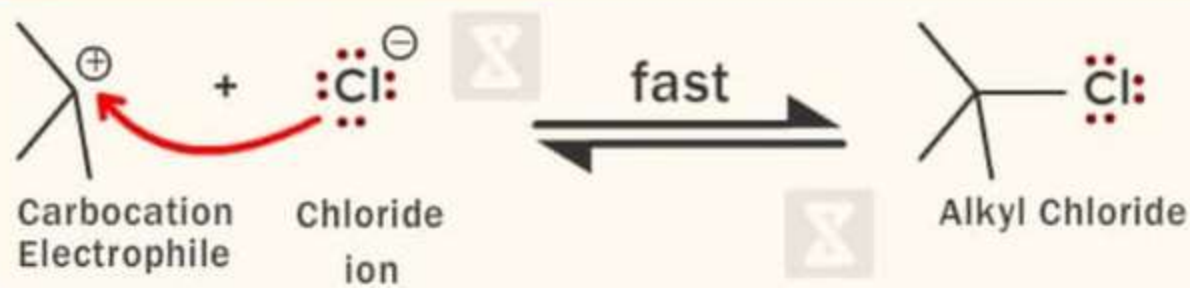
DISSOCIATION OF WATER



RATE - DETERMINING STEP

- Unimolecular
- Nucleophilic reaction
- " SN_1 " mechanism

CAPTURE OF CARBOCATION



Protic solvent



3° or 2° carbon



1° carbon



SUBSTITUTION REACTION (SN_2)

Part II



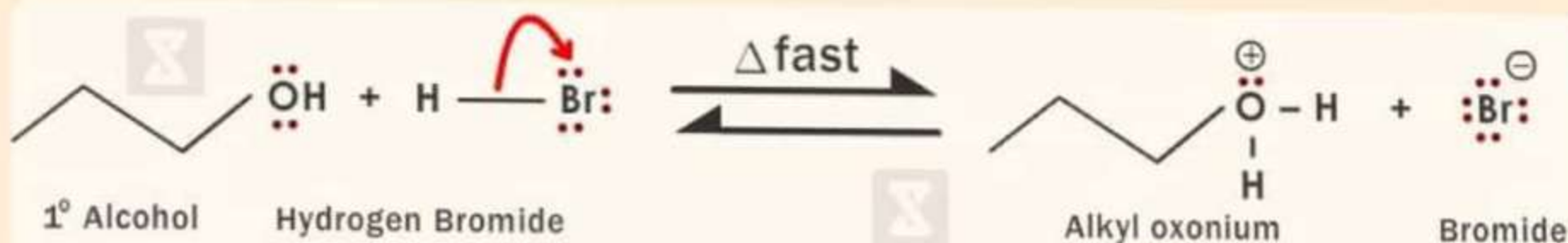
Black cat forces the brown cat out from the pod. Similarly in SN_2 , strong nucleophile kicks out the weaker nucleophile.

WHAT IS SN_2 REACTION?

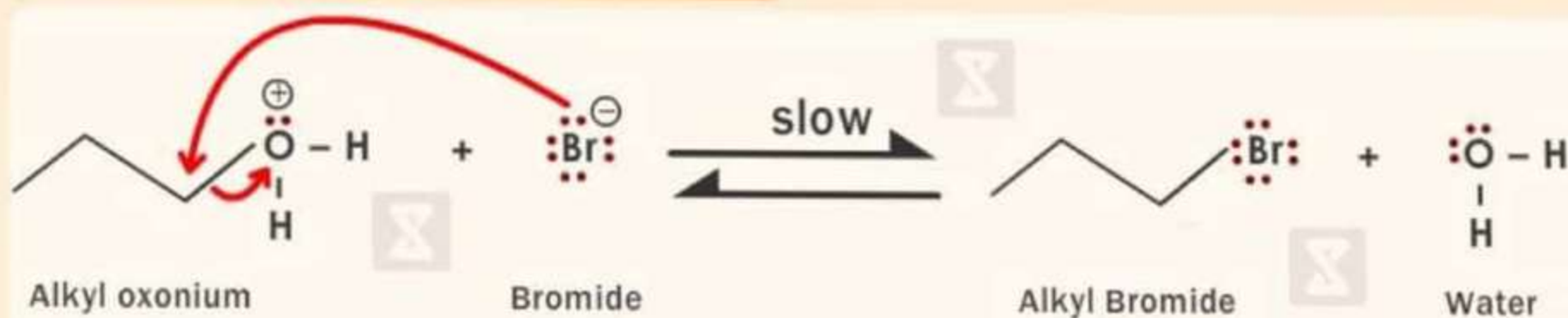
SN_2 indicates the **bimolecular nucleophilic substitution** reaction in organic chemistry. In this mechanism, separation of leaving group and formation of new bond happen synchronously. Therefore, two molecular species are involved in the rate determining step and this leads to the term bimolecular nucleophilic substitution reaction or SN_2 . The rate of the SN_2 reaction can be expressed as $\text{rate} = k [R-LG] [Nu^-]$. In inorganic chemistry, this reaction is also called **associative substitution** or **interchange mechanism**.

SN_2 MECHANISM ($R-OH \rightarrow R-X$)

PROTONATE ALCOHOL



NUCLEOPHILE (HALIDE) SEPARATES WATER



Here, nucleophile attacks from the opposite direction of the leaving group. Thus, SN_2 reaction always leads to an **inversion of stereochemistry**. This reaction works best **with methyl and primary halides** because bulky alkyl groups block the backside attack of the nucleophile. In addition, the stability of the leaving group as an anion and the strength of its bond to the carbon atom both affect the rate of reaction.

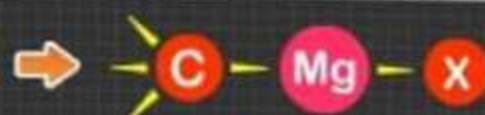
GRIGNARD REAGENT

WHAT IS GRIGNARD REAGENT?



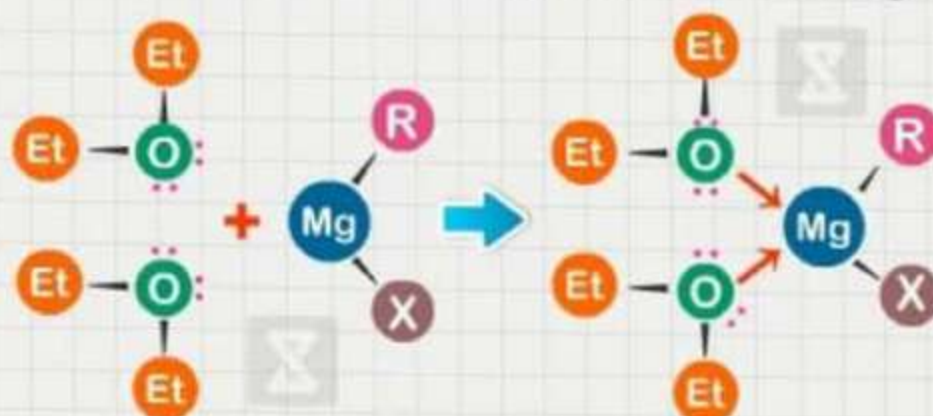
→ If the metal attached to the carbon is Magnesium, then it is called Grignard Reagent.

→ Victor Grignard first synthesized this compound.



WHY GRIGNARD REAGENT IS PLACED IN DRY ETHER ?

Ether provides stability to Grignard reagent by completing the octet on Mg.



WHAT ARE ORGANO METALLICS ?

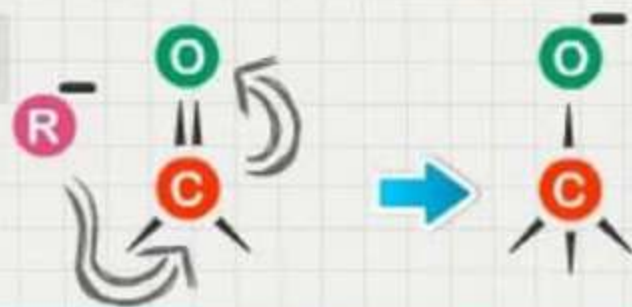
Organo metallic compounds are organic compounds in which metal atom is directly attached to Carbon atom



CHEMICAL REACTIVITY OF GRIGNARD REAGENT

Grignard reagent produces nucleophile therefore it attacks at electrophilic centers

ATTACK ON ALDEHYDES



ATTACK ON ALKYNES



REACTIVITY

Carbon has high electronegativity than Mg.
So on breaking this bond Produces a C^- nucleophile

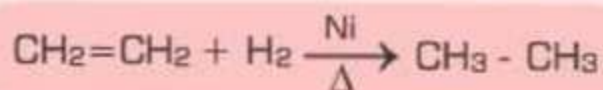


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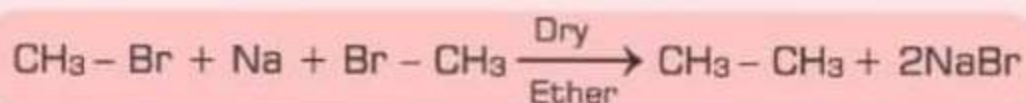
FACTS ABOUT ALKANES

Preparation of Alkanes

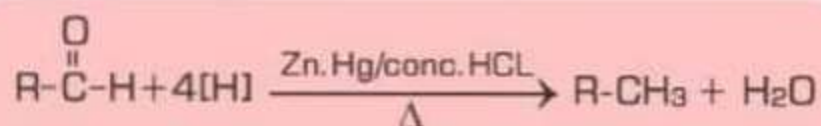
- From unsaturated Hydrocarbons in the presence of catalyst 'Ni' or 'Pt'



- By Wurtz Reaction:



- By the Reduction of Aldehydes and Ketones



- Grignard's Reagent



Density of alkanes is less than water therefore they float over it.

Solubility

Alkanes do not dissolve in water. They form a layer on top of water. However alkanes dissolve in non-polar organic solvents like Toulene, Benzene



Example of Alkanes

Methane



Ethane

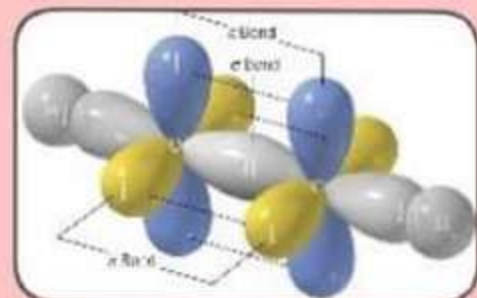


Propane



Physical State	
Carbon Count	State
1 - 4	Gases
5 - 15	Liquid
15 - more	Solid

C - in sp³ Hybridisation



General Formula :- $\text{C}_n\text{H}_{2n+2}$

Uses

Lighter alkanes are used in natural gas.

Propane and Butane are used in LPG cylinders



Boiling

Boiling point depends on Vander waal forces.



More Carbon atoms



High Vander waal forces

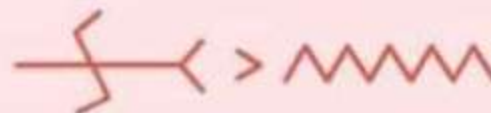


High boiling point



Melting

Melting point depends on packing of compound.



More branching



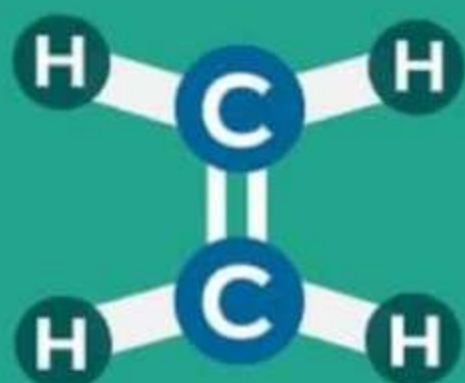
Close packing



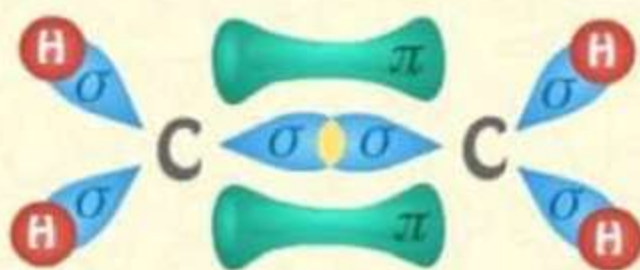
High melting point



ALKENES



SP² hybridisation



Physical State

Carbon Count	State
1 - 3	Gases
4 - 20	Liquid
> 20	Solid

Acts as a Nucleophile

In organic chemistry, π -bond is considered as a nucleophile. Therefore alkenes participate in addition reactions

Polymerisation

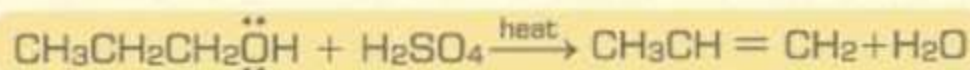
Ethene undergoes polymerisation and forms products like polyethene



Preparation

1. Dehydration of alcohols

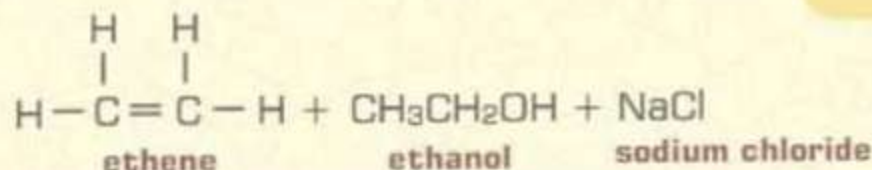
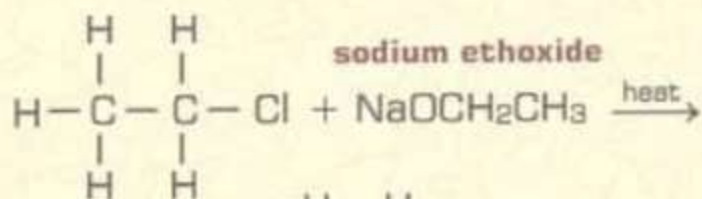
A molecule of water is eliminated from an alcohol molecule by heating the alcohol in the presence of a strong mineral acid.



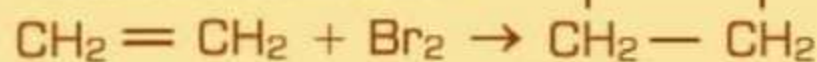
2. Dehydrohalogenation of alkyl halides

The **dehydrohalogenation** of alkyl halides, another β elimination reaction, involves the loss of a hydrogen and a halide from an alkyl halide (RX).

ethyl chloride

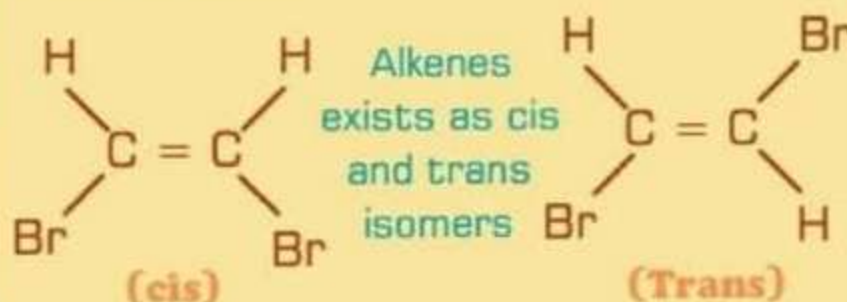


Test for Alkenes



If you add bromine water to alkene, it decolourises the liquid because bromine reacts with alkene, whereas with alkanes it cannot react.

Isomerism



Dipole Moments

cis > 0, trans = 0

Melting Point

trans > cis

Boiling Point

cis > trans

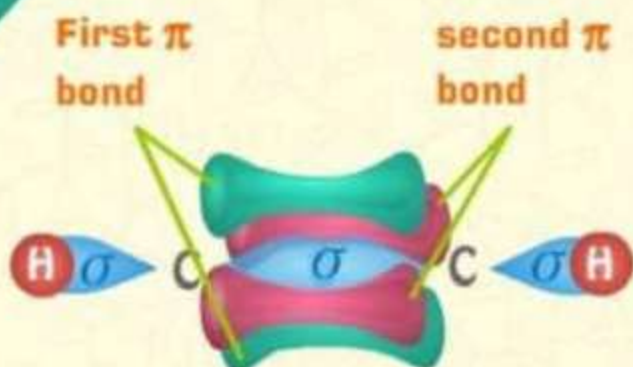


More dipole moment $\Rightarrow$ More polarity $\Rightarrow$ More solubility in polar solvents

More polarity $\Rightarrow$ More interaction between compounds $\Rightarrow$ Higher boiling point

ALKYNES

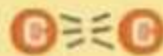
SP hybridisation



Physical Properties



Alkynes are gases at room temperature.



192 kcal/mol energy is required to break the triple bond.

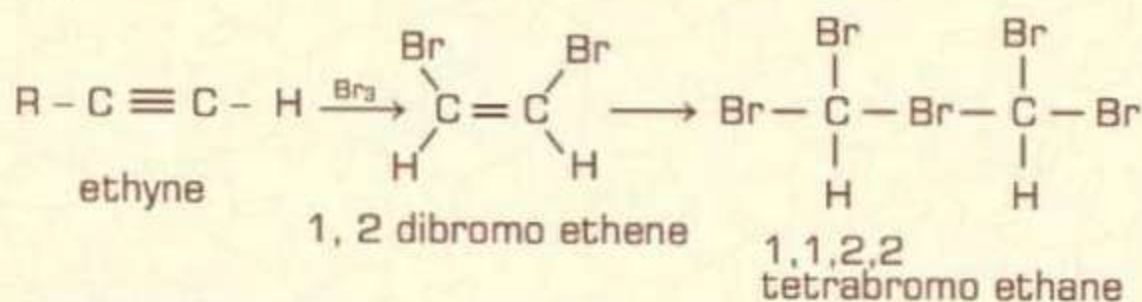


Shortest bond length is 120 Å.

Test for Alkynes

Reaction with Bromine

The alkynes react slowly with bromine water to decolourise it, and this reaction can be used to distinguish between alkenes and alkynes. Alkenes decolourise bromine water very rapidly, but alkynes take several minutes.



Acidic Hydrogen



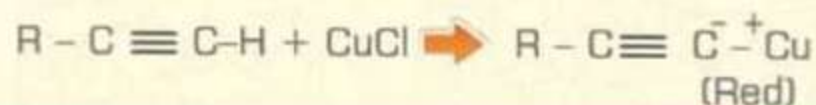
Order of electronegativity is



Due to large electronegativity of sp carbon, terminal hydrogen becomes acidic and reacts with bases and undergoes neutralization.

For Terminal Alkynes

Terminal alkynes have acidic hydrogen, therefore by reacting with CuCl in NH_4OH , acidic hydrogen is replaced with Cu giving red colour.



Uses

Alkynes don't have any commercial use. Acetylene is used in oxy-acetylene flame.



Nucleophile: Like alkenes, they also act as nucleophiles, due to presence of 2π bonds

Preparation: Alkynes are prepared by **hydrolysing carbides**.





IONIC EQUILIBRIUM



IONIC CONDUCTORS OR ELECTROLYTES

Conductors in which the current passes through them due to the presence of free ions are called **Ionic Conductors** or **Electrolytes** or **Electrolytic conductors**. Ionic conductors are further divided into two types on the basis of their strengths :-

STRONG ELECTROLYTES

- Those substance which almost completely ionize into ions in their aqueous solution are called **strong electrolytes**.
- Degree of ionisation (α) for strong electrolyte is equal to 1.



HCl



KOH



NaOH

WEAK ELECTROLYTES

- Those substance which ionize to a small extent in their aqueous solution are known **weak electrolytes**.
- Degree of ionization (α) for weak electrolytes is less than 1.



H₂O



HCN



NH₄OH

CONCEPT OF ELECTROLYTE

STRONG ELECTROLYTE	WEAK ELECTROLYTE
$A^+ B^- \rightarrow A^+ + B^-$	$A B \rightleftharpoons A^+ + B^-$
$\begin{matrix} a & 0 & 0 \\ 0 & a & a \end{matrix}$	$\begin{matrix} a & 0 & 0 \\ a(1-\alpha) & a\alpha & a\alpha \end{matrix}$
$\alpha = 1$	$\alpha < 1$
No equilibrium	Equilibrium
Irreversible	Reversible

OSTWALD'S DILUTION LAW

According to Ostwald, the degree of dissociation (α) of a weak electrolyte is inversely proportional to the square root of the molar concentration of the solution.

$$K = \frac{C\alpha^2}{1-\alpha}$$

If α is very small
 $1 - \alpha \approx 1 \Rightarrow K_a = C\alpha^2$

$$\text{or } \alpha = \sqrt{\frac{K_a}{C}} \Rightarrow \alpha \propto \frac{1}{\sqrt{C}}$$

Here K_a is dissociation constant and C is molar concentration of the solution.

DEGREE OF IONISATION OR DEGREE OF DISSOCIATION (α)

- It is the fraction of the total number of molecules which ionise (dissociate) into constituent ions.
- $$\alpha = \frac{\text{number of molecules ionised or dissociated}}{\text{total numbers of molecules taken}}$$
- Values of the degree of dissociation (α) depends upon the following factors

Nature of solute

Nature of solvent

Concentration

Temperature

SALT HYDROLYSIS

Hydrolysis is defined as the process in which a salt reacts with water to give back the acid and the base. It is reverse of neutralisation.



Salt hydrolysis is an endothermic reaction therefore on increasing the temperature, the extent of hydrolysis increases.

Hydrolysis at a glance

SALT	NATURE	DEGREE	HYDROLYSIS CONSTANT	pH
Strong acid + Strong Base (NaCl)	Neutral	No Hydrolysis	—	—
Weak acid + Strong base (CH ₃ COONa)	Base	$h = \sqrt{\frac{K_w}{CK_a}}$	$K_h = \frac{K_w}{K_a}$	$\text{pH} = 1/2[\text{pK}_w + \text{pK}_a + \log C]$
Strong acid + Weak base (NH ₄ Cl)	Acidic	$h = \sqrt{\frac{K_w}{CK_b}}$	$K_h = \frac{K_w}{K_b}$	$\text{pH} = 1/2[\text{pK}_w - \text{pK}_b - \log C]$
Weak acid + Weak base (CH ₃ COONH ₄)	—	$h = \sqrt{\frac{K_w}{(K_a + K_b)}}$	$K_h = \frac{K_w}{K_a \cdot K_b}$	$\text{pH} = 1/2[\text{pK}_w + \text{pK}_a - \text{pK}_b]$

Solution

Basic solution



NaOH
strong base

H₂CO₃
weak acid

Neutral solution



NaOH
strong base

H₂SO₄
strong acid

Acidic solution

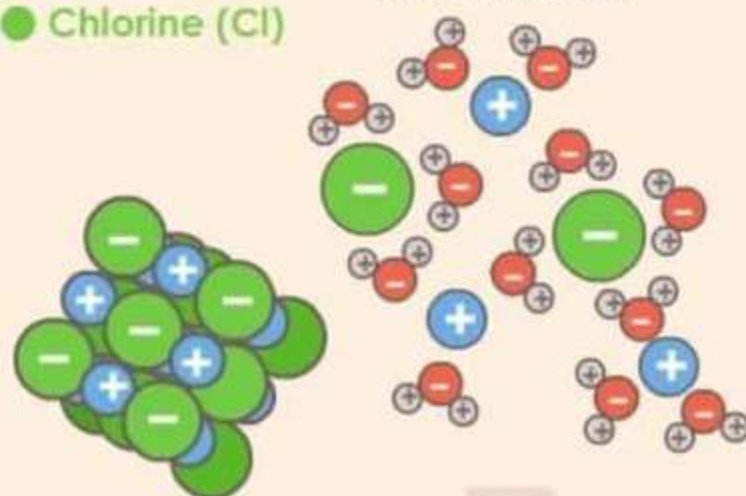


NH₃
weak base

HNO₃
strong acid

● Sodium (Na)
● Chlorine (Cl)

NaCl in water



Use of Hydrolysis in the "Real World"

Life processes requires energy for processes like **respiration**, **muscle contraction**, **distribution of hormone** etc. The energy that is converted from food, is stored into **ATP molecules (Adenosine Triphosphate)**. To release the energy stored in the bonds of ATP molecules, hydrolysis must occur to break a phosphate group off of an ATP molecule, thus releasing energy from the bonds. ATP now becomes **ADP (Adenosine Diphosphate)** by losing a phosphate group through hydrolysis.



ACID-BASE INDICATORS

Acid-Base indicators are organic substances which have one colour in acid solution while an altogether different colour in alkaline solution.

THEORIES OF ACID-BASE INDICATORS

OSTWALD'S THEORY (1891)

According to this theory:

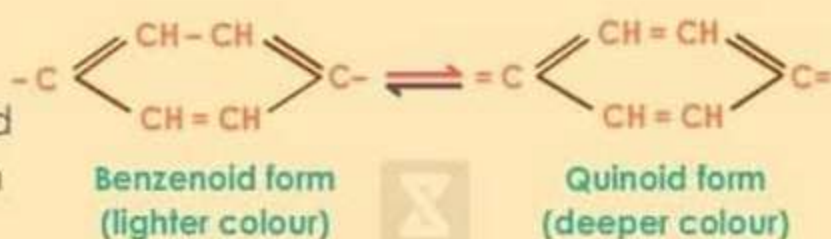
- Acid-base indicators are weak organic acids or bases.
- They have different colours in ionised and non-ionised states.
- The colour of the indicator depends on the relative proportion of the unionised indicator molecules and its ions.



MODERN QUINOID THEORY

Main postulates of this theory are:

- The indicators used in **acid-alkali titrations** are aromatic organic compounds which are equilibrium mixtures of at least **two tautomeric forms**, ordinarily one form is **benzenoid** while the other is **quinoid**.
- The two forms have different colours. The **quinoid** form is usually **deeper in colour** than the benzenoid form. Out of these one form exists in acidic solution while other in alkaline solution.
- Change in **pH** causes the transition of benzenoid form to quinoid form and vice versa and consequently a change in colour. This theory explains the action of **phenolphthalein**, **methyl orange** and other acid base indicators.

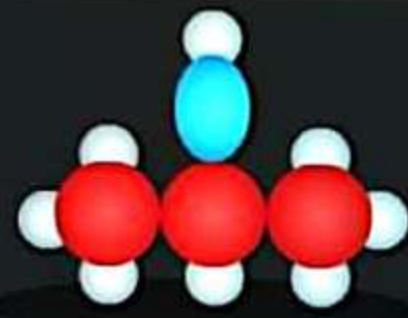


CHOICE OF INDICATORS

Nature of Titration	pH Jump at the Equivalence point	Suitable Indicator
Strong acid and strong base	4-10	Any Indicator (Methyl orange, Methyl red Phenolphthalein, Bromo thymol blue etc.)
Weak acid and strong base	6.5-10	Phenolphthalein, thymol blue, thymolphthalein
Strong acid and Weak base	4-7.5	Methyl orange, Methyl red, Bromocresol green
Weak acid and Weak base	6.5-7.5	No Indicator is suitable



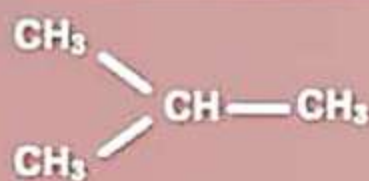
TYPES OF STRUCTURAL ISOMERISM



CHAIN ISOMERISM



Butane



Isobutane

Same molecular formula, difference in arrangement. Chain of minimum 4 carbons is necessary.

RING CHAIN ISOMERISM



Propene



Cyclopropane

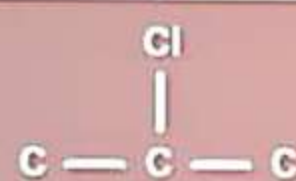
Mode of chain formation differs in open and close chain formation.



POSITION ISOMERISM



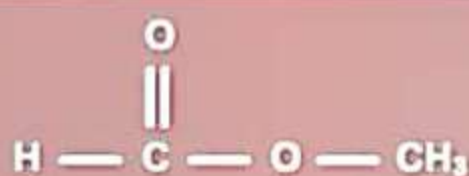
1 - Chloropropane



2 - Chloropropane

Same molecular formula, difference in the position of either the functional group or the multiple bond or the branched chain.

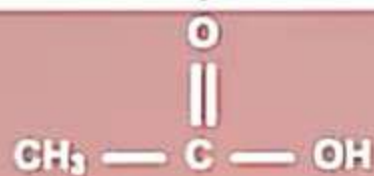
FUNCTIONAL ISOMERISM



Ester



functional isomer



Carboxylic acid

Same molecular formula but different in functional group. Other two functional group isomers

Alcohol & Ether.

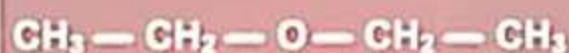
• Aldehydes & Ketones.



METAMERISM



Methyl propyl ether



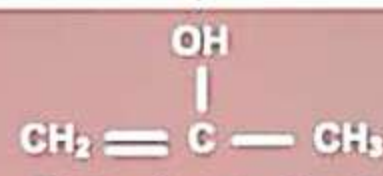
Diethyl ether

Compound having same molecular formula but differ from the nature of alkyl group directly attached with polyvalent atom or polyvalent functional group.

TAUTOMERISM



Dimethyl ketone



Prop - 1 - en - 2 - ol

Compound having same molecular formula but different due to oscillation of an atom (usually H^+) are known as tautomers.





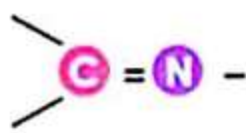
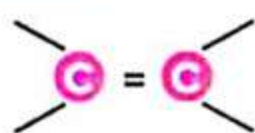
GEOMETRICAL ISOMERISM

Isomers which possess the same molecular and structural formula but differ in the arrangement of atoms or groups in space due to restricted rotation are known as **geometrical isomers** and the phenomenon is known as **geometrical isomerism**.



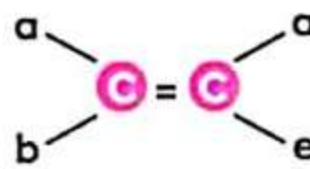
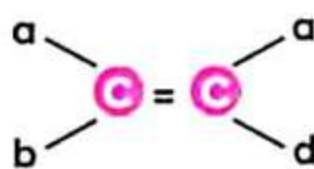
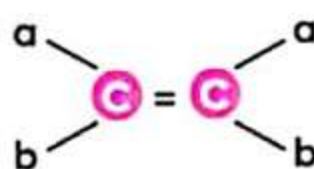
CONDITION OF GEOMETRICAL ISOMERISM

PRESENCE OF A DOUBLE BOND OR A RING STRUCTURE



-N = N - or ring structure

DIFFERENT GROUPS SHOULD BE ATTACHED AT EACH DOUBLY BONDED ATOM



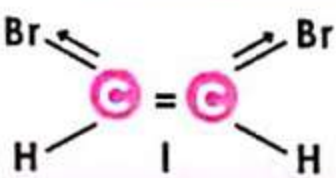
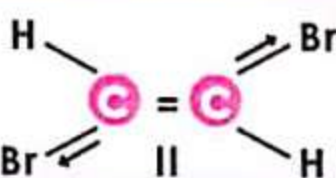
CONFIGURATIONAL NOMENCLATURE IN GEOMETRICAL ISOMERISM

Configuration	Criteria	Remarks
cis / trans	Similarity of groups	If the two similar groups are on same side of restricted bond , the configuration is cis otherwise trans.
E/Z	Seniority of groups	If the two senior groups are on same side of restricted bond, the configuration is Z (Z = zusammen = together) otherwise E (E = entgegen = opposite).

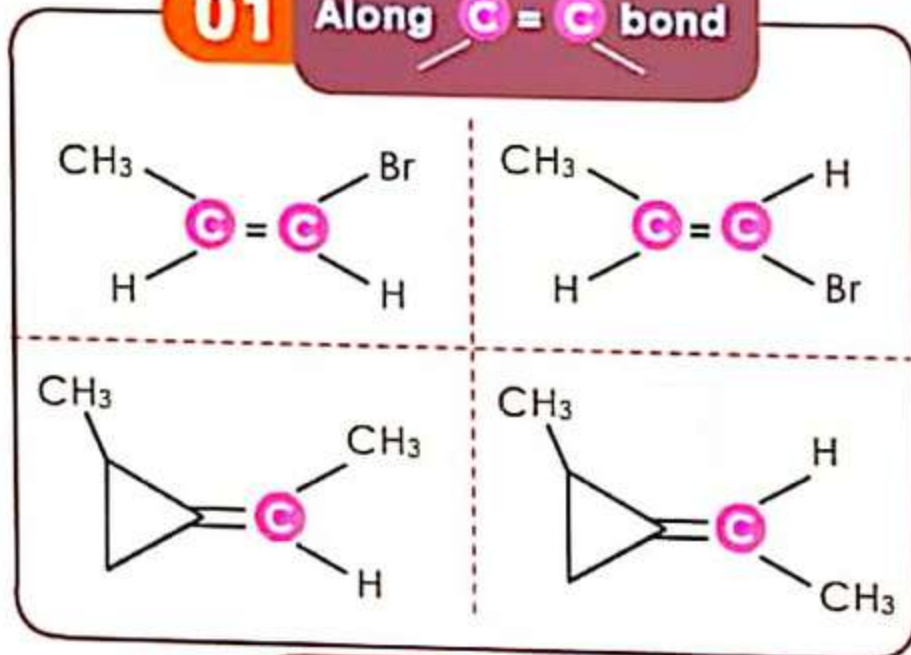
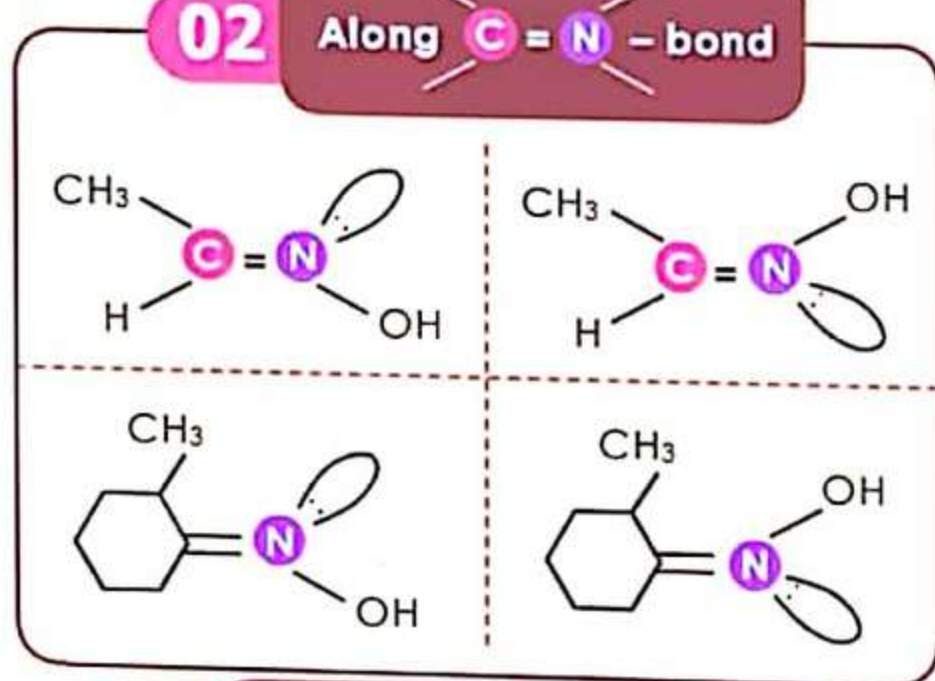
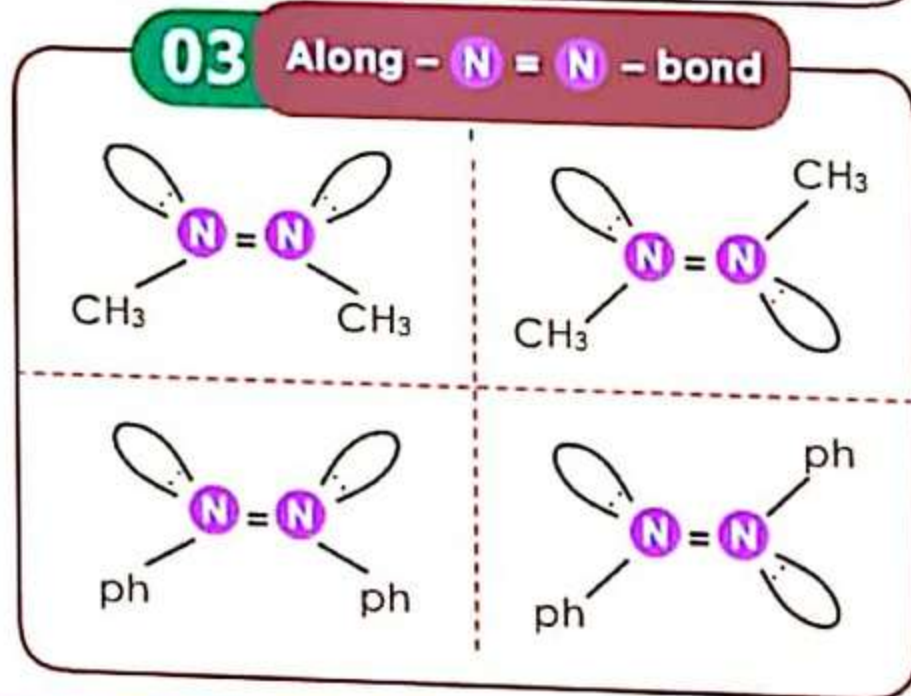
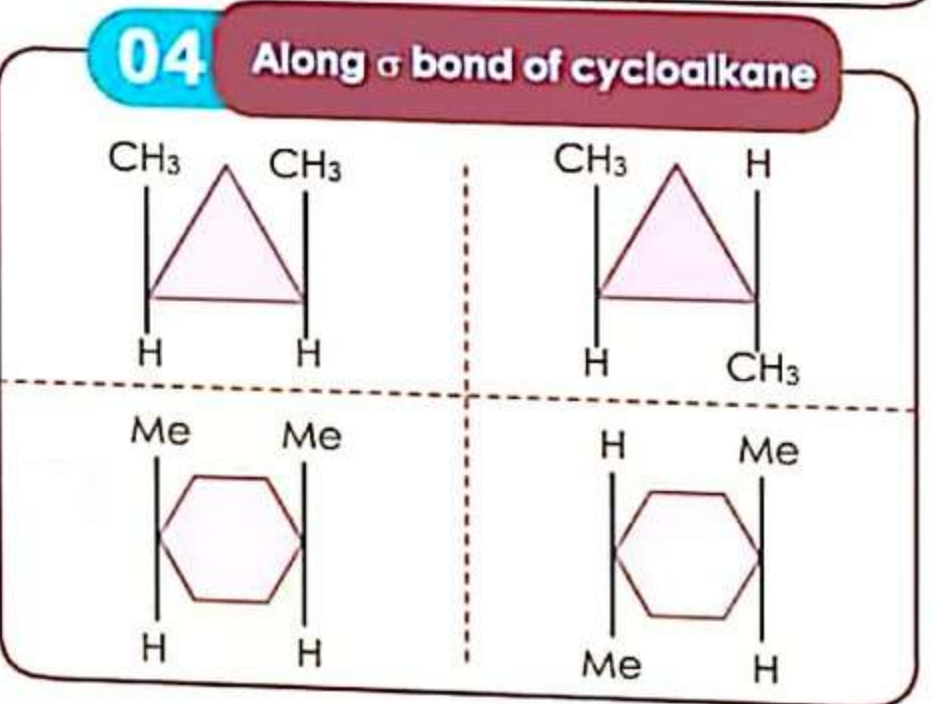
NUMBER OF GEOMETRICAL ISOMERS

Number of geometrical isomers can be found by calculating the number of **stereocentres** in the compound. **Stereocentre** is defined as an atom or bond bearing groups of such nature that an interchange of any two group will produce a **stereoisomer**.

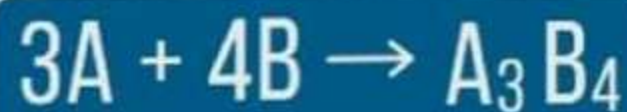
Nature of compound	No. of G.I. (n = no. of stereocentres)	Example	No. of Isomers
Compound with dissimilar ends	2^n	$\text{CH}_3\text{-CH=CH-CH=CH-C}_2\text{H}_5$	4
Compound with similar ends with even stereocentres	$2^{n+1} + 2^{\frac{n}{2}-1}$	$\text{CH}_3\text{-CH=CH-CH=CH-CH}_3$	3
Compound with similar ends with odd stereocentres	$2^{n-1} + 2^{\frac{n-1}{2}}$	$\text{CH}_3\text{-CH=CH-CH=CH-CH=CH-CH}_3$	6

Physical properties	 	Example
Dipole moment	$I > II$	Only cis - isomer has dipole
Boiling point	$I > II$	High boiling point due to larger intermolecular force of attraction
Solubility (in H_2O)	$I > II$	Polar molecules are more soluble in H_2O
Melting point	$I < II$	More symmetric isomers , higher melting points
Stability	$I < II$	More vander waal strain , less stable molecule

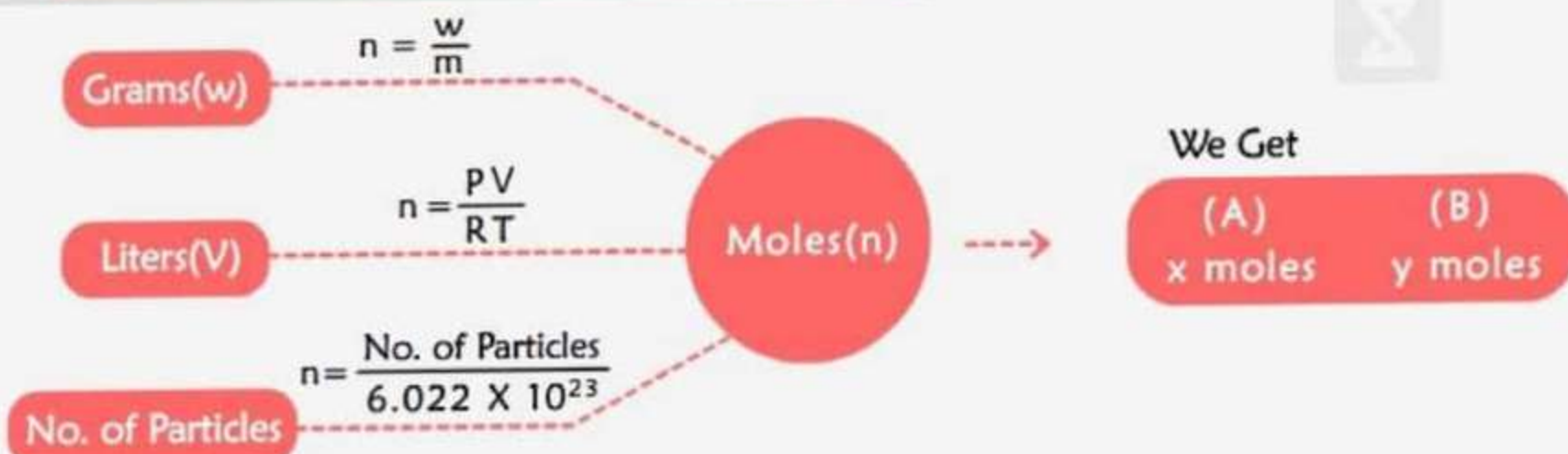
● EXAMPLES OF GEOMETRICAL ISOMERS ●

01 Along **C = C** bond

02 Along **C = N** - bond

03 Along **- N = N -** bond

04 Along σ bond of cycloalkane


HOW TO SOLVE ? STOICHIOMETRY PROBLEM

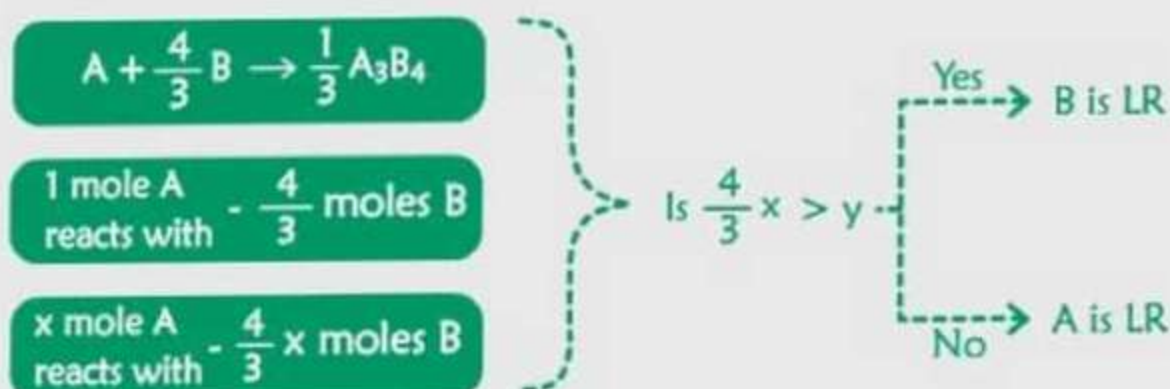


1 Converting reactants into moles

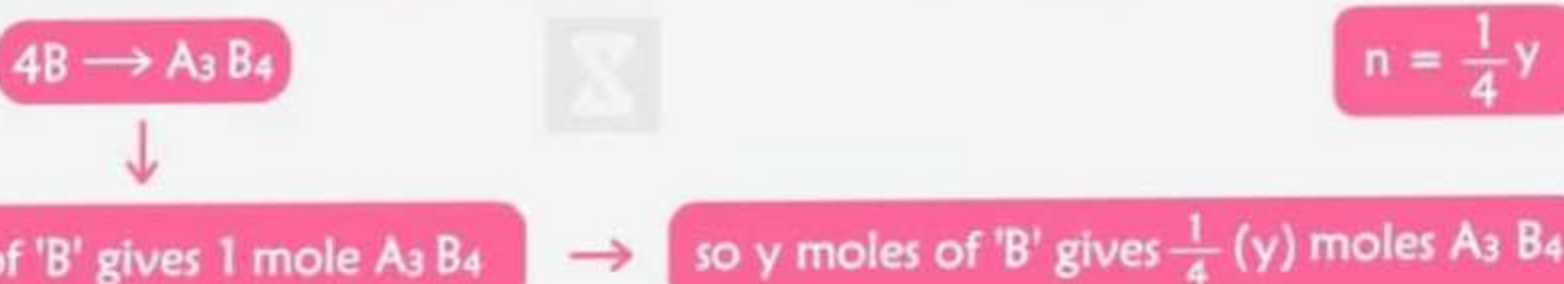


2 Finding Limiting Reagent (L.R)

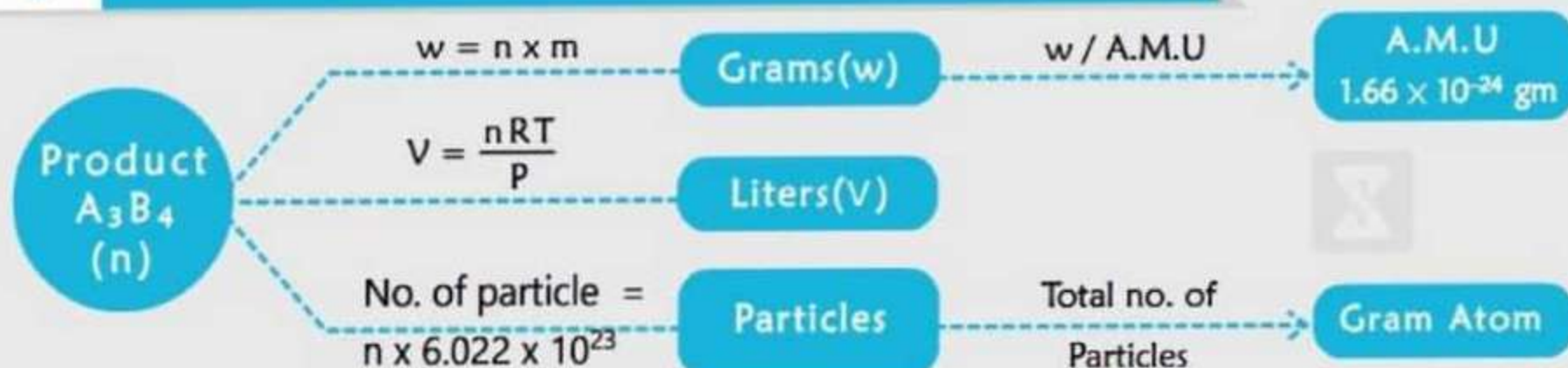
Divide by Coefficient of any Reactant, in above reaction its 'A' i.e. 3



3 Find Moles of Product using L.R (assume L.R = B)



4 Converting Product Moles into required quantity



Concentration Terms



Molarity (M)

Molarity is no. of moles of a solute per liter of solution

$$M = \frac{\text{moles of solute}}{\text{volume of solution (in liter)}}$$



Molality (m)

Molality is the no. of moles of solute per kilogram of solvent.

$$m = \frac{\text{moles of solute}}{\text{weight of solvent (in kg.)}}$$



Normality (N)

Normality is the gram equivalent weight per liter of solution.

$$N = \frac{\text{gram equivalents of solute}}{\text{volume of solution (in liter)}}$$



Formality (f)

Formality is the no. of gram formula masses of the ionic solute dissolved per liter of solution.

$$f = \frac{\text{weight in gram}}{\text{formula weight} \times \text{volume of solution (liter)}}$$



Mole Fraction

Mole Fraction is equal to the moles of one component divided by total moles in the solution or mixture.

$$X_A = \frac{n_A}{n_A + n_B}$$



Parts per million (ppm)

Parts per million is value that represent the part of whole no. in units of 1/1000000

$$\text{ppm} = \frac{\text{mass of solute}}{\text{mass of solvent}} \times 10^6$$

CONCENTRATION OF SOLUTION

(1) % by weight

4.9 gm
 H_2SO_4

(4.9% H_2SO_4 by weight) : $\Rightarrow$

100gm of solution contains 4.9gm of H_2SO_4



Solvent



Solution

(2) % by volume

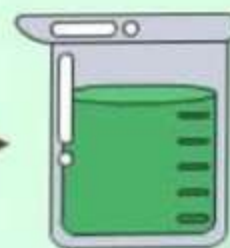
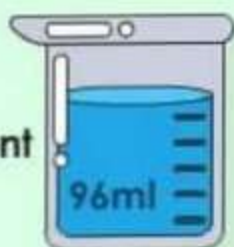
4ml
 H_2SO_4

(4% H_2SO_4 by Volume) : $\Rightarrow$

100ml of solution contains 4ml of H_2SO_4



Solvent



Solution

(3) % weight by volume

(4% H_2SO_4 w/v) : $\Rightarrow$

4ml H_2SO_4 present in 100ml of solution.



4gm H_2SO_4



Solvent



Solution

(4) % volume by weight

(4% H_2SO_4 v/w) : $\Rightarrow$

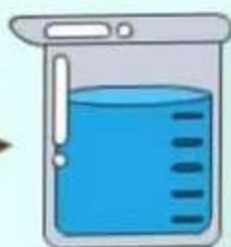
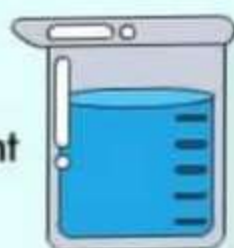
4ml H_2SO_4 present in 100gm solution.



4ml
 H_2SO_4



Solvent



Solution
100gm



EQUIVALENT MASS

Definition

The mass of a substance especially in grams that combines with or is chemically equivalent to **eight grams of oxygen** or **one gram of hydrogen**

$$\text{Equivalent Mass} = \frac{\text{Molar Mass}}{n\text{-Factor}}$$

EQUIVALENT MASS

Element

$$E = \frac{\text{Atomic mass of element}}{\text{Valence of element}}$$

Ion

$$E = \frac{\text{Formula mass of ion}}{\text{Charge on ion}}$$

Salt

$$E = \frac{\text{Formula mass of salt}}{\text{Total positive or negative charge on cationic or anionic part}}$$

Acid

$$E = \frac{M_{\text{acid}}}{\text{Basicity}}$$

Base

$$E = \frac{M_{\text{Base}}}{\text{Acidity}}$$

Acid Salt

$$E = \frac{M}{\text{Replaceable 'H' left in the salt}}$$

Redox Change

$$E = \frac{M_{\text{oxidant or reductant}}}{\text{Number of electrons lost or gained by one molecule of oxidant or reductant}}$$

CHEMICAL REACTIONS

COMBUSTION



COMBUSTION or burning is a high temperature exothermic redox chemical reaction between a fuel and an oxidant usually atmospheric oxygen.



ADDITION

ADDITION reaction is a reaction in which one molecule combines with another to form a large molecule with no other product.



DECOMPOSITION

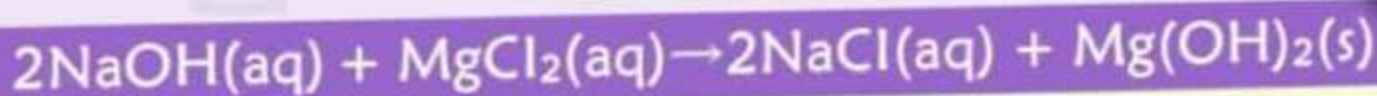


DECOMPOSITION reaction involves the breakdown of chemical compound into its elements or simpler compounds. These reactions often involve an energy source such as heat, light, or electricity that breaks apart the bonds of compounds.



PRECIPITATION

PRECIPITATION reaction is the one in which aqueous compounds react to form an insoluble solid, called a precipitate. Whether or not a reaction will form a precipitate is dictated by solubility rules for ionic compounds.



NEUTRALISATION



Salt + water

NEUTRALISATION reaction is the one in which an acid and a base react to form a salt and water. Neutralisation reactions do not necessarily result in a pH of 7, resultant pH is dependent on the strengths of the acid and base.



CHEMICAL REACTIONS

CONDENSATION



CONDENSATION reaction is the one in which two molecules combine to form a larger molecule with a small molecule. The small molecule lost is usually water, but not always. It can be considered to be the opposite of hydrolysis.



HYDROLYSIS

HYDROLYSIS reaction involves the breaking of chemical bonds by the addition of water to a substance. In some cases, this addition can cause both the substance and the water molecule to split into two parts.



DISPLACEMENT

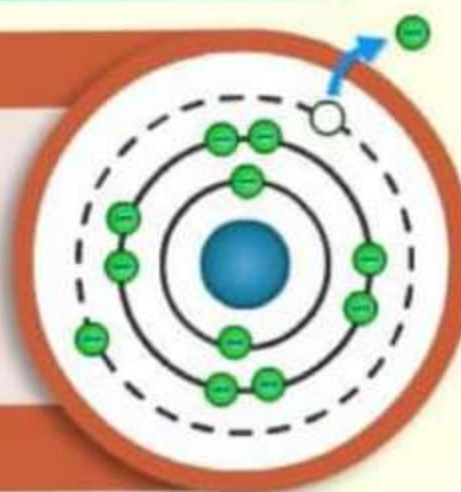
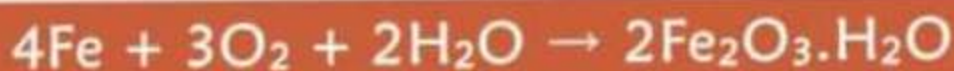


DISPLACEMENT reaction is chemical reaction in which a more reactive element displaces a less reactive element from its compound. Both metals and non-metals take part and more reactive displaces the less reactive.



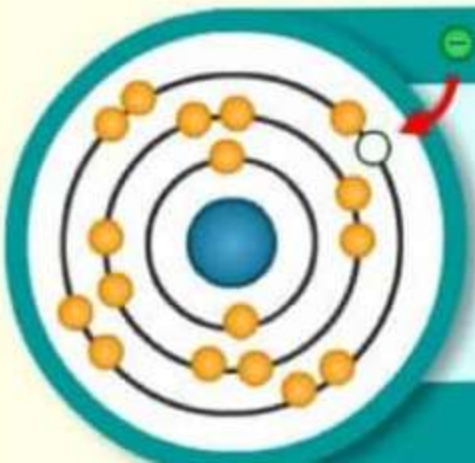
OXIDATION

OXIDATION reactions are sometimes defined as the reactions in which an element forms bonds with oxygen atoms. Generally, oxidation can be defined as a reaction in which atoms of an element **lose electrons**.



REDUCTION

REDUCTION reactions are sometimes defined as reactions in which other molecules lose oxygen atoms. Generally, reduction can be defined as a reaction in which atoms of an element **gain electrons**.



BORON FAMILY



ELECTRONIC CONFIGURATION



ns^2np^1
Valence
electrons = 3

PHYSICAL STATE

Boron is a non metal.
Aluminium, Indium,
Gallium are metalloids.



Gallium has the **second lowest melting point** after mercury. It remains in liquid phase even at high temperatures.

AMPHOTERIC

$Al(OH)_3$ is amphoteric

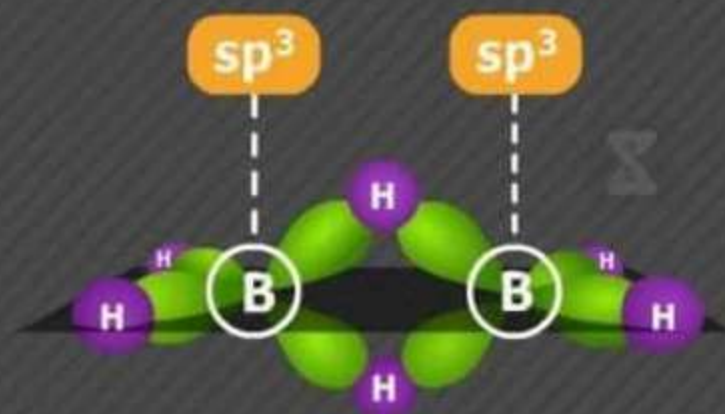


Strength between Al - O
and O - H is same.

Bond strength Al - O = Bond strength O - H



DIBORONE (B_2H_6)



BH_3 is an electron deficient molecule. It has six electrons and a vacant orbital. So, it is used as an electrophile in organic reactions.

Aluminium is used in tins, aeroplanes and even in Iphone

Borax ($Na_2B_4O_7$) is used
for cleaning and washing
purposes.



Aluminium was classified as precious metal during 19th century. **Napoleon III** gave aluminium cutlery to his most valuable guests, all other guests had to eat with gold cutlery.



Indium is used in display
screens of television
and mobiles.



Thallium is used as rat
poison.

CARBON



Part-I

6	14	32	50	82
C	Si	Ge	Sn	Pb
12.0107	28.09	72.81	118.71	207.2

FAMILY

DO
you?
know?

- Carbon is a non-metal
- Silicon and germanium are metalloid
- Lead and tin are metal

Carbon has three main solid state allotropes: Graphite, Diamond and Fullerenes (the most commonly known of which, buckminster fullerene, is also known as a "bucky-ball").

Germanium



Germanium is a rare element used in the manufacturing of **semi-conductor** devices. The physical and chemical properties of germanium are very similar to those of silicon. Germanium is **grey-white** in color and forms crystal structures.

Silicon



Tablet

Silicon is the second most common element in the earth's crust (after oxygen) and it is the backbone of the mineral world. Silicon is used extensively as a semiconductor in **solid-state** devices in the computer and microelectronics industries.

Tin



Tin is **malleable**, **ductile**, and **crystalline**. It is a superconductor at low temperatures. Tin reacts with bases, acid salts and strong acids. Tin chlorides are good reducing agents and often used to reduce iron ores. Tin fluoride is often the anticavity **"fluoride"** additive in toothpastes.

Lead



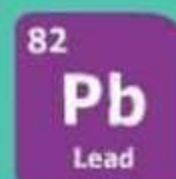
It is a soft, malleable metal with a low **melting point**. Lead is toxic to humans, especially children. Even low levels of exposure can cause nervous system damage and can prevent proper production of **haemoglobin**. Its oxides have many industrial uses as oxidizing agents, such as cathodes in lead-acid storage cells.

Point to Remember

Carbon has the highest melting/sublimation point of the elements. The melting point of diamond is **3550°C**, with the sublimation point of carbon is around **3800°C**

CARBON FAMILY

Part-II



ELECTRONIC CONFIGURATION



ns^2np^2

Valence Electrons = 4

REACTIVITY

The carbon family elements tend to be fairly unreactive. The elements tend to form covalent compounds, though tin and lead also form ionic compounds.

CARBIDES

Carbon combines with other elements and forms carbides Al_4C_3 , Be_2C , SiC and B_4C are considered as the hardest compounds.



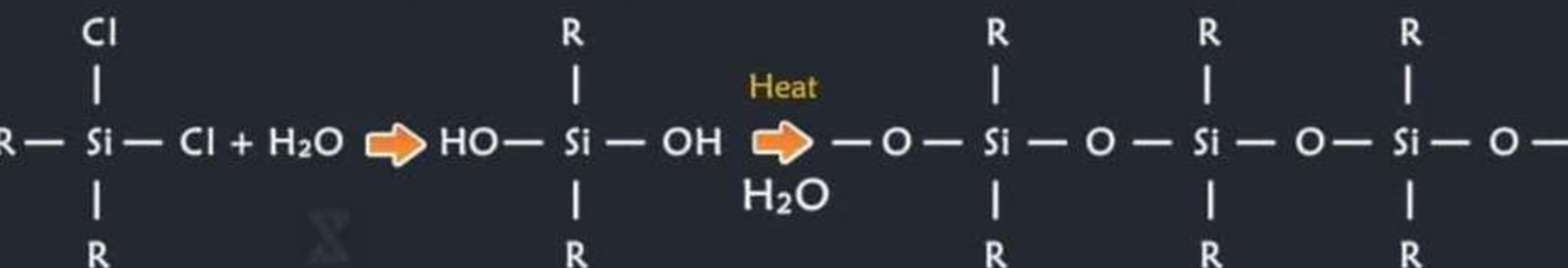
DON'T STORE JUICES IN TIN CONTAINERS

Generally juices have metallic taste if we store them in tin containers. Juices are mostly acidic and they react with tin. Tin reduces the acid by absorbing oxygen.



SILICONES

Silicones are organo silicon polymers and are formed by hydrolysis of R_2SiCl_2



Linear Silicones

CARBON-GODLY ELEMENT

Every human being and everything around us consists of carbon.



TIN-CANS

Initially we used Tin in making cans, later it was replaced by Aluminium.



SILICON

Silicon is used in semiconductor devices, which are used in modern day computers.

LEAD PENCILS

Lead Pencils which we use do not contain Lead. It contains Graphite (an allotrope of carbon).



NON STICK PAN

Nonstick pan has a Teflon layer on its surface. Teflon ($-CF_2-CF_2-$)



NITROGEN FAMILY



ELECTRONIC CONFIGURATION



Valence Electron = 5

BORAZONE

Borazone(cBN) has graphite like structure and it is harder than diamond. So, it is used for cutting the diamond.



Half life of Bismuth is very high. If 100 gm of Bismuth had been present at the beginning of universe, 14 billion years ago. About 99.9999 gm would be left today.

INERT NATURE OF N_2 ($N \equiv N$)

Triple bond exists between Nitrogen, which is very hard to break. It reacts only when suitable conditions are met for reaction.

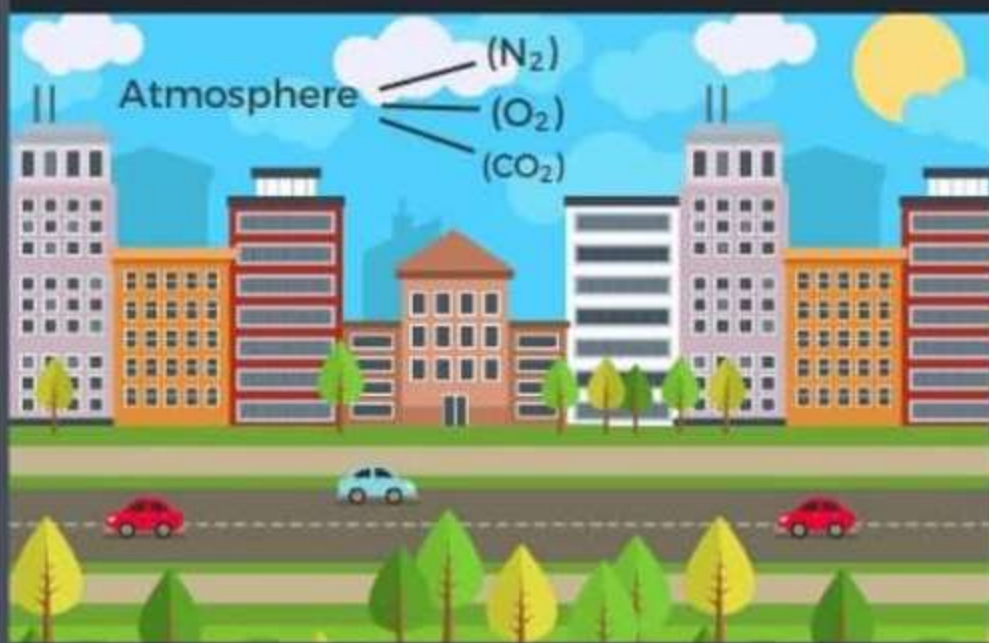


TEST OF PROTEINS

Nitric acid attacks the protein and forms Xanthoprotein which is yellow in colour.

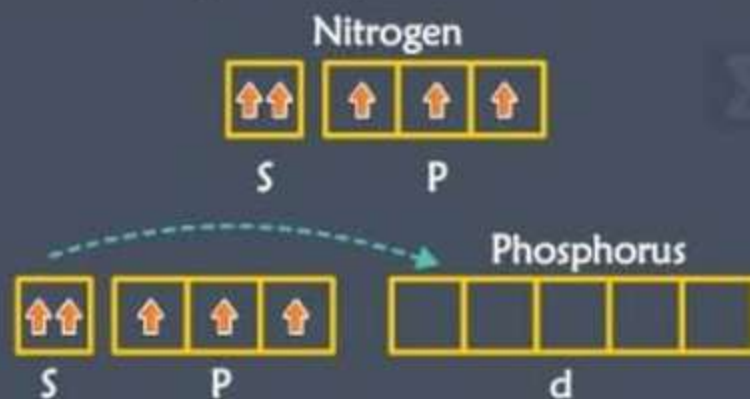


EARTH ATMOSPHERE CONTAINS 71% NITROGEN



PENTAHALIDES

Nitrogen does not show pentahalides because of the absence of d-orbitals. Phosphorous shows pentahalides by exciting s electrons into d-orbitals.



TNT

Tri Nitro Toulene is used in explosives.

USES

Red phosphorus is used in matchbox.

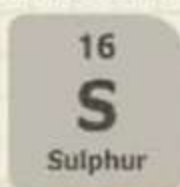


Arsenic is a poisonous gas therefore, it is used as pesticides for trees.

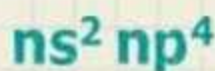
P_2O_5 is used as a drying agent, but it is corrosive to use outside the labs.



CHALCOGENS



ELECTRONIC CONFIGURATION



Valence Electrons = 6

OZONE LAYER (O₃)



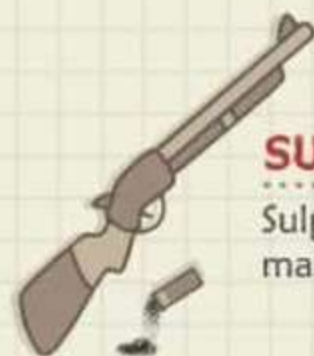
Ozone layer is filled with ozone molecules.

Ozone is the **BEST OXIDISING AGENT**

REACTION WITH HYDROGEN

Hydrogen with Oxygen forms water (H₂O)

Hydrogen with sulphur forms H₂S which gives foul smell.



SULPHUR

Sulphur is used in making Gun powder.



TELLURIUM

Tellurium is used in CD's and Blue-ray discs.



SELENIUM

Selenium is used in Glass making.

TAILING OF MERCURY

Mercury loses its meniscus and sticks to the glass due to formation of mercurous oxide.



Ozone

Mercurous oxide



POLONIUM

Radioactive and deadly. It is rumoured that Yasar Arafat died due to poisoning of Polonium.

CAREFUL WITH H₂SO₄

H₂SO₄ is the strongest and most corrosive acid. It can cause serious burns on the skin.



MELTING POINT

Oxygen -219°C Selenium 221°C Tellurium 450°C



Sulphur 120°C

Polonium 250°C



OXYGEN

Liquid oxygen is used as Rocket fuel.

HALOGENS

1 Physical State



3 REACTIVITY

Reactivity decreases down the group as it is harder to add electrons.

F is the most reactive



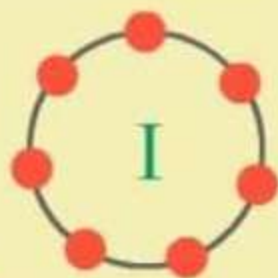
5 INTER HALOGEN

Halogens combine with each other to form series of compounds like **AX AX₃ AX₅ AX₇**

Large molecules can accommodate more halogens like in **IF₇**

Small fluorine can't accommodate

many halogens like in **FCl**



7 VARIABLE OXIDATION STATES

Halogens exhibit variable oxidation states, starting from **-1 to 7**.

Fluorine can't show positive oxidation state.

9 USES

Fluorine is used in Tooth Paste



Chlorine is used as a Bleaching Agent



Iodine is used in Disinfectants



Bromine is used for developing Photographic films



2 ELECTRONIC CONFIGURATION



They have **seven** valence electrons

GENERAL CONFIGURATION
 $ns^2 np^5$

4 ACIDITY



Acidity decreases down the group

HF is the strongest acid which can dissolve glass also.

6 CHLORINE IN WAR FARE

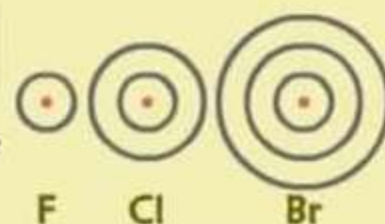


Germany used chlorine in world war- I against France.

Chlorine gas **destroys** respiratory organs.

8 ELECTRONEGATIVITY

Atomic size increases down the group, therefore electronegativity decreases down the group.



NOBLE GASES



Also known as **inert gases** as their atoms don't combine with other atoms in chemical reaction

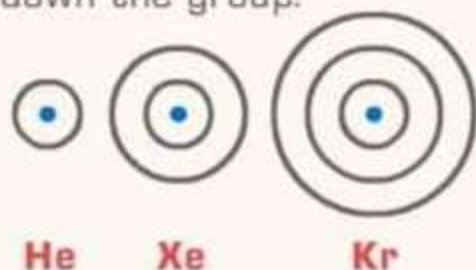


First Noble Gas Compound - Xenon

Bartlett surprised every one by synthesizing **XePtF₆**

Atomic Size

Size increases as you go down the group.



Helium

Helium is used in **ballons**. Initially hydrogen was used which is much lighter than Helium but due to its **flammability**, we shifted to **Helium**.



Electronic Configuration

These elements have **8 electrons** in valence orbital which is considered stable.

Exception : Helium has two valence **electrons**
General electronic configuration



Krypton

Light bulb containing **krypton gas** can produce a bright white light useful for photography and runaway **lights**.

Neon Light

Neon Lights which we use in our streets are produced by **ionising inert** gas by applying voltage.



Argon

Do you know?

There are 22 known isotopes of argon ranging from **argon-31** to **argon-51** and **argon-53**.



Radon

Radon is radioactive and can enter homes through basements after **radioactive** decay of rocks below the **earth**.

Reactivity of Xenon

- 1 Xenon acts as fluorinating agent



- 2 Xenon acts as fluoride acceptor

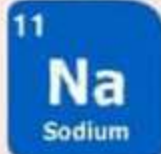


- 3 Xenon acts as fluorine donar



GROUP - 1 THE ALKALI METALS

Group-1 elements are Shiny, Soft, and highly reactive metals, none of them occur as a natural free element



MELTING POINTS



Li (180.5°C)



Na (97.7°C)



K (63.4°C)



Rb (39.5°C)



Cs (25.4°C)



Fr (27.0°C)



All of the **Group-1 Metals** have **one** Valence Electron

The reactivity of the **GROUP-1 METALS** increases down the group as the outer electron gets further from the nucleus & becomes easier to remove.

The alkali metals react with water to form

METAL HYDROXIDES

LiOH

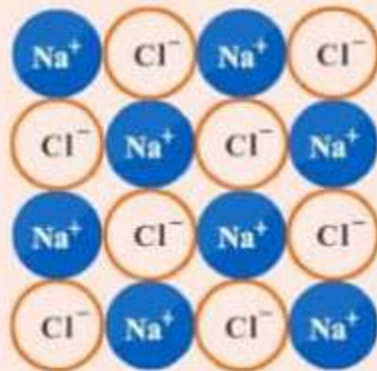
NaOH

KOH

RbOH

CsOH

Alkali metals react with **oxygen** to form **METAL OXIDES**



Alkali metals react with **halogens** to form **IONIC SALTS**

PROPERTIES OF ALKALI METALS



Silver-like lustre



Low melting point



High Ductility



High malleability



Excellent conductor of heat and electricity

Properties of Alkali Metals

FLAME TEST COLORS



VERY SOFT



- Alkali metals can be easily cut with a knife
- Among all alkali metals lithium is hardest.

ELECTRONIC CONFIGURATION


 ns^1

- They have one valence shell electron.
- General valence electronic configuration

USES

LITHIUM

- Anti depressants
- Batteries



SODIUM

- Street lamps
- Salt



POTASSIUM

- Fertilizers



CAESIUM

- Atomic Clocks



ATOMIC SIZE



Li

Na

K

Rb

Size increases down the group due to added extra orbit.

ELECTROPOSITIVE

It is the ability to remove an electron

 Cs


- Electropositivity increases down the group.
- Caesium** has the highest electropositive character.

STORED IN KEROSENE



Alkali metals react with air easily to form oxide layer therefore they are stored in kerosene.

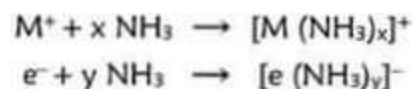
REACTS WITH WATER

 H_2O

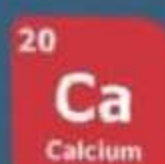

- They react violently with water and form hydroxides.
- Don't even dare to go near when **caesium** reacts with water.

REACTS WITH AMMONIA

On dissolving NH_3 forms Ammoniated cation and electron. Solution turns blue



ALKALINE EARTH METALS



ELECTRONIC CONFIGURATION



Valence
Electrons
= 2

DO YOU KNOW ?

Kidney stones generally consist of calcium oxalate, $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ which dissolves in dilute strong acids but remains insoluble in bases.



ATOMIC SIZE

Size increases down the group due to added extra orbit.

ELECTROPOSITIVITY

Strong electropositive elements due to large size, electropositivity increases down the group.

USES

● BERYLLIUM

Corrosion resistant alloys

● MAGNESIUM

Present in chlorophyll, helps in photosynthesis

● CALCIUM

Hydrated CaCl_2 used for melting ice on roads

● STRONTIUM

Glass for colour television cathode ray tubes

● BARIUM

Nuclear Medicine

FLAME COLOUR TEST

Group II Element	Flame Colour
Beryllium	Colourless
Magnesium	Colourless
Calcium	Brick red
Strontium	Crimson red
Barium	Apple green



They are commonly called alkaline earth metals because their oxides are alkaline in nature and are found in earth's crust.

REACTION WITH WATER

Be does not react even with boiling water and **Ba** react vigorously even with cold water. Thus increasing order of reactivity with water is



REACTION WITH NITROGEN

These metals react with nitrogen to form nitrides of the types M_3N_2 which are hydrolysed with water to evolve NH_3 .



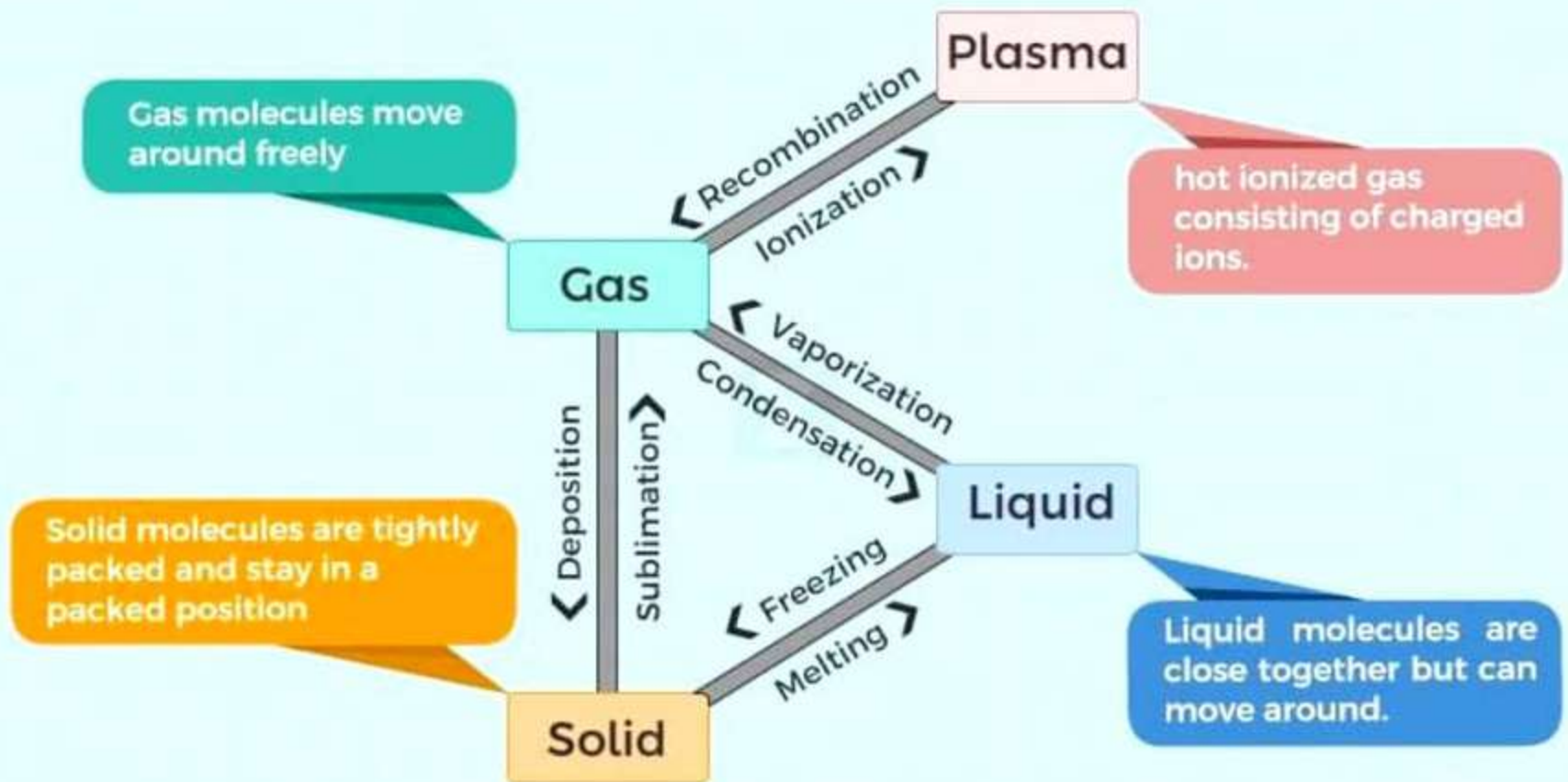
ALKALI METALS

DIFFERENCE
? BETWEEN

ALKALINE EARTH METALS

PROPERTIES	ALKALI METALS	ALKALINE EARTH METALS
Physical properties	Soft, Low melting point, Paramagnetic. 	Comparatively harder. High melting point, Diamagnetic
Valency	Monovalent	Bivalent
Electropositive nature	More electropositive	Less electropositive 
Hydroxides 	Strong base, highly soluble and stable towards heat.	Weak base, less soluble and decomposes on heating.
Bicarbonates	These are known in solid state.	These are not known in free state. Exist only in solution
Carbonates	Soluble in water. Do not decomposes on heating (LiCO_3 is an exception)	Insoluble in water. Decomposes on heating.
Action of carbon	Do not directly combine with carbon 	Directly combine with carbon to form carbides 
Solubility of salts	Sulphates, phosphates, fluorides, chromates, oxides etc are soluble in water.	Sulphates, phosphates, fluorides, chromates, oxalates etc are insoluble in water
Reducing power	Stronger as ionization potential values are low and oxidation potential values are high	Weaker as ionization potential values are high and oxidation potential values are low.
Electronic configuration 	One electron is present in the valence shell. The configuration is ns^1 (monovalent)	Two electrons are present in the valence shell. The configuration is ns^2 (bivalent)

STATES OF MATTER



PHYSICAL STATE

SOLID



The molecules that make up a solid are arranged in regular, repeating pattern. They are held firmly in place but can vibrate within a limited area.

LIQUID



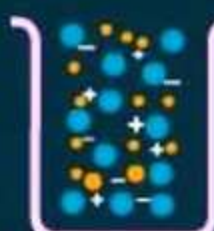
The molecules that make up a liquid flow easily around one another. They are kept from flying apart by attractive forces between them. Liquids assume the shape of their containers.

GAS



The molecules that make up a gas fly in all directions at great speeds. They are so far apart that the attractive forces between them are insignificant.

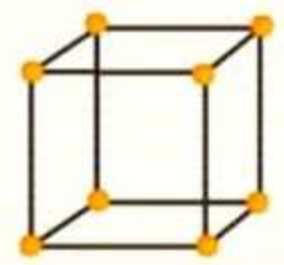
PLASMA



At very high temperatures of stars, atoms lose their electrons. The mixture of electrons and nuclei that results is the plasma state of matter.



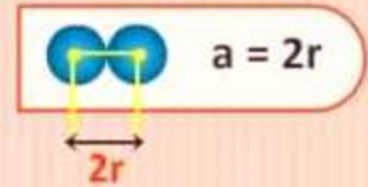
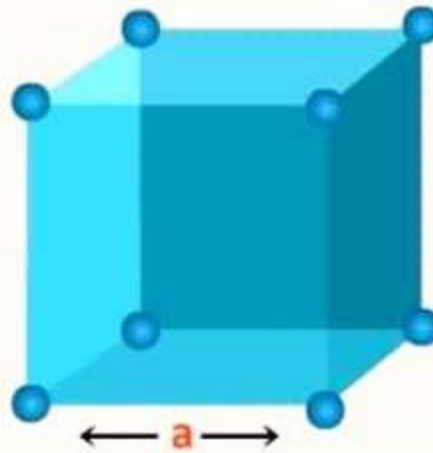
Bravais Lattices Of Crystals



Primitive Cube

8 - Corner atoms = $8 \times \frac{1}{8} = 1$

Total Number of atoms = 1



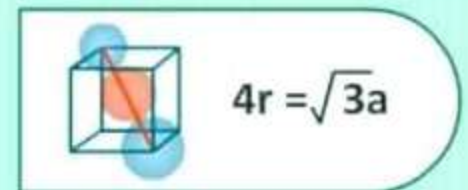
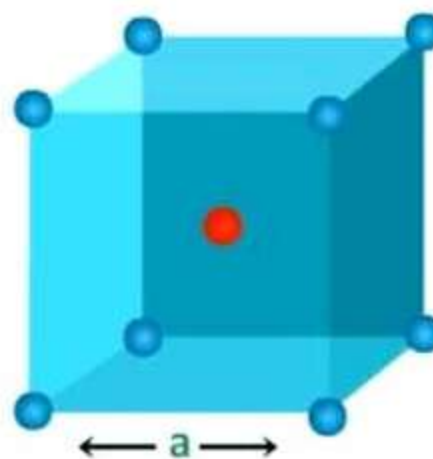
Packing Fraction = $\frac{\text{Volume of atoms}}{\text{Volume of cube}}$
= 0.52

Body Centered Cube

8 - Corner atoms = $8 \times \frac{1}{8} = 1$

Center atoms = 1

Total Number of atoms = 2



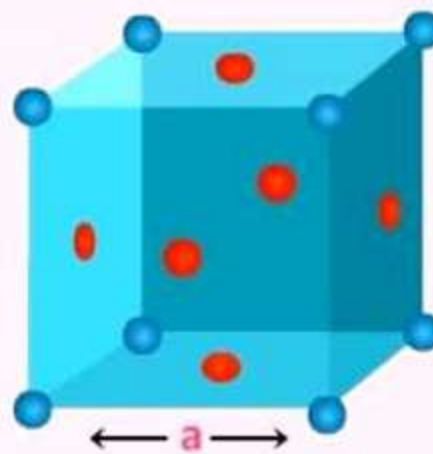
Packing Fraction = $\frac{\text{Volume of atoms}}{\text{Volume of cube}}$
= 0.68

Face Centered Cube

8 - Corner atoms = $8 \times \frac{1}{8} = 1$

6 - Face atoms = $6 \times \frac{1}{2} = 3$

Total Number of atoms = 4



Packing Fraction = $\frac{\text{Volume of atoms}}{\text{Volume of cube}}$
= 0.74

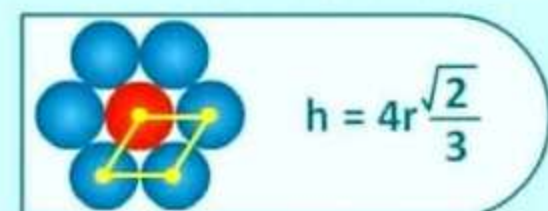
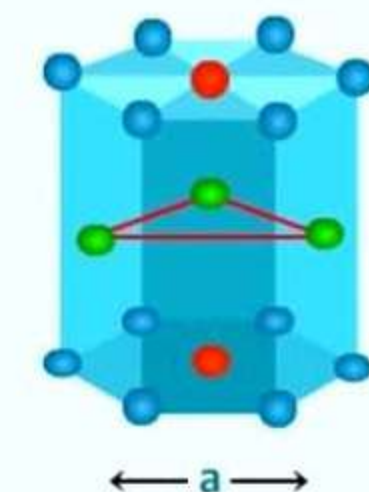
Hexagonal Close Packed

12 - Side corner = $12 \times \frac{1}{6} = 2$

2 - Face side atoms = $2 \times \frac{1}{2} = 1$

3 - atoms inside $3 \times 1 = 3$

Total Number of atoms = 6



Packing Fraction = $\frac{\text{Volume of atoms}}{\text{Volume of hexagonal}}$
= 0.74

RAOULT'S LAW

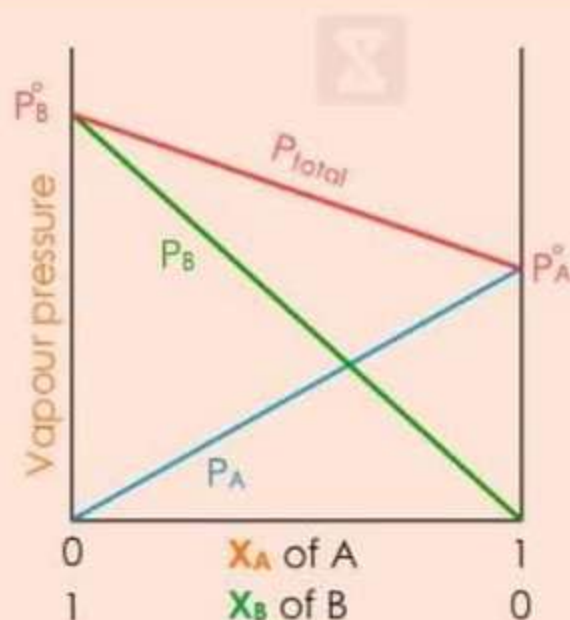
The vapour pressure of a solution of a non-volatile solute is equal to the vapour pressure of the pure solvent at that temperature multiplied by its mole fraction.

$$P_{\text{SOLN}} = X_{\text{SOLVENT}} P^{\circ}_{\text{SOLVENT}}$$

P_{SOLN} = VAPOUR PRESSURE OF THE SOLUTION

X_{SOLVENT} = MOLE FRACTION OF THE SOLVENT

$P^{\circ}_{\text{SOLVENT}}$ = VAPOUR PRESSURE OF THE PURE SOLVENT

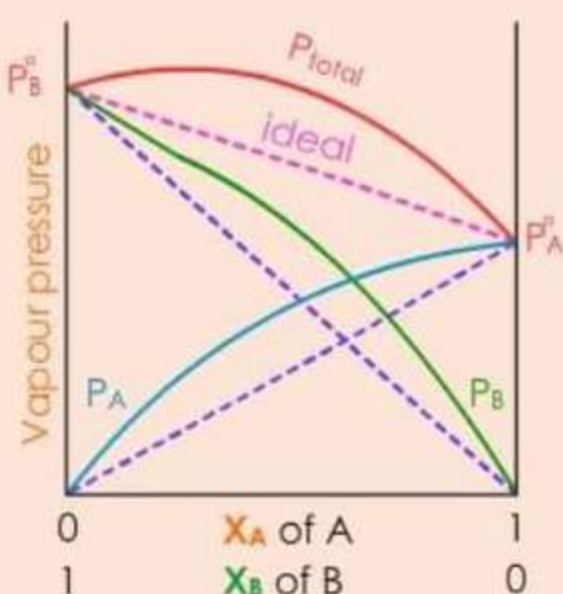


$$P_A = X_A P_A^{\circ}$$

$$P_B = X_B P_B^{\circ}$$

$$P_{\text{Total}} = P_A + P_B$$

Raoult's law

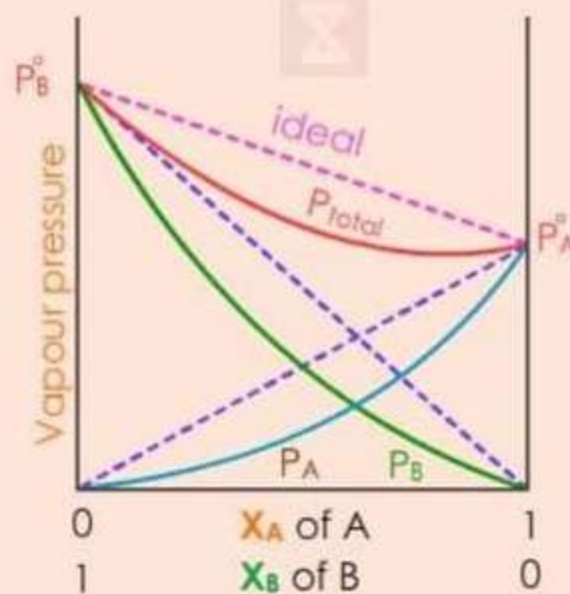


$$P_A > X_A P_A^{\circ}$$

$$P_B > X_B P_B^{\circ}$$

$$P_{\text{Total}} > P_{\text{Ideal}}$$

Positive deviation from Raoult's law



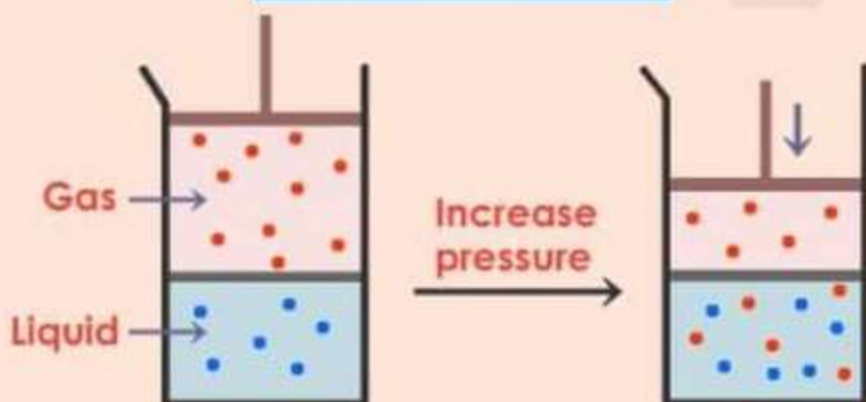
$$P_A < X_A P_A^{\circ}$$

$$P_B < X_B P_B^{\circ}$$

$$P_{\text{Total}} < P_{\text{Ideal}}$$

Negative deviation from Raoult's law

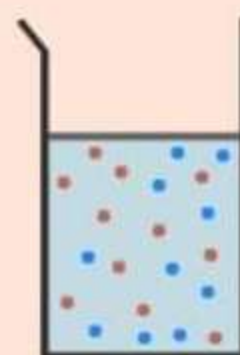
HENRY'S LAW



$$P = K_H \cdot C$$

At a constant temperature, the amount of a given gas that dissolves in a given type and volume of liquid is directly proportional to the partial pressure of that gas in equilibrium with that liquid.

VAPOUR PRESSURE



$$P_{\text{Total}} = P_A + P_B$$

$$P_A = X_A P_A^{\circ}$$

$$P_B = X_B P_B^{\circ}$$

Each substance has a lower vapour pressure from that of a pure substance !
Even though it looks like the mix may be "50/50", "A" particles have a higher vapour pressure (evaporate more easily) than the "B" substance.



COLLIGATIVE PROPERTY

A property that depends on the concentration of solute particles but not on the identity of the solute

Colligative properties include:

- Vapour pressure lowering
- Boiling point elevation
- Freezing point depression



VAPOUR PRESSURE

- Vapour pressure is the pressure of the vapour over a liquid at equilibrium.
- A non-volatile solute will lower the vapour pressure of a solvent



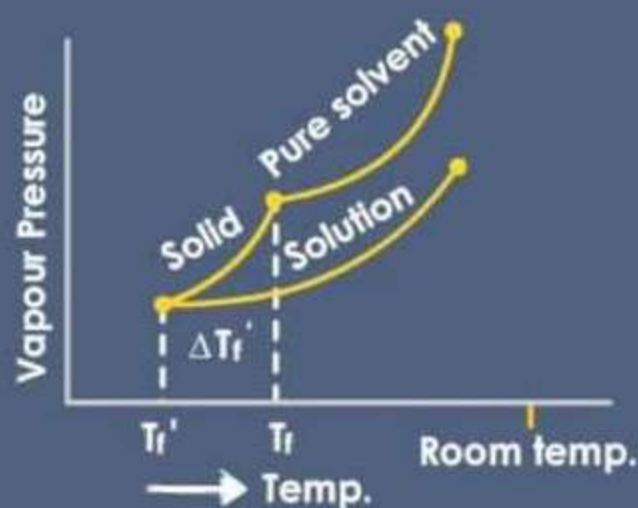
FREEZING POINT DEPRESSION

Freezing Point Depression is the decrease in the freezing point of a solvent due to the presence of solute particles.



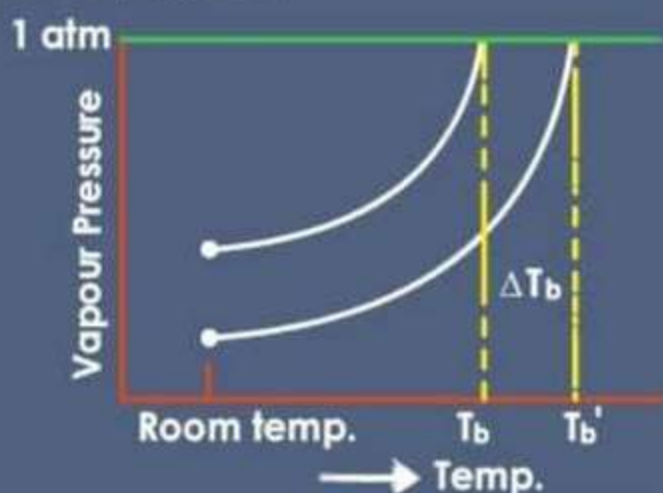
Solute

$$\Delta T_f = K_f \cdot m$$



BOILING POINT ELEVATION

Boiling Point elevation is the increase in the boiling point of a solvent due to the presence of solute particles.



$$\Delta T_b = K_b \cdot m$$



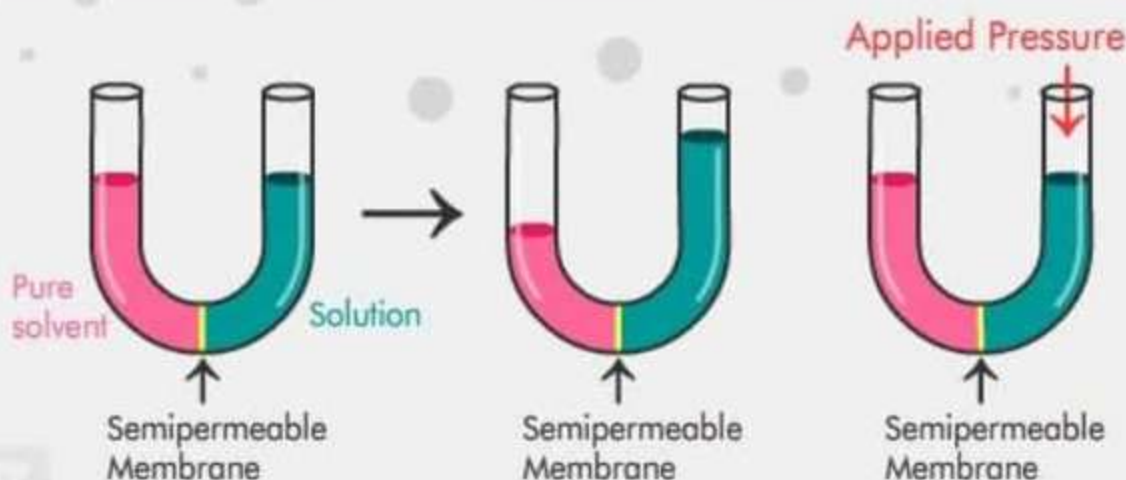
OSMOTIC PRESSURE



VAN'T HOFF FACTOR

WHAT IS? OSMOTIC PRESSURE

The minimum pressure that stops the osmosis is equal to the osmotic pressure of the solution.



$$\pi = iMRT$$

π = Osmotic pressure

i = Van't Hoff factor

M = Molar concentration of solution (mol/L)

R = Ideal gas constant (0.08206 L atm Mol⁻¹ K⁻¹)

T = Temperature in Kelvin (K)

VAN'T HOFF FACTOR

Colligative properties are directly proportional to the number of mole of solute, dissociation or association may cause some abnormal increase or decrease in the measured values of colligative properties. Van't Hoff factor is a factor that takes into account this abnormal behaviour and can be defined as,

$$i = \frac{\text{Observed magnitude of any colligative property}}{\text{Normal magnitude of the same colligative property}}$$

Since, colligative properties are inversely related to the molar mass of the solute, hence, one can write,

$$i = \frac{\text{Normal molar mass}}{\text{Observed molar mass (obtained from a colligative property)}}$$

Thus, the value of i depends upon the state of solute in the solution. Following cases are possible:

- When, $i = 1$ then the solute remains **unaffected** (i.e., normal) in solution.
- When, $i > 1$ then the solute undergo **dissociation** in solution.
- When, $i < 1$ then the solute undergo **association** in solution.

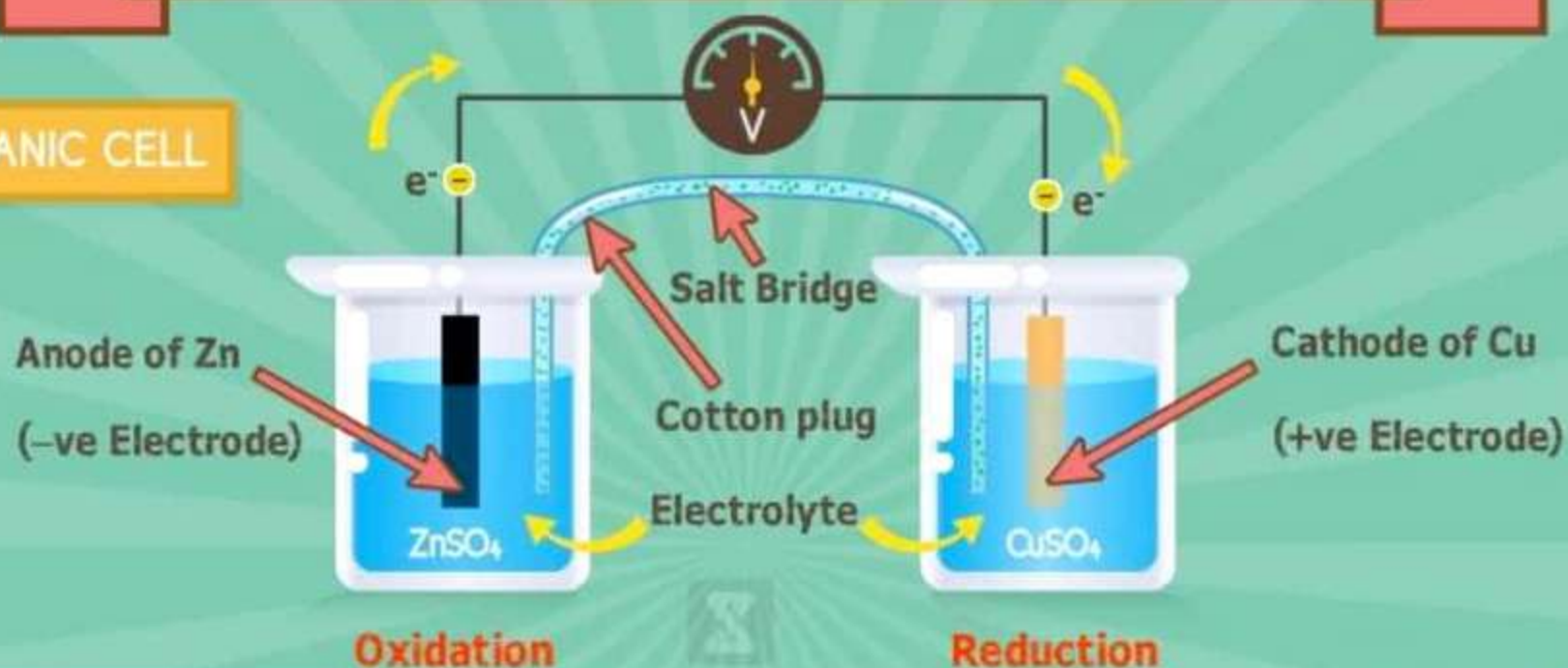


ELECTROCHEMISTRY

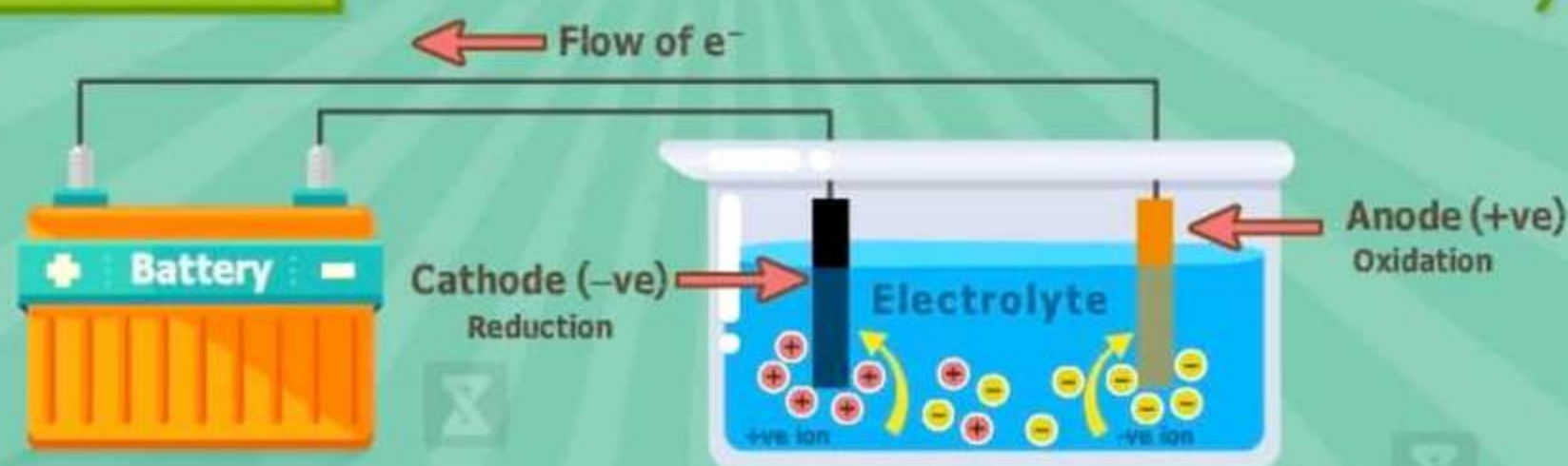


COMPARING GALVANIC AND ELECTROLYTIC CELL

GALVANIC CELL



ELECTROLYTIC CELL

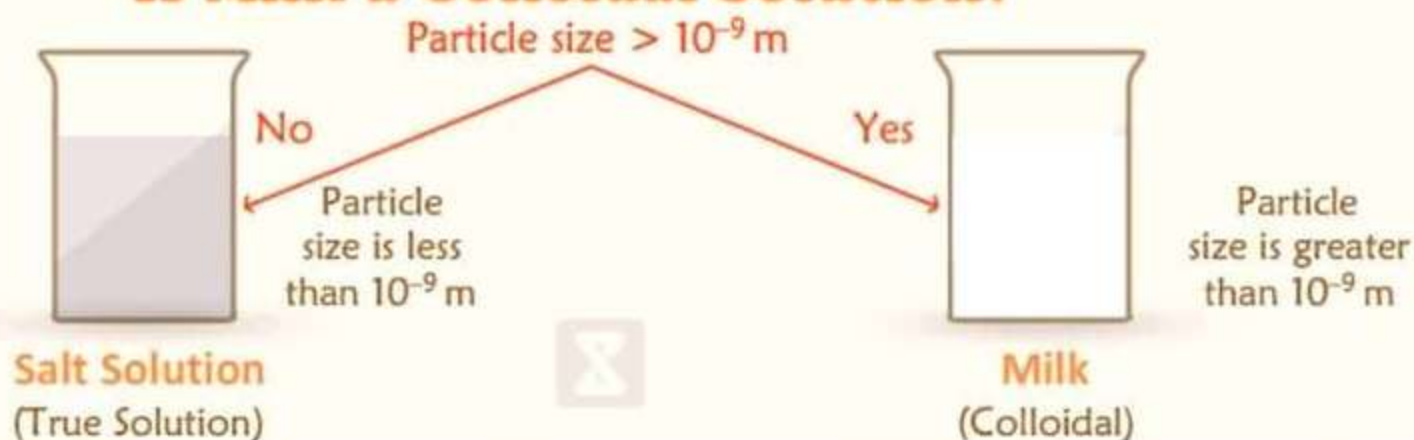


PROPERTY	ELECTROLYTIC CELL	GALVANIC CELL
Basic structure	Two half cells connected by salt bridge.	Both electrodes placed in same container
Energy conversion	Electrical energy $\longrightarrow$ Chemical energy	Chemical energy $\longrightarrow$ Electrical energy
Transfer of electrons on the electrode	Positive(anode) $\longrightarrow$ Negative(Cathode)	Negative(anode) $\longrightarrow$ Positive(Cathode)
Anode	Positively charged electrode	Negative terminal cell
Cathode	Negatively charged electrode	Positive terminal cell



WHAT ARE COLLOIDS ?

Is Milk a Colloidal Solution?

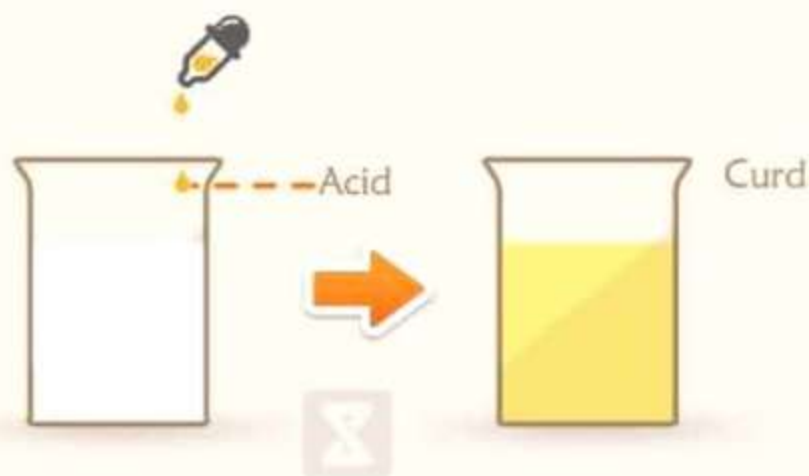


Colloids are everywhere around us

Colloid = dispersed phase + dispersed medium

How Curd is Formed ?

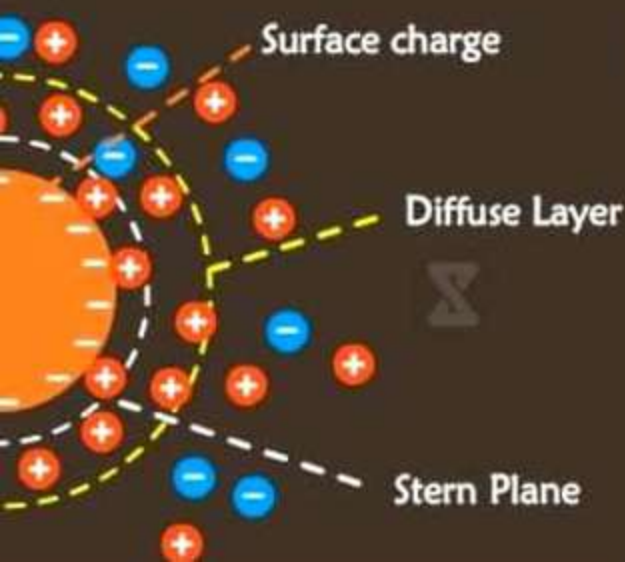
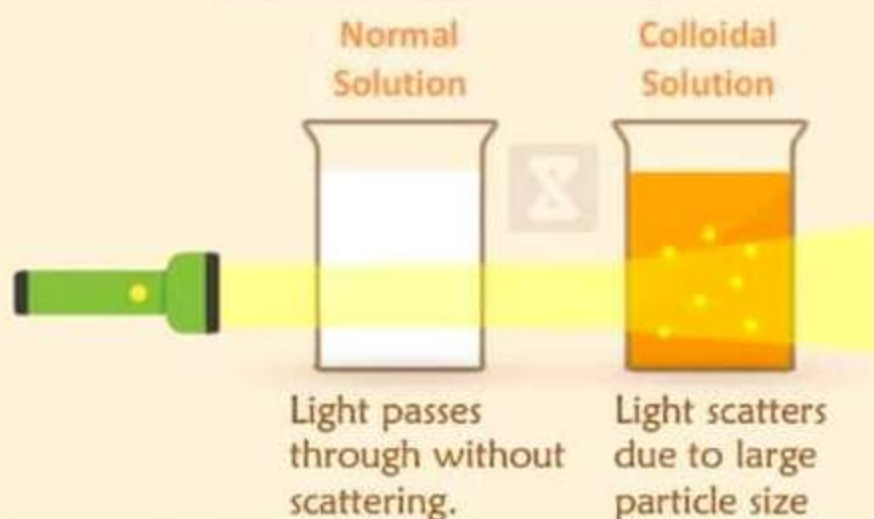
Coagulation : It is a process of forming precipitate by adding electrolyte to a colloidal solution.



Adding electrolytes to colloidal solution causes opposite charge to combine and form precipitate. Here acid is the electrolyte added and curd is the precipitate formed.

Tyndall Effect

Tyndall effect is observed in morning, where sunlight scatters due to colloidal solution i.e. Fog.



Stability of Colloids

Colloids are generally unstable. They are stabilised by forming a electric double layer.

Surface Charge :- Formed by absorbing charged ions on particle.

Diffuse Layer :- Diffuse layer consists of both the ions with higher concentration of counter ions.

Stern Layer :- Layer formed due to electrostatic attraction of opposite ions to the particle surface

12 AMAZING COLLOIDS USED IN OUR DAILY LIFE

Colloid is a homogeneous non-crystalline substance consisting of large molecules or ultra-microscopic particles of one substance dispersed through a second substance



Solid



Liquid



Gas



Paint



Dispersed
Phase



Dispersed
Medium



Ice Cream



Dispersed
Phase



Dispersed
Medium



Tooth Paste



Dispersed
Phase



Dispersed
Medium



Smoke



Dispersed
Phase



Dispersed
Medium



Jam



Dispersed
Phase



Dispersed
Medium



Sponge



Dispersed
Phase



Dispersed
Medium



Milk



Dispersed
Phase



Dispersed
Medium



Clouds



Dispersed
Phase



Dispersed
Medium



Butter



Dispersed
Phase



Dispersed
Medium



Hair Gel



Dispersed
Phase



Dispersed
Medium



Blood



Dispersed
Phase



Dispersed
Medium



Shaving Cream



Dispersed
Phase

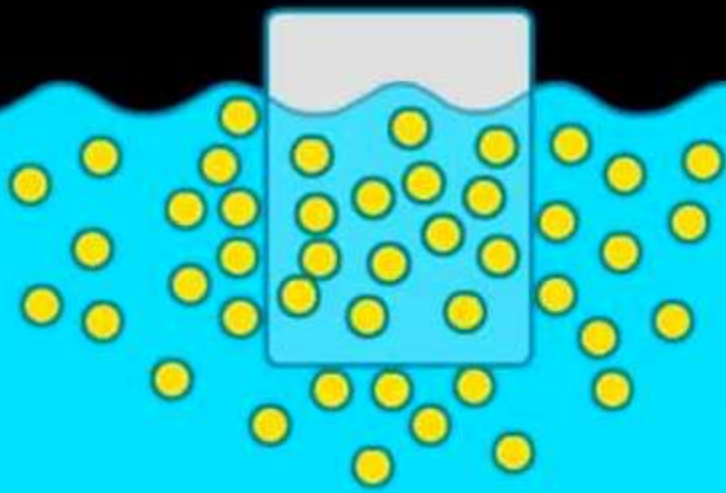


Dispersed
Medium

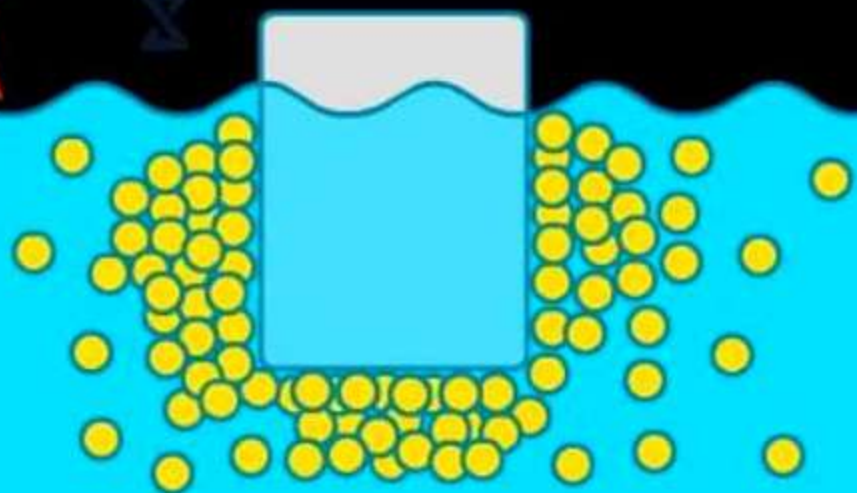
Absorption

Adsorption

VS



Absorption



Adsorption

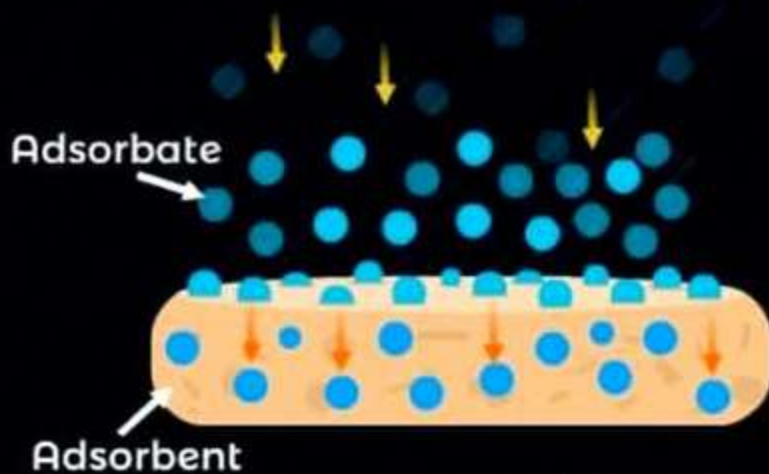
Absorption is the distribution of a certain substance through out the volume of another substrate.

Absorbed materials can be separated between different phases due to their chemical characteristics.

Adsorption is the attachment of a certain substance onto the surface of another substrate.

Adsorption can be used to separate out certain material by adsorbing them onto one surface.

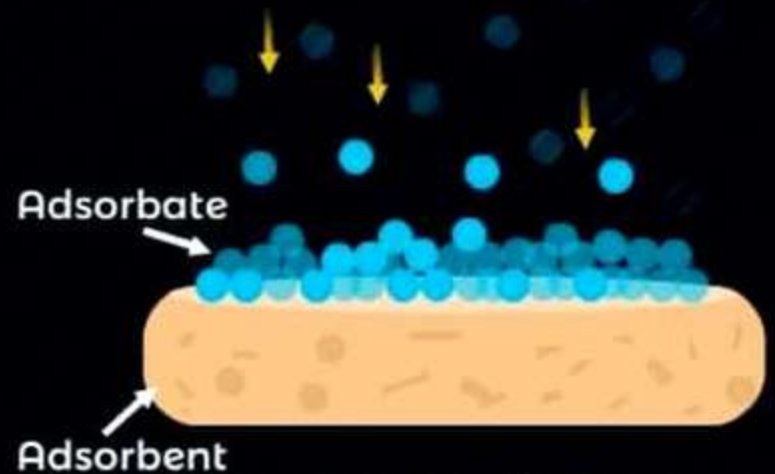
ILLUSTRATION



Materials get absorbed into other material through the availability of space and also due to the properties of each material.

Absorbed materials remain within the space of the absorbent without having a chemical relationship being perfectly intact as when it is free

ILLUSTRATION



The vacant surface of the adsorbent has extra energy that stimulates the attachment of foreign molecules on to it.

Adsorbates bind onto the surface of the adsorbent most commonly through van der waals interactions or through covalent bonds.



THERMODYNAMICS

Thermodynamics deals with energy interactions between two bodies & its effect on the properties of matter.

SCOPE OF THERMODYNAMIC

● Feasibility of a process

● Extent of a process



● Efficiency of a process

SYSTEM

The part of the universe under thermodynamical observation is called system.

SURROUNDINGS

All the part of the universe except system is called surroundings.

BOUNDARY

The part which separates system and surroundings is called boundary. It may be rigid or flexible.

TYPES OF THERMODYNAMIC PROCESSES

● QUASI-STATIC PROCESS

Arbitrarily slow process such that the system always stays arbitrarily close to thermodynamic equilibrium.



● REVERSIBLE PROCESS

Any changes induced by the process in the universe (system + environment) can be removed by retracing its path.

Reversible processes must be quasi-static.

● IRREVERSIBLE PROCESS

Any process in which a part or whole of process is not reversible.

E.g. : any process involving friction, free expansion of gas etc.



BASIC THERMODYNAMIC PROCESS

● ISOBARIC: Constant P

$$W = p\Delta V$$

● ISOTHERMAL: Constant T

$$\Delta U = 0 \text{ (for ideal gases)}$$

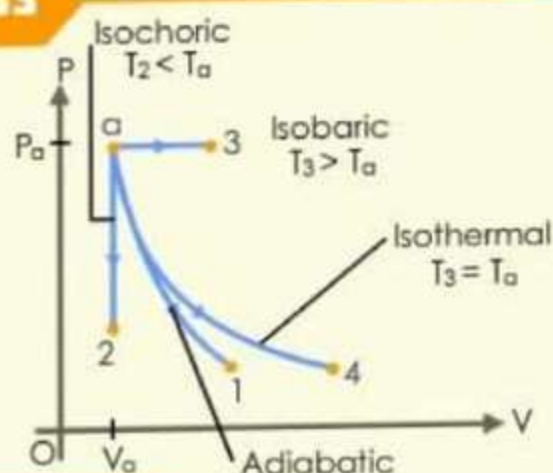
● ISOCHORIC: Constant V

$$W = 0$$

● ADIABATIC: No heat exchange

$$Q = 0$$

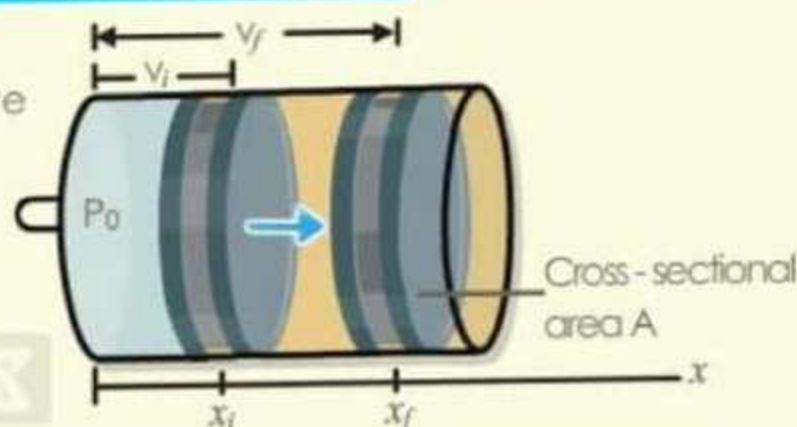
● There are an infinite number of other processes without any special name!



WORK DONE IN THERMODYNAMIC PROCESS

Work done compressing a system is defined to be positive

$$W_{V_i - V_f} = \int_{V_i}^{V_f} P dV = P_0 \int_{V_i}^{V_f} P dV = P_0 \cdot W_{V_i - V_f}$$



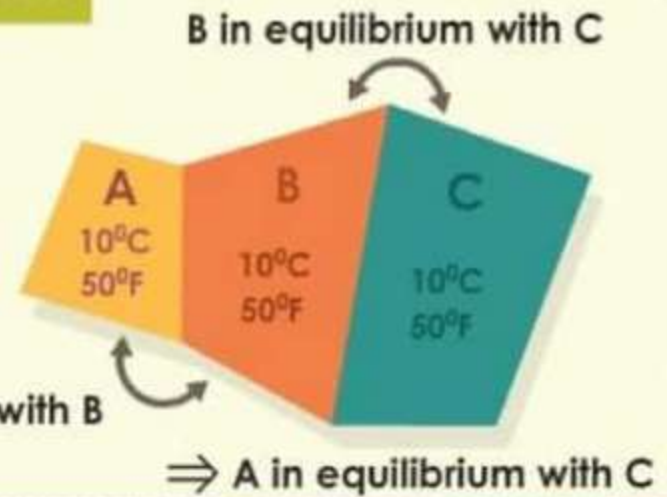


THERMODYNAMICS LAWS

The branch of **physical science** that deals with the relations between heat and other forms of energy (such as mechanical, electrical or chemical energy) and by extension of relationship between them.

ZEROth LAW OF THERMODYNAMICS

If two systems are in **thermal equilibrium** with a third system, then they all are in **thermal equilibrium** with each other. This law helps define the notion of temperature.



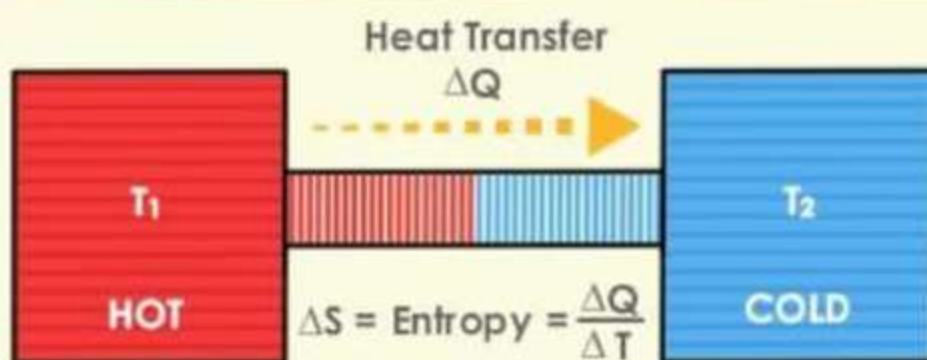
THE FIRST LAW OF THERMODYNAMICS

The first Law of thermodynamics states that overall amount of energy is **Conserved**. Therefore, **energy cannot be created or destroyed**, only lost to an outside system.

$$\Delta U = Q - W$$



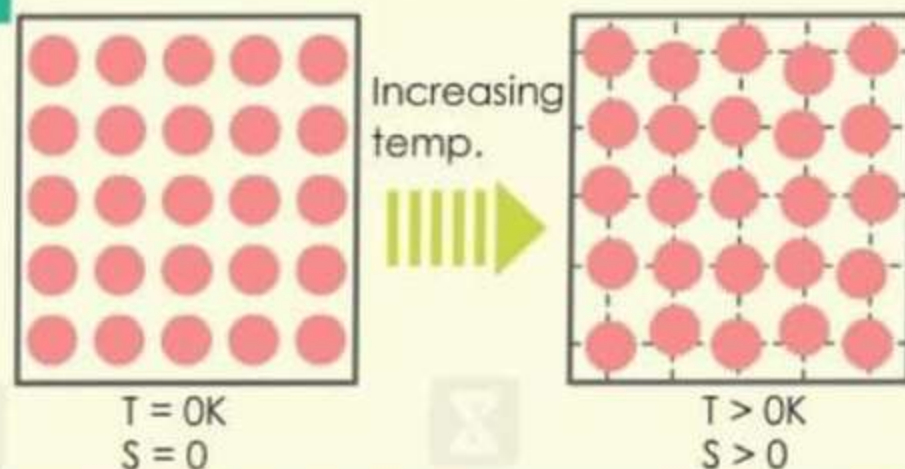
THE SECOND LAW OF THERMODYNAMICS



This law states that **energy naturally flows** from hotter objects to cooler objects. In order for energy to flow from a cooler object to a hotter object, work must be done. When heat is converted into work, the **efficiency** or output of usable work will always be less than 100%.

THE THIRD LAW OF THERMODYNAMICS

The **entropy** of a system approaches a constant value as the temperature approaches **absolute zero**. With the exception of non-crystalline solids (glasses), the entropy of a system at absolute zero is typically close to **zero** and is equal to the logarithm of the product of the **quantum ground states**.



EXOTHERMIC

An exothermic reaction occurs when the temperature of a system increases due to the evolution of heat.



VS

ENDOTHERMIC

An endothermic reaction occurs when the temperature of an isolated system decreases while the surroundings of a non-isolated system gains heat.



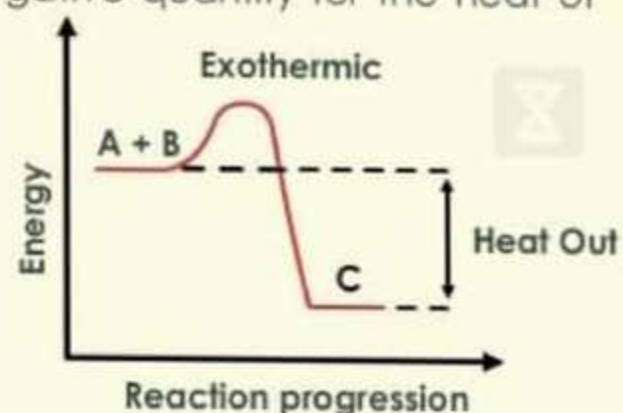
EXOTHERMIC

Heat is released into the surroundings, resulting in an overall negative quantity for the heat of reaction.

An exothermic reaction has a negative ΔH by convention, because the enthalpy of the products is lower than the enthalpy of the reactants of the system.



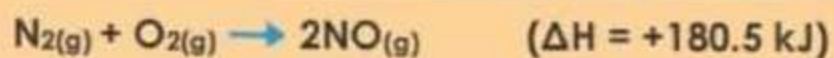
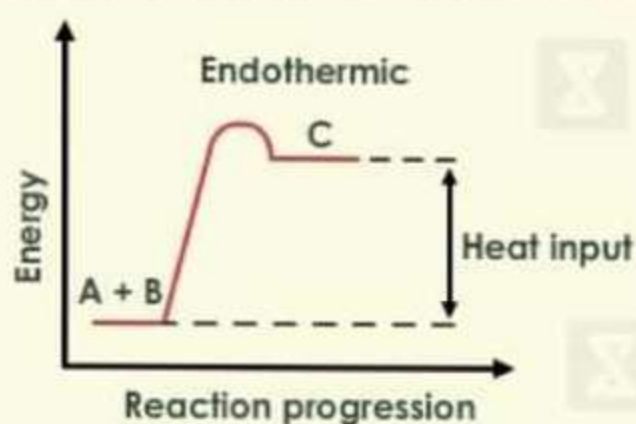
The enthalpies are less than zero.



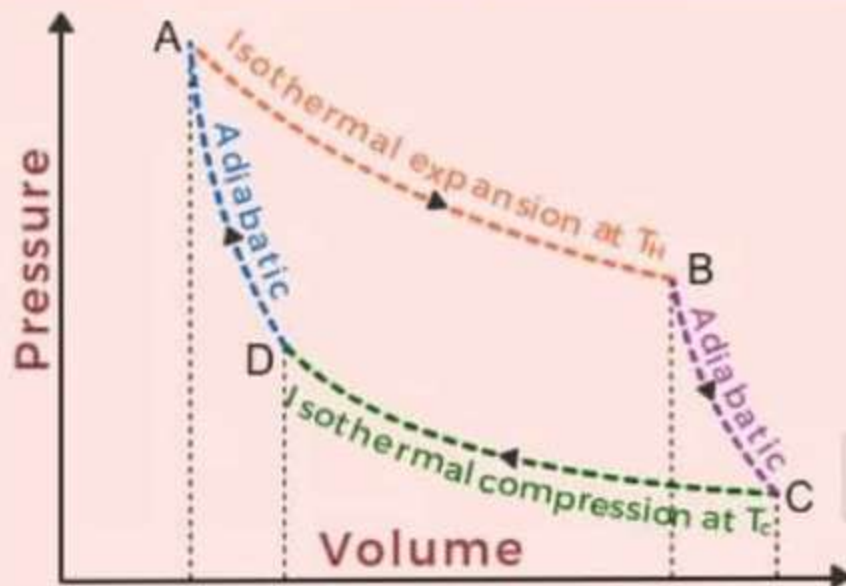
ENDOTHERMIC

Endothermic reactions result in an overall positive heat of reaction.

An endothermic reaction has a positive ΔH , because the enthalpy of the products is higher than the enthalpy of the reactants of the system.



EXOTHERMIC	ENDOTHERMIC
Making ice cubes	Melting ice cubes
Formation of snow in clouds	Conversion of frost to water vapour
Condensation of rain from water vapour	Evaporation of water
A candle flame	Forming a cation from an atom in the gas phase
Mixing sodium sulphite and bleach	Baking bread
Rusting iron	Cooking an egg
Burning sugar	Producing sugar by photosynthesis
Forming ion pairs	Separating ion pairs
Combining atoms to make a molecule in the gas phase	Splitting a gas molecule apart
Mixing water and strong acids	Mixing water and ammonium nitrate
Mixing water with an anhydrous salt	Making an anhydrous salt from a hydrate
Crystallizing liquid salts (as in sodium acetate in chemical handwarmers)	Melting solid salts

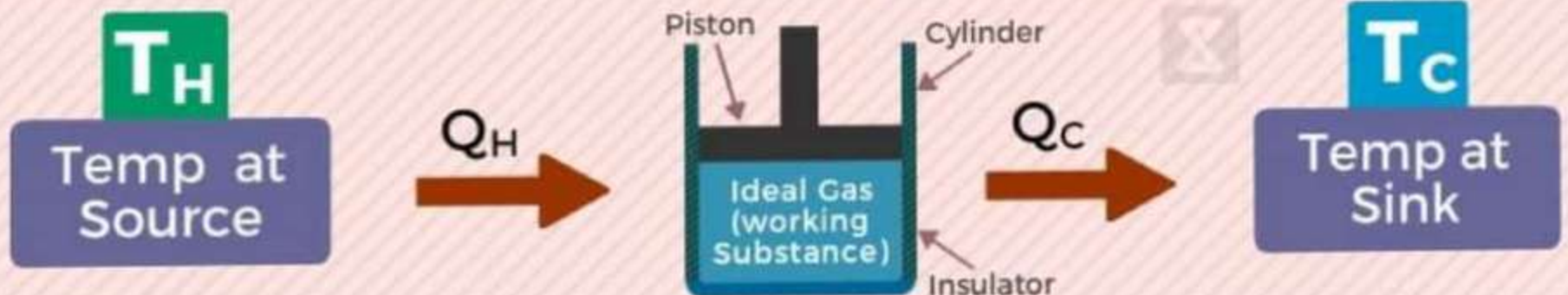


A Carnot heat engine is an engine that operates on the **reversible Carnot cycle**. The basic model for this engine was developed by **Nicolas Léonard Sadi Carnot** in 1824.

It is an ideal **heat engine** whose efficiency is less than 100%.

$$\text{Efficiency of Carnot Engine} = \frac{T_H - T_C}{T_H} \times 100\%$$

PARTS OF CARNOT ENGINE



Carnot engine diagram shows that an amount of heat ' Q_H ' flows from a **high temperature** ' T_H ' furnace through the fluid of the "**working body**" (ideal gas) and the remaining heat ' Q_C ' flow into the cold sink ' T_C ', thus forcing the working substance to do mechanical work ' W ' on the surroundings, via cycles of contractions and expansions.

CYLINDER

It is a hollow cylinder whose walls are bad conductors of heat, and its base is a good conductor of heat.

PISTON

It is a movable piston which is fixed in a hollow cylinder. We neglect the friction force between the piston and walls of the cylinder.

SINK

It is a low-temperature reservoir; system rejects heat to the sink during iso-thermal compression. The thermal capacity of the sink is infinity.

SOURCE

It is a perfect insulator in which thermal conductivity is zero. System is placed on an insulator during adiabatic expansion and adiabatic compression.

INSULATOR

It is a high-temperature reservoir; system absorbs heat from the source during iso-thermal expansion. The thermal capacity of the source is infinity.